Group A: Early progression

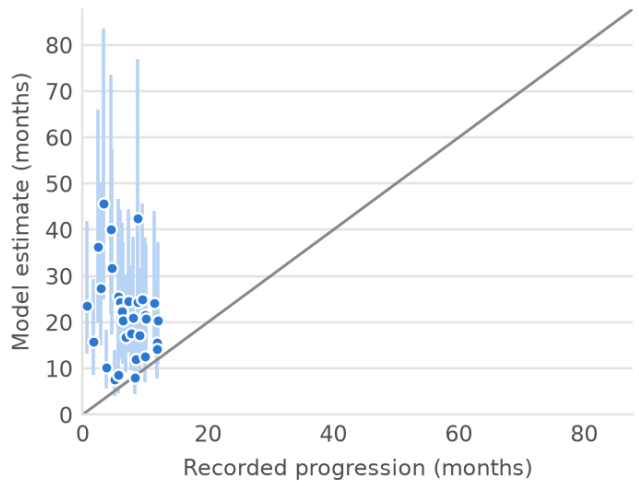
PrimeKG Patient Reasoning · Precision Oncology Research

Group A · 32 patients

Report date 14 Oct 2026

Group A · Early progression recorded progression-free survival: ≤ 12 months

Patients	Median model risk percentile	Recorded progression	Model window agreement
32	**80**	**7.1 mo**	**16%**
3.0% of the cohort	cohort-wide risk ranking, out-of-fold	median; range 0.7–12.0 mo	estimate in this window · median |error| 13 mo



Most frequent path end-points in this group	Patients
Neutrophil degranulation	4
Interleukin-4 and Interleukin-13 signaling	3
PI3K Cascade	3
Olfactory Signaling Pathway	3
Extra-nuclear estrogen signaling	3
PI-3K cascade:FGFR1	3
PI-3K cascade:FGFR4	2
SUMOylation of DNA damage response and repair proteins	2

Rules firing most often in this group	HR	Patients
F1-R7: M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	0.36	7
F4-R4: 26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	0.38	6
F4-R3: Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	0.36	5
F5-R1: M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	0.32	4
F5-R2: M stage = M0 AND 26.5 ≤ Age < 83.5	0.35	4
F1-R2: N stage = N0 AND Age ≥ 31.5	0.36	4

Most common driver alterations

TP53 mutated (41%), PIK3CA mutated (31%), ERBB2 amplified (22%), PTEN deleted (16%), MYC amplified (16%), CDH1 mutated (12%), PTEN mutated (12%), GATA3 mutated (12%), CCND1 amplified (12%), FGFR1 amplified (9%)

Patient index

Patient	Age	Stage	Subtype	Recorded	Model estimate	Window	Risk pct	Report file
TCGA-D8-A73W	79	III	LumA	0.7 mo	23 mo	✗	80	A_progression_within_12m/TCGA-D8-A73W.pdf
TCGA-E2-A152	56	I	Her2	1.8 mo	16 mo	✗	50	A_progression_within_12m/TCGA-E2-A152.pdf
TCGA-E2-A10A	41	II	LumA	2.5 mo	36 mo	✗	28	A_progression_within_12m/TCGA-E2-A10A.pdf
TCGA-E9-A1NF	60	II	LumB	3.0 mo	27 mo	✗	41	A_progression_within_12m/TCGA-E9-A1NF.pdf
TCGA-A2-A04P	36	III	Basal	3.4 mo	46 mo	✗	97	A_progression_within_12m/TCGA-A2-A04P.pdf
TCGA-UU-A93S	63	IV	Her2	3.8 mo	10 mo	✓	99	A_progression_within_12m/TCGA-UU-A93S.pdf
TCGA-C8-A137	34	II	Her2	4.5 mo	40 mo	✗	64	A_progression_within_12m/TCGA-C8-A137.pdf
TCGA-AO-A0JA	36	III	LumA	4.7 mo	32 mo	✗	70	A_progression_within_12m/TCGA-AO-A0JA.pdf
TCGA-E2-A1LK	84	III	Basal	5.1 mo	8 mo	✓	91	A_progression_within_12m/TCGA-E2-A1LK.pdf
TCGA-A7-A3RF	79	II	LumA	5.7 mo	25 mo	✗	8	A_progression_within_12m/TCGA-A7-A3RF.pdf
TCGA-AC-A23H	90	II	Her2	5.7 mo	9 mo	✓	80	A_progression_within_12m/TCGA-AC-A23H.pdf
TCGA-Z7-A8R5	61	III	LumA	6.0 mo	24 mo	✗	67	A_progression_within_12m/TCGA-Z7-A8R5.pdf
TCGA-E2-A14Z	64	I	LumA	6.2 mo	23 mo	✗	4	A_progression_within_12m/TCGA-E2-A14Z.pdf
TCGA-LL-A73Z	55	IV	Normal	6.3 mo	22 mo	✗	100	A_progression_within_12m/TCGA-LL-A73Z.pdf
TCGA-E2-A10B	67	II	LumA	6.4 mo	20 mo	✗	16	A_progression_within_12m/TCGA-E2-A10B.pdf
TCGA-EW-A1P8	58	III	Basal	6.8 mo	17 mo	✗	87	A_progression_within_12m/TCGA-EW-A1P8.pdf
TCGA-BH-A18T	70	II	Basal	7.4 mo	24 mo	✗	61	A_progression_within_12m/TCGA-BH-A18T.pdf
TCGA-AC-A2QJ	48	III	missing	7.8 mo	18 mo	✗	91	A_progression_within_12m/TCGA-AC-A2QJ.pdf
TCGA-A8-A091	61	II	LumA	8.0 mo	21 mo	✗	4	A_progression_within_12m/TCGA-A8-A091.pdf
TCGA-A2-A0T2	66	IV	Basal	8.4 mo	8 mo	✓	99	A_progression_within_12m/TCGA-A2-A0T2.pdf
TCGA-LL-A5YM	88	III	LumB	8.5 mo	12 mo	✓	90	A_progression_within_12m/TCGA-LL-A5YM.pdf
TCGA-AO-A126	39	II	LumA	8.8 mo	42 mo	✗	30	A_progression_within_12m/TCGA-AO-A126.pdf
TCGA-AR-A1AR	50	III	Basal	8.8 mo	24 mo	✗	88	A_progression_within_12m/TCGA-AR-A1AR.pdf
TCGA-OL-A97C	67	II	Normal	9.0 mo	17 mo	✗	25	A_progression_within_12m/TCGA-OL-A97C.pdf
TCGA-A7-A26H	72	II	LumA	9.6 mo	25 mo	✗	24	A_progression_within_12m/TCGA-A7-A26H.pdf
TCGA-AR-A5QQ	68	III	Basal	9.9 mo	21 mo	✗	92	A_progression_within_12m/TCGA-AR-A5QQ.pdf
TCGA-D8-A1Y1	80	III	LumB	9.9 mo	13 mo	✗	80	A_progression_within_12m/TCGA-D8-A1Y1.pdf
TCGA-E9-A243	52	II	Basal	10.0 mo	21 mo	✗	59	A_progression_within_12m/TCGA-E9-A243.pdf
TCGA-AC-A62V	58	IV	missing	11.4 mo	24 mo	✗	98	A_progression_within_12m/TCGA-AC-A62V.pdf
TCGA-E2-A1LE	71	III	missing	11.9 mo	16 mo	✗	94	A_progression_within_12m/TCGA-E2-A1LE.pdf
TCGA-B6-A0I9	62	IV	Her2	11.9 mo	14 mo	✗	100	A_progression_within_12m/TCGA-B6-A0I9.pdf
TCGA-BH-A1EV	45	III	Her2	12.0 mo	20 mo	✗	96	A_progression_within_12m/TCGA-BH-A1EV.pdf

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

Sample: **TCGA-D8-A73W-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.3** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

80th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 27%

Model-estimated progression time

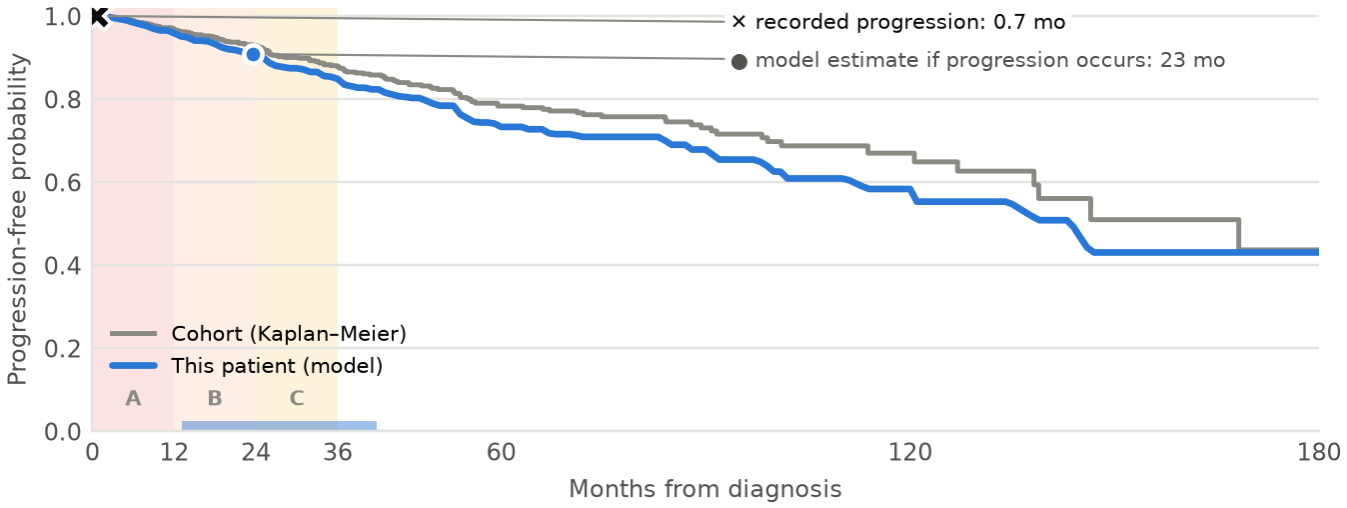
≈ 23 months

Window 12–24 months · IQR 13–42 mo
if progression occurs

Recorded progression

0.7 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 23 mo



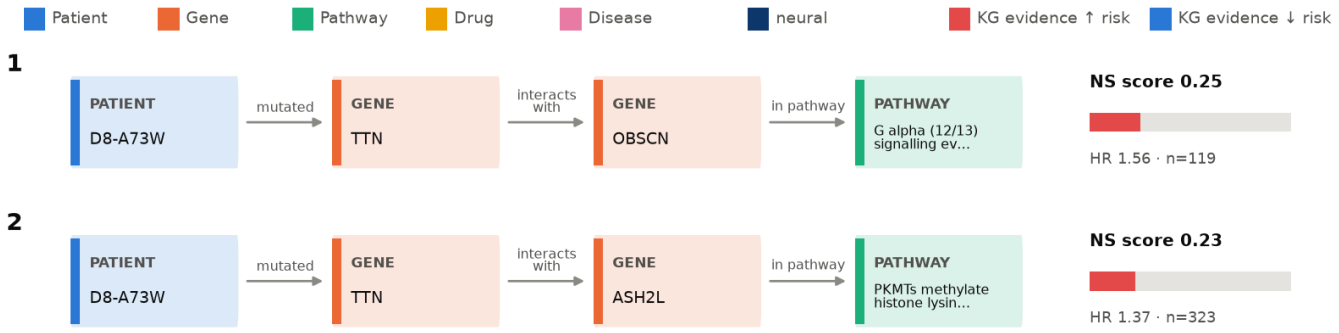
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **80th risk percentile** (high tier). The most significant rule covering them (*40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TTN** to **G alpha (12/13) signalling events**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.56). Progression was recorded at 0.7 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

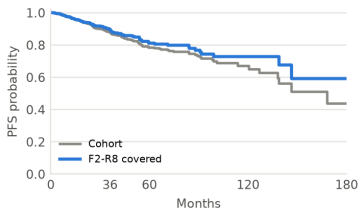


None of the 25 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-D8-A73W** mutated → **TTN** interacts with → **OBSCN** in pathway → **G alpha (12/13) signalling events**	0	↑ faster progression HR 1.56, z +1.9, n = 119 carriers of “G alpha (12/13) signalling events”	**0.25**
2	**TCGA-D8-A73W** mutated → **TTN** interacts with → **ASH2L** in pathway → **PKMTs methylate histone lysines**	0	↑ faster progression HR 1.37, z +1.7, n = 323 carriers of “PKMTs methylate histone lysines”	**0.23**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

Rule	IF …	Covered	HR vs rest	Median PFS
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TTN	Mutation	p.E18412A	2	0.30



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3KD	47	III	LumA	0.70	no progression, followed 39.6 mo
TCGA-AC-A3W6	90	III	missing	0.68	no progression, followed 19.8 mo
TCGA-AC-A2QI	76	III	missing	0.67	no progression, followed 19.3 mo
TCGA-BH-A1FR	73	III	LumA	0.66	progressed at 54.0 mo (group D)
TCGA-LQ-A4E4	73	III	missing	0.65	progressed at 22.4 mo (group B)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**56**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A152-01** · Aneuploidy score: **25** · TMB (nonsyn., /Mb): **4.13** · MSIsensor score: **0.43** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

50th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 18%

Model-estimated progression time

≈ 16 months

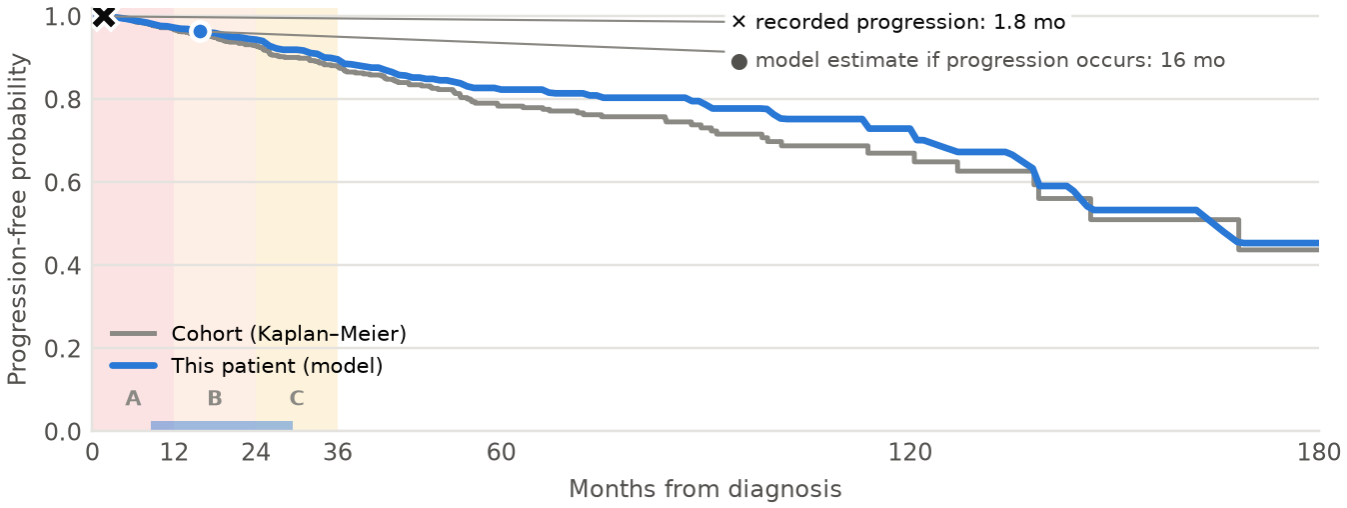
Window 12–24 months · IQR 9–29 mo if progression occurs

Recorded progression

1.8 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 14 mo



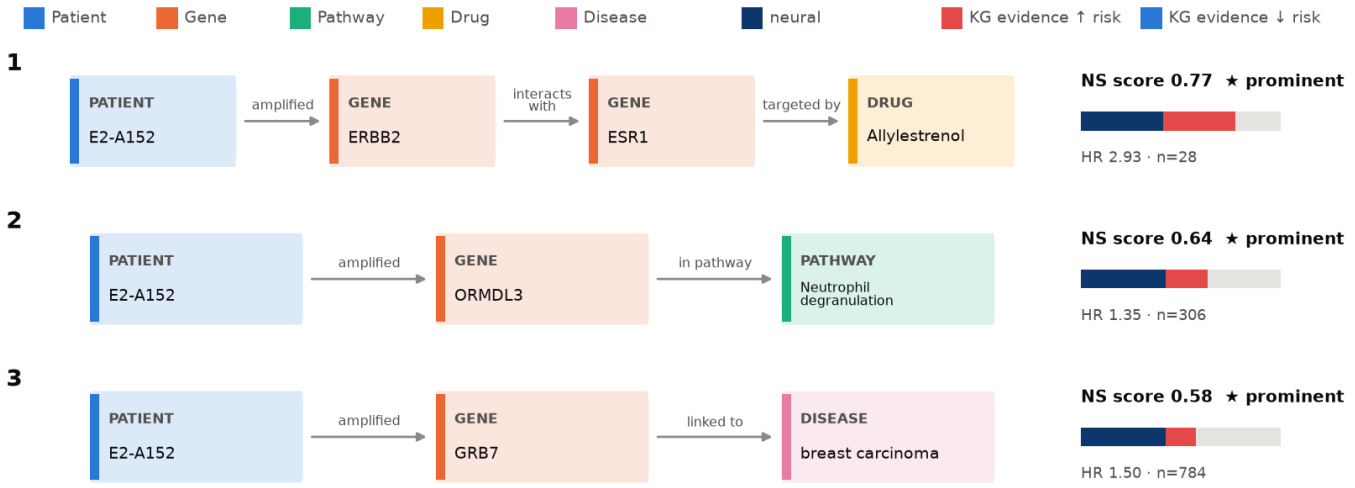
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 56-year-old patient has a Her2 tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **50th risk percentile** (intermediate tier). The most significant rule covering them (*N stage = N0 AND 41.5 ≤ Age < 78.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **Allylestrenol**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.93). Progression was recorded at 1.8 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

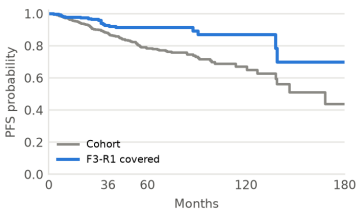


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A152** amplified → **ERBB2** interacts with → **ESR1** targeted by → **Allylestrenol**	69	↑ faster progression HR 2.93, z +2.7, n = 28 carriers of “Allylestrenol”	**0.77**
2 ★	**TCGA-E2-A152** amplified → **ORMDL3** in pathway → **Neutrophil degranulation**	71	↑ faster progression HR 1.35, z +1.6, n = 306 carriers of “Neutrophil degranulation”	**0.64**
3 ★	**TCGA-E2-A152** amplified → **GRB7** linked to → **breast carcinoma**	71	↑ faster progression HR 1.50, z +1.1, n = 784 carriers of “breast carcinoma”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

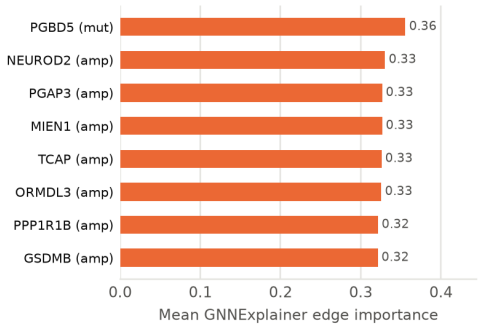
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R1 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	351 pts 22 events	**0.30** p 5.2e-08	not reached
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R2 ↓ lower risk	N stage = N0	418 pts 33 events	**0.37** p 4.8e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.K132E	57	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
PIK3CA · driver	Mutation	p.X764_splice	70	0.32
PGBD5	Mutation	p.*42*	0	0.36
NEUROD2	High-level amplification	GISTIC ±2	0	0.33
PGAP3	High-level amplification	GISTIC ±2	0	0.33
MIEN1	High-level amplification	GISTIC ±2	0	0.33
TCAP	High-level amplification	GISTIC ±2	1	0.33
ORMDL3	High-level amplification	GISTIC ±2	2	0.33
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
GRB7	High-level amplification	GISTIC ±2	12	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 49 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A1AX	64	I	LumA	0.84	no progression, followed 86.4 mo
TCGA-B6-A0RH	51	II	Her2	0.82	progressed at 71.2 mo (group D)
TCGA-D8-A1J9	48	I	LumB	0.82	no progression, followed 17.5 mo
TCGA-A2-A3XZ	46	I	Her2	0.81	no progression, followed 50.4 mo
TCGA-E2-A1IN	60	I	LumA	0.80	no progression, followed 22.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**41**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**—**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

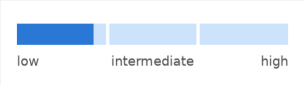
Sample: **TCGA-E2-A10A-01** · Aneuploidy score: **6** · TMB (nonsyn., /Mb): **0.73** · MSIsensor score: **0.26** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

28th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

≈ 36 months

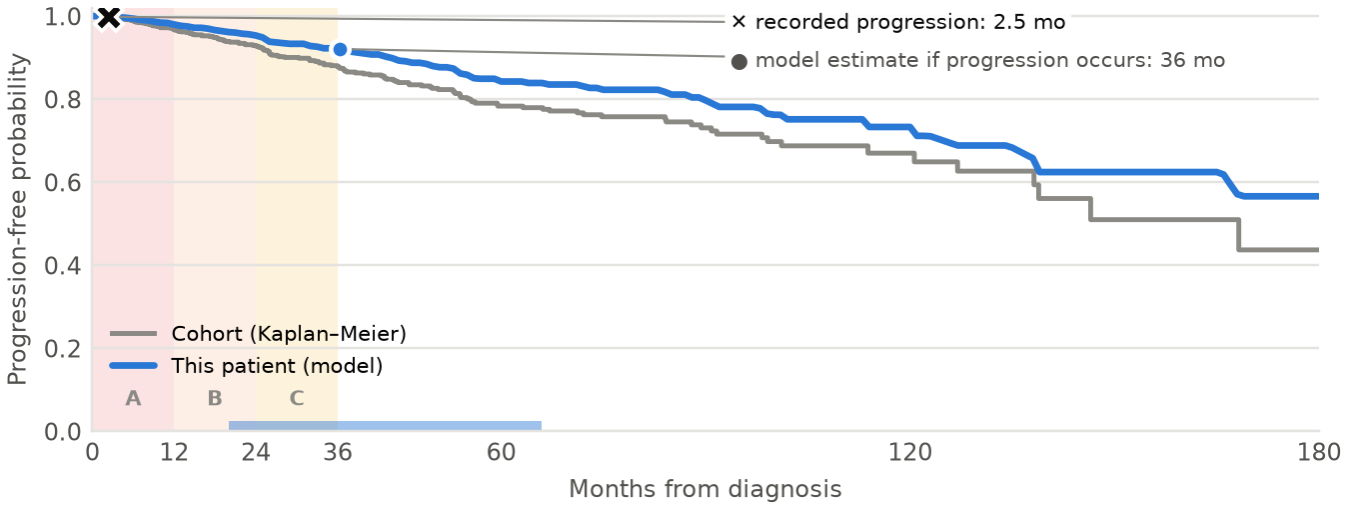
Window > 36 months · IQR 20–66 mo if progression occurs

Recorded progression

2.5 months

■ **Group A** · Early progression

✗ recorded window differs · |error| 34 mo



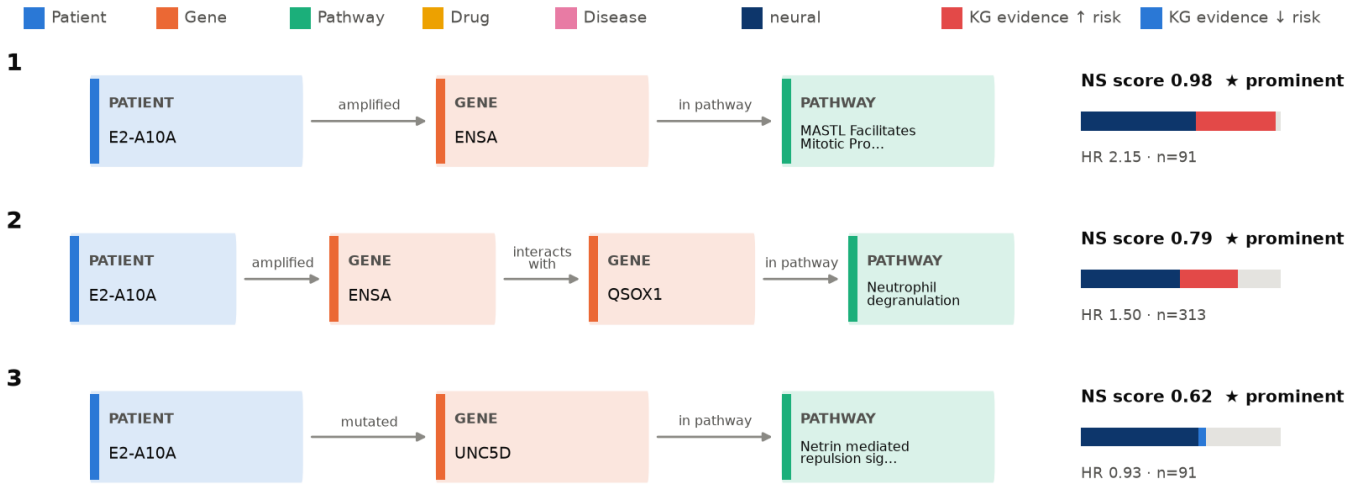
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 41-year-old patient has a LumA tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **28th risk percentile** (low tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ENSA** to **MASTL Facilitates Mitotic Progression**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.15). Progression was recorded at 2.5 months; the model's estimate given progression was 36 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

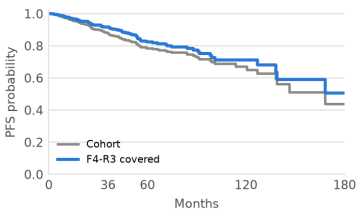


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A10A** amplified → **ENSA** in pathway → **MASTL Facilitates Mitotic Progression**	96	↑ **faster progression** HR 2.15, z +3.3, n = 91 carriers of “MASTL Facilitates Mitotic Progression”	**0.98**
2 ★	**TCGA-E2-A10A** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Neutrophil degranulation**	83	↑ **faster progression** HR 1.50, z +2.2, n = 313 carriers of “Neutrophil degranulation”	**0.79**
3 ★	**TCGA-E2-A10A** mutated → **UNC5D** in pathway → **Netrin mediated repulsion signals**	98	↓ **slower progression** HR 0.93, z -0.3, n = 91 carriers of “Netrin mediated repulsion signals”	**0.62**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

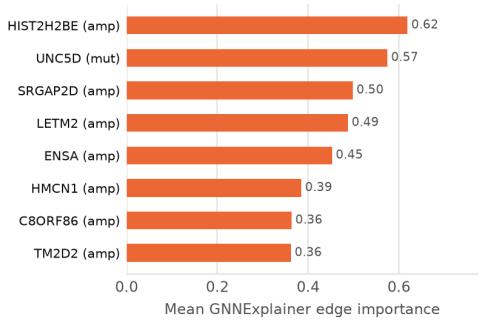
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R1 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.38 AND Age ≥ 40.5	355 pts 26 events	**0.39** p 7.7e-06	not reached
F4-R2 ↓ lower risk	N stage = N0 AND 34.5 ≤ Age < 83.5	386 pts 32 events	**0.43** p 2.5e-05	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P425Afs*82	13	0.32
ESR1 · driver	Mutation	p.P29Sfs*79	141	0.31
MAP3K1 · driver	Mutation	p.G172Rfs*7	13	0.31
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.31
HIST2H2BE	High-level amplification	GISTIC ±2	0	0.62
UNC5D	Mutation	p.C354F	1	0.57
SRGAP2D	High-level amplification	GISTIC ±2	0	0.50
LETM2	High-level amplification	GISTIC ±2	0	0.49
ENSA	High-level amplification	GISTIC ±2	1	0.45
HMCN1	High-level amplification	GISTIC ±2	0	0.39
C8ORF86	High-level amplification	GISTIC ±2	0	0.36
TM2D2	High-level amplification	GISTIC ±2	0	0.36



49 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 137 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A1-A0SD	59	II	LumA	0.71	no progression, followed 14.4 mo
TCGA-OL-A5D8	40	II	LumA	0.71	no progression, followed 32.0 mo
TCGA-B6-A0IM	75	II	LumA	0.69	no progression, followed 127.3 mo
TCGA-A8-A090	74	II	LumA	0.68	no progression, followed 0.0 mo
TCGA-AN-A0AM	56	II	LumB	0.66	no progression, followed 0.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E9-A1NF · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**60**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8522/3**

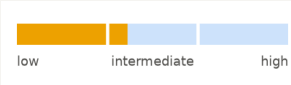
Sample: **TCGA-E9-A1NF-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **1.7** · MSIsensor score: **0.77** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

41st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

9% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

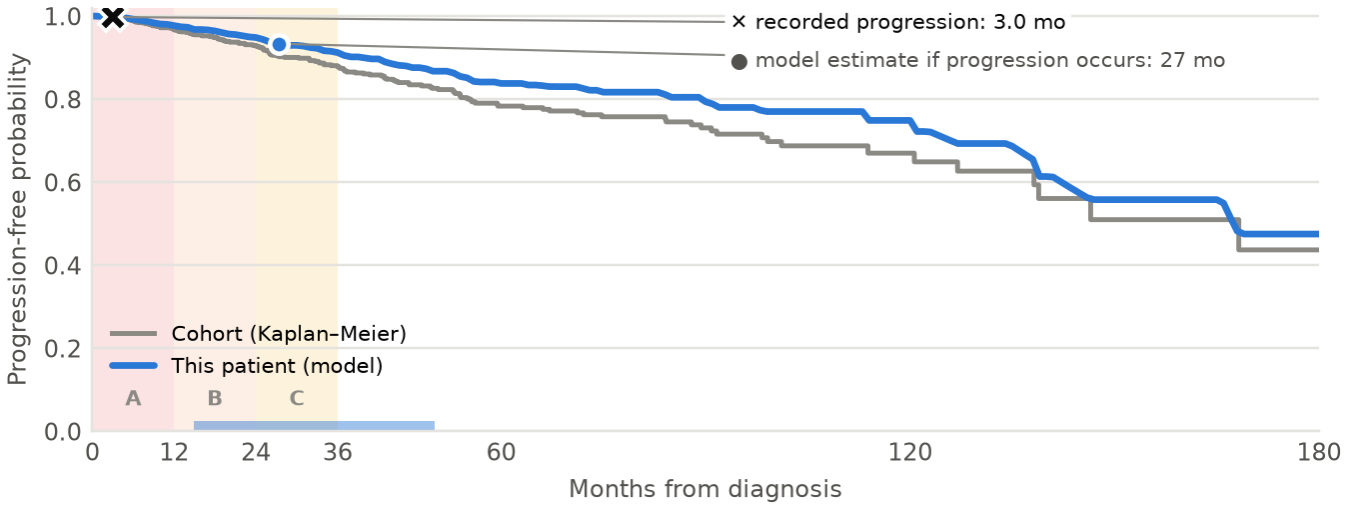
≈ 27 months

Window 24–36 months · IQR 15–50 mo
if progression occurs

Recorded progression

3.0 months

Group A · Early progression
✗ recorded window differs · |error| 24 mo



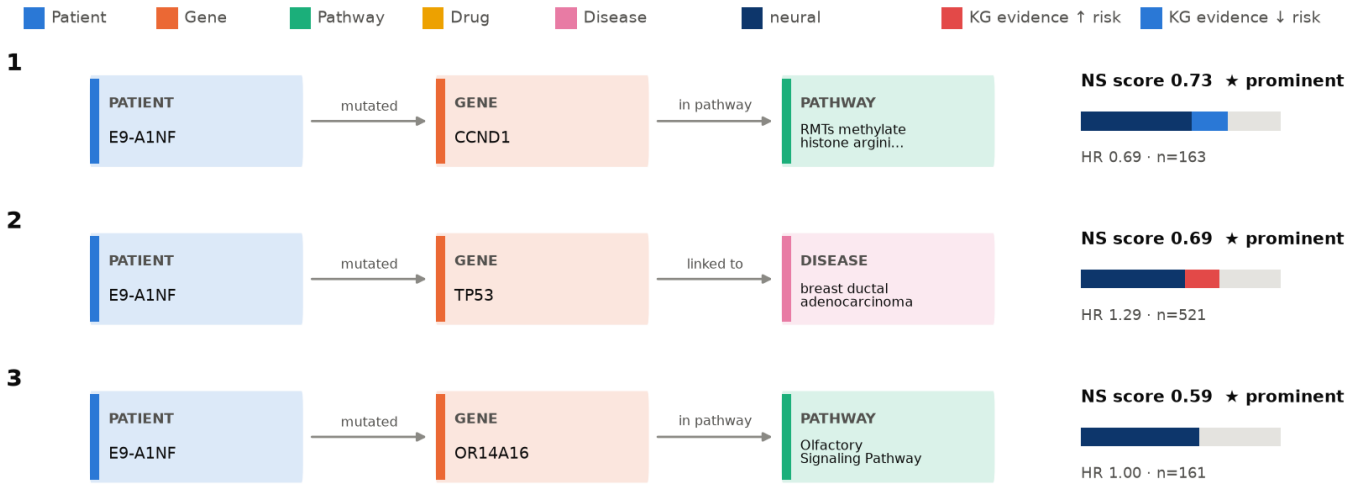
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 60-year-old patient has a LumB tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **41st risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CCND1** to **RMTs methylate histone arginines**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.69). Progression was recorded at 3.0 months; the model's estimate given progression was 27 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

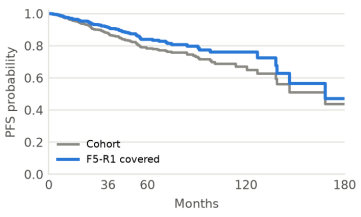


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E9-A1NF** mutated → **CCND1** in pathway → **RMTs methylate histone arginines**	92	↓ **slower progression** HR 0.69, z -1.3, n = 163 carriers of “RMTs methylate histone arginines”	**0.73**
2 ★	**TCGA-E9-A1NF** mutated → **TP53** linked to → **breast ductal adenocarcinoma**	87	↑ **faster progression** HR 1.29, z +1.3, n = 521 carriers of “breast ductal adenocarcinoma”	**0.69**
3 ★	**TCGA-E9-A1NF** mutated → **OR14A16** in pathway → **Olfactory Signaling Pathway**	99	↑ **faster progression** HR 1.00, z +0.0, n = 161 carriers of “Olfactory Signaling Pathway”	**0.59**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

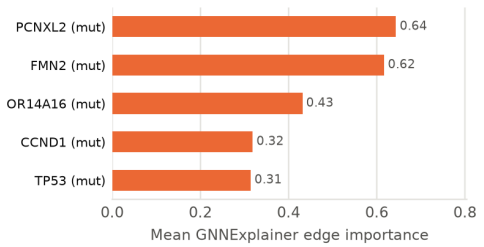
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	Mutation	p.L32=	26	0.32
TP53 · driver	Mutation	p.F113C	57	0.31
PCNXL2	Mutation	p.X897_splice	0	0.64
FMN2	Mutation	p.L959F, p.P814S, p.S747F	0	0.62
OR14A16	Mutation	p.A196T	1	0.43



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 2 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A109	64	II	LumB	0.78	no progression, followed 46.6 mo
TCGA-AO-A1KS	69	II	LumB	0.77	no progression, followed 11.5 mo
TCGA-A2-A4S1	66	II	Basal	0.77	no progression, followed 27.0 mo
TCGA-AR-A0TV	66	II	LumB	0.75	no progression, followed 75.2 mo
TCGA-EW-A2FW	52	II	LumB	0.73	no progression, followed 22.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A04P · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**36**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3C**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-A2-A04P-01** · Aneuploidy score: **9** · TMB (nonsyn., /Mb): **4.27** · MSIsensor score: **1.12** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

97th

High risk tier · fused neuro-symbolic model



Probability of progression

25% by 36 mo

12 mo 6% · 24 mo 14% · 60 mo 44%

Model-estimated progression time

≈ 46 months

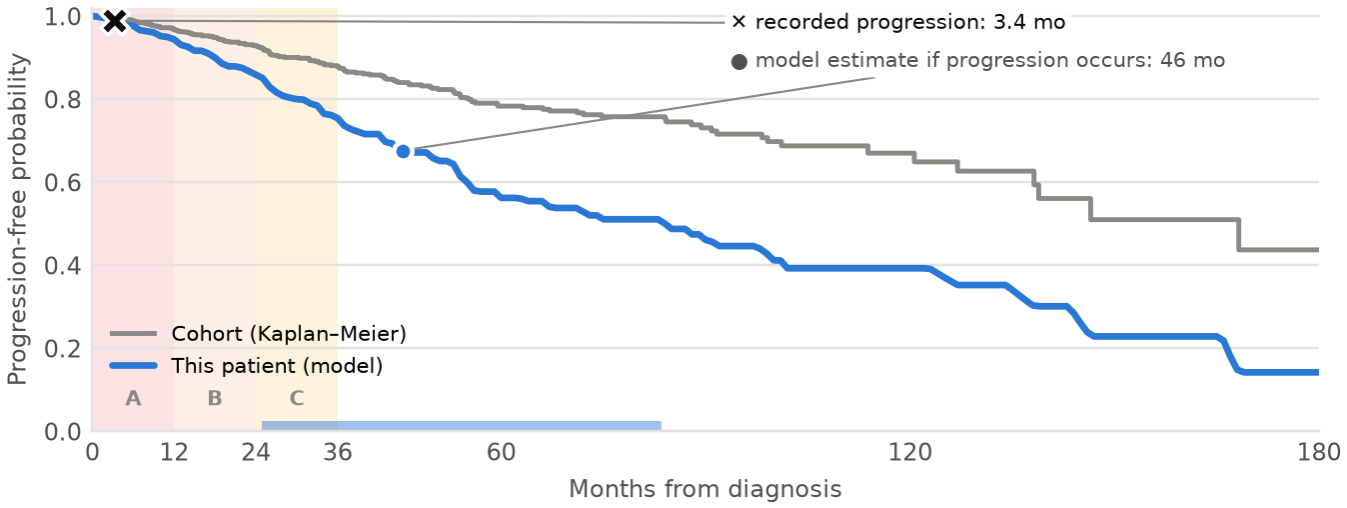
Window > 36 months · IQR 25–84 mo if progression occurs

Recorded progression

3.4 months

Group A · Early progression

✗ recorded window differs · |error| 42 mo



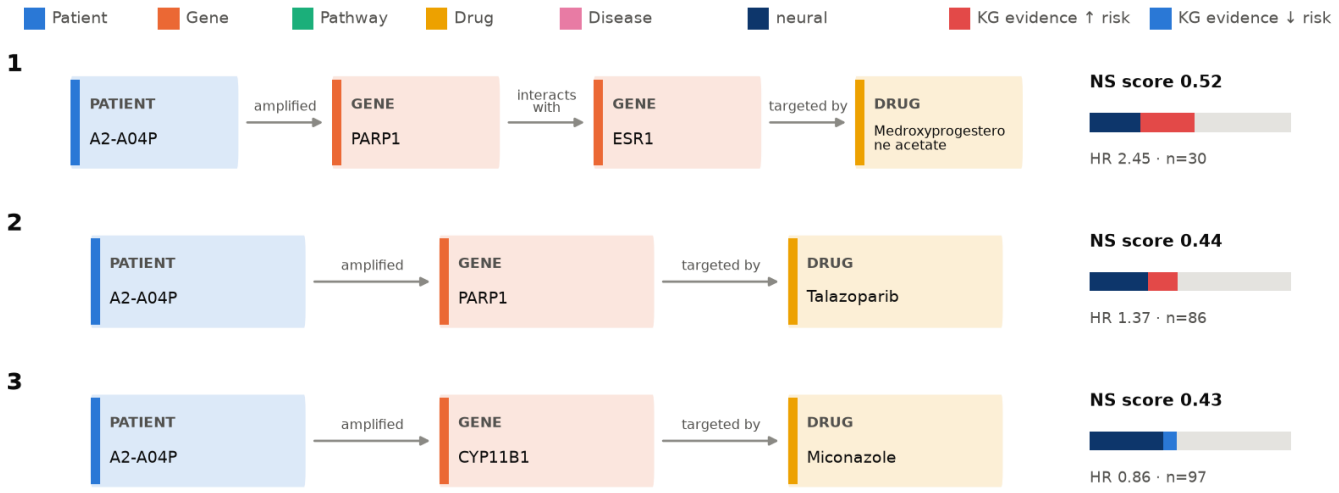
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 36-year-old patient has a Basal tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **97th risk percentile** (high tier). The most significant rule covering them (*N stage = N3 AND 4.0 ≤ Aneuploidy score < 26.5*) marks a group with a progression hazard 5.0× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PARP1** to **Medroxyprogesterone acetate**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.45). Progression was recorded at 3.4 months; the model's estimate given progression was 46 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

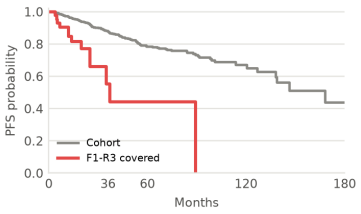


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A2-A04P** amplified → **PARP1** interacts with → **ESR1** targeted by → **Medroxyprogesterone acetate**	42	↑ **faster progression** HR 2.45, z +2.0, n = 30 carriers of "Medroxyprogesterone acetate"	**0.52**
2	**TCGA-A2-A04P** amplified → **PARP1** targeted by → **Talazoparib**	48	↑ **faster progression** HR 1.37, z +1.1, n = 86 carriers of "Talazoparib"	**0.44**
3	**TCGA-A2-A04P** amplified → **CYP11B1** targeted by → **Miconazole**	61	↓ **slower progression** HR 0.86, z -0.5, n = 97 carriers of "Miconazole"	**0.43**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

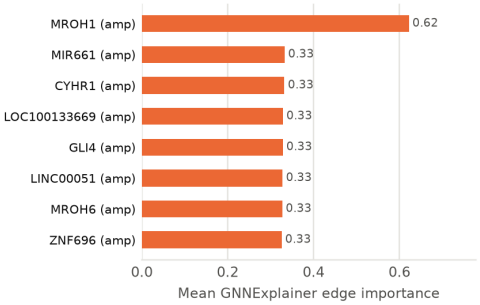
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R3 ↑ **higher risk**	N stage = N3 AND 4.0 ≤ Aneuploidy score < 26.5	46 pts 13 events	**4.96** p 9.3e-08	37 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	High-level amplification	GISTIC ±2	141	0.32
PIK3CA · driver	High-level amplification	GISTIC ±2	70	0.32
PIK3CA · driver	Mutation	p.R88Q	70	0.32
BRCA2 · driver	Mutation	p.Q3206E	14	0.31
RUNX1 · driver	High-level amplification	GISTIC ±2	18	0.31
TP53 · driver	Mutation	p.G108Vfs*15	57	0.31
MROH1	High-level amplification	GISTIC ±2	0	0.62
MIR661	High-level amplification	GISTIC ±2	0	0.33
CYHR1	High-level amplification	GISTIC ±2	0	0.33
LOC100133669	High-level amplification	GISTIC ±2	0	0.33
GLI4	High-level amplification	GISTIC ±2	0	0.33
LINC00051	High-level amplification	GISTIC ±2	0	0.33



149 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 225 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A1L7	40	III	LumB	0.73	progressed at 53.2 mo (group D)
TCGA-GM-A3XL	49	II	Basal	0.73	no progression, followed 69.3 mo
TCGA-AR-A0U4	54	II	Basal	0.72	no progression, followed 107.2 mo
TCGA-AR-A0TS	46	II	Basal	0.72	no progression, followed 84.1 mo
TCGA-AR-A251	51	III	Basal	0.72	no progression, followed 99.6 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-UU-A93S · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**63**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T4D**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

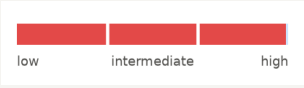
Sample: **TCGA-UU-A93S-01** · Aneuploidy score: **15** · TMB (nonsyn., /Mb): **4.63** · MSIsensor score: **0.48** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

64% by 36 mo

12 mo 23% · 24 mo 46% · 60 mo 89%

Model-estimated progression time

≈ 10 months

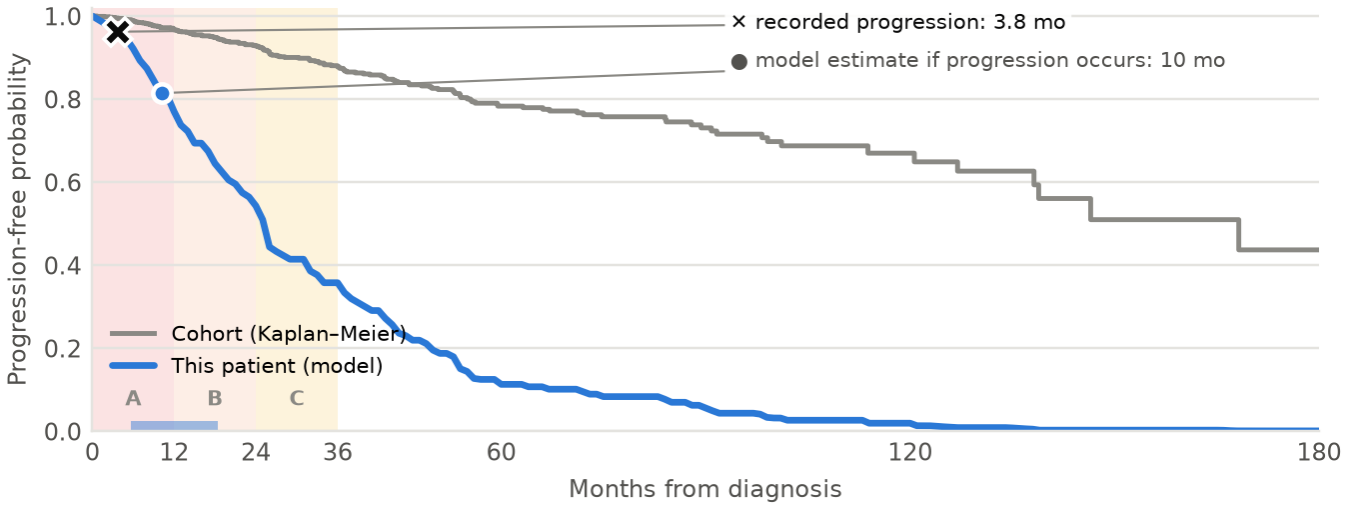
Window ≤ 12 months · IQR 6–18 mo if progression occurs

Recorded progression

3.8 months

■ **Group A** · Early progression

✓ **estimate in recorded window** · |error| 6 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 63-year-old patient has a Her2 tumour, AJCC stage IV (T4, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **MIA3** to **Cargo concentration in the ER**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.60). Progression was recorded at 3.8 months; the model's estimate given progression was 10 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

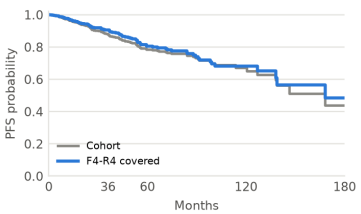


4 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-UU-A93S** mutated → **MIA3** in pathway → **Cargo concentration in the ER**	62	↑ **faster progression** HR 1.60, z +1.9, n = 88 carriers of “Cargo concentration in the ER”	**0.62**
2 ★	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **BRCA1** in pathway → **Presynaptic phase of homologous DNA pairing and strand exchange**	61	↑ **faster progression** HR 1.44, z +1.6, n = 167 carriers of “Presynaptic phase of homologous DNA pairing and strand exchange”	**0.58**
3 ★	**TCGA-UU-A93S** mutated → **TP53** interacts with → **ASH2L** in pathway → **Formation of the beta-catenin:TCF transactivating complex**	60	↑ **faster progression** HR 1.40, z +1.7, n = 186 carriers of “Formation of the beta-catenin:TCF transactivating complex”	**0.58**
4 ★	**TCGA-UU-A93S** amplified → **GATA3** interacts with → **PIK3R1** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	58	↓ **slower progression** HR 0.72, z -1.7, n = 553 carriers of “PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

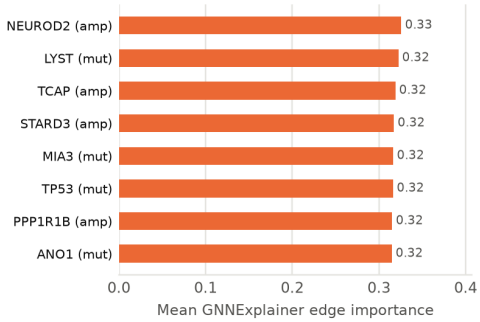
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.Q136P	57	0.32
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NEUROD2	High-level amplification	GISTIC ±2	0	0.33
LYST	Mutation	3'UTR	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
MIA3	Mutation	p.S1706*	3	0.32
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
ANO1	Mutation	p.*37*	2	0.32
ZP4	Mutation	p.Y359N	1	0.31
IKZF3	High-level amplification	GISTIC ±2	0	0.31



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259) … and 39 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A8-A08J	52	IV	Her2	0.70	progressed at 37.1 mo (group D)
TCGA-E2-A14P	79	III	Her2	0.65	no progression, followed 41.0 mo
TCGA-A8-A08X	43	III	Her2	0.61	progressed at 12.0 mo (group B)
TCGA-B6-A0I9	62	IV	Her2	0.61	progressed at 11.9 mo (group A)
TCGA-BH-A0B7	42	II	Her2	0.60	no progression, followed 84.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-C8-A137 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**34**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**—**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-C8-A137-01** · Aneuploidy score: **5** · TMB (nonsyn., /Mb): **1.43** · MSIsensor score: **0.24** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

64th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 21%

Model-estimated progression time

≈ 40 months

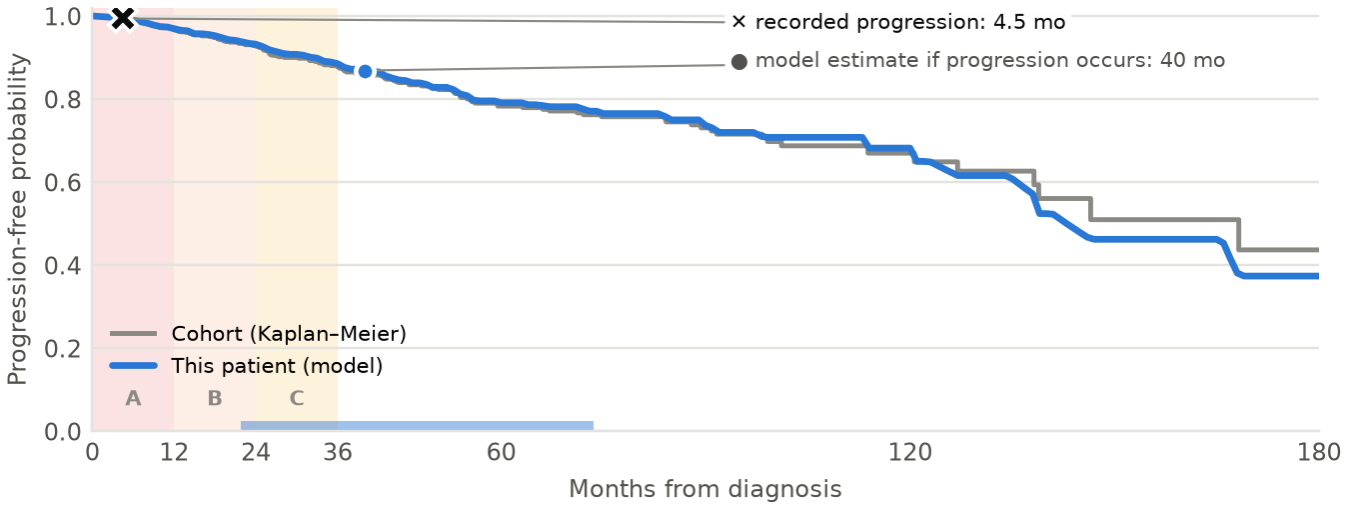
Window > 36 months · IQR 22–74 mo if progression occurs

Recorded progression

4.5 months

Group A · Early progression

✗ recorded window differs · |error| 36 mo



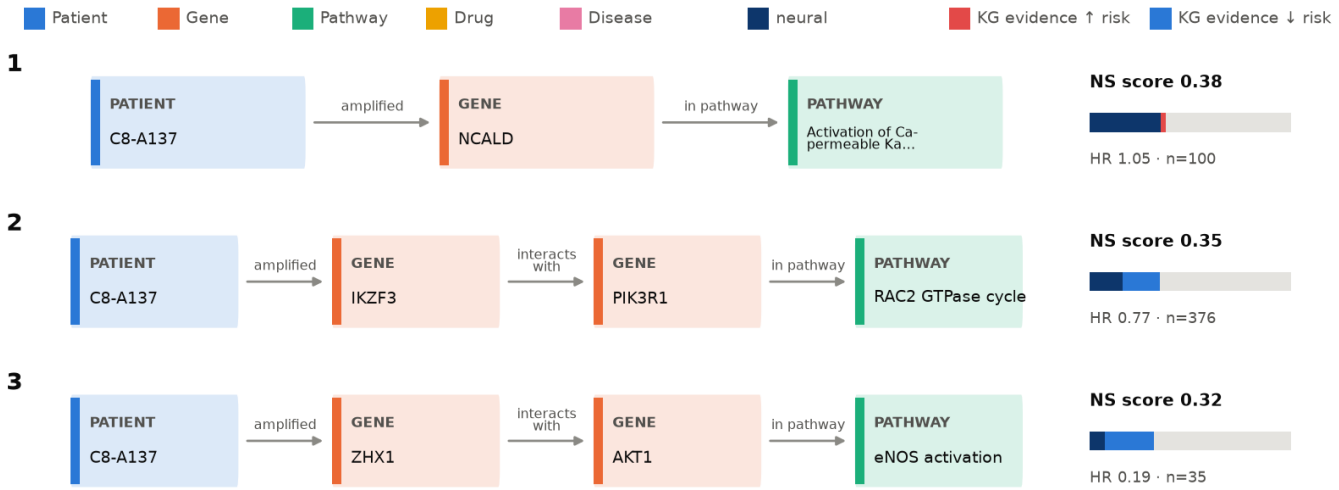
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 34-year-old patient has a Her2 tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **64th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **NCALD** to **Activation of Ca-permeable Kainate Receptor**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.05). Progression was recorded at 4.5 months; the model's estimate given progression was 40 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

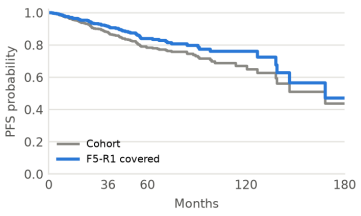


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-C8-A137** amplified → **NCALD** in pathway → **Activation of Ca-permeable Kainate Receptor**	59	↑ **faster progression** HR 1.05, z +0.2, n = 100 carriers of “Activation of Ca-permeable Kainate Receptor”	**0.38**
2	**TCGA-C8-A137** amplified → **IKZF3** interacts with → **PIK3R1** in pathway → **RAC2 GTPase cycle**	27	↓ **slower progression** HR 0.77, z -1.4, n = 376 carriers of “RAC2 GTPase cycle”	**0.35**
3	**TCGA-C8-A137** amplified → **ZHX1** interacts with → **AKT1** in pathway → **eNOS activation**	12	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of “eNOS activation”	**0.32**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

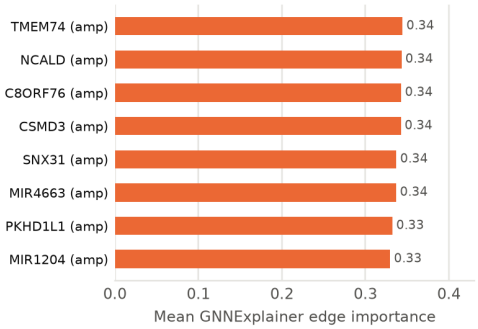
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ **lower risk**	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ **lower risk**	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
TMEM74	High-level amplification	GISTIC ±2	0	0.34
NCALD	High-level amplification	GISTIC ±2	1	0.34
C8ORF76	High-level amplification	GISTIC ±2	0	0.34
CSMD3	High-level amplification	GISTIC ±2	0	0.34
SNX31	High-level amplification	GISTIC ±2	0	0.34
MIR4663	High-level amplification	GISTIC ±2	0	0.34
PKHD1L1	High-level amplification	GISTIC ±2	0	0.33
MIR1204	High-level amplification	GISTIC ±2	0	0.33
AARD	High-level amplification	GISTIC ±2	0	0.33



148 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916) … and 38 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E2-A15S	34	II	LumB	0.81	no progression, followed 14.1 mo
TCGA-C8-A135	64	II	Her2	0.80	no progression, followed 12.9 mo
TCGA-AN-A04C	51	II	Her2	0.79	no progression, followed 1.8 mo
TCGA-C8-A12Z	45	II	Her2	0.78	no progression, followed 12.6 mo
TCGA-AR-A24U	47	II	Her2	0.77	no progression, followed 102.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**36**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**AFR_ADMIX**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AO-A0JA-01** · Aneuploidy score: **10** · TMB (nonsyn., /Mb): **0.7** · MSIsensor score: **0.11** · AJCC edition: —

2 · Progression assessment

Model risk (cohort percentile)

70th

High risk tier · fused neuro-symbolic model



Probability of progression

12% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

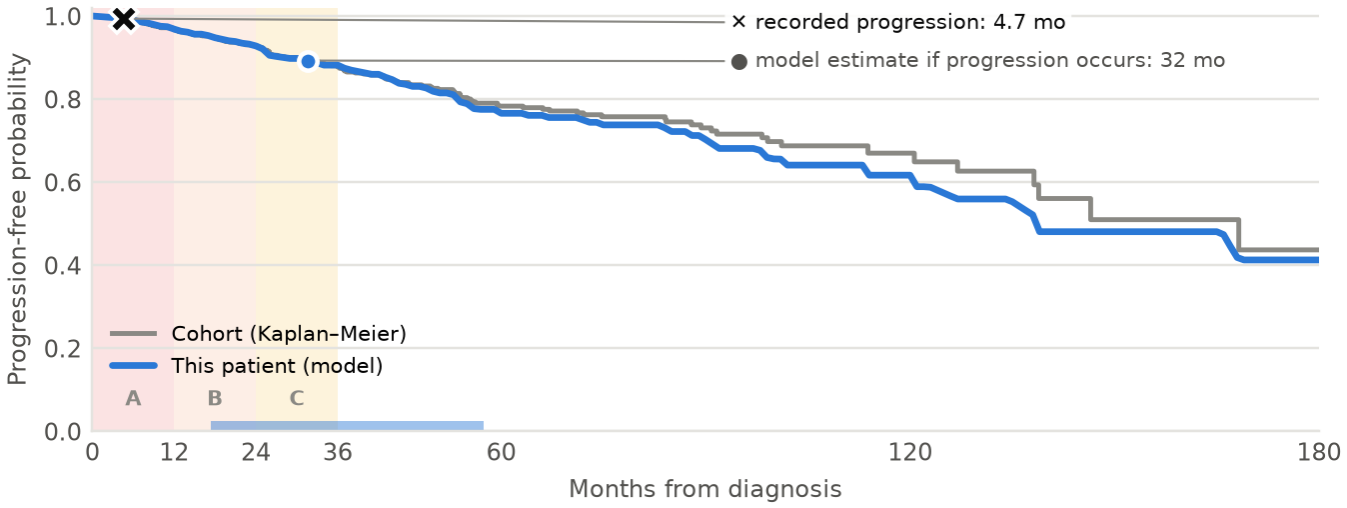
≈ 32 months

Window 24–36 months · IQR 17–57 mo
if progression occurs

Recorded progression

4.7 months

Group A · Early progression
✗ recorded window differs · |error| 27 mo



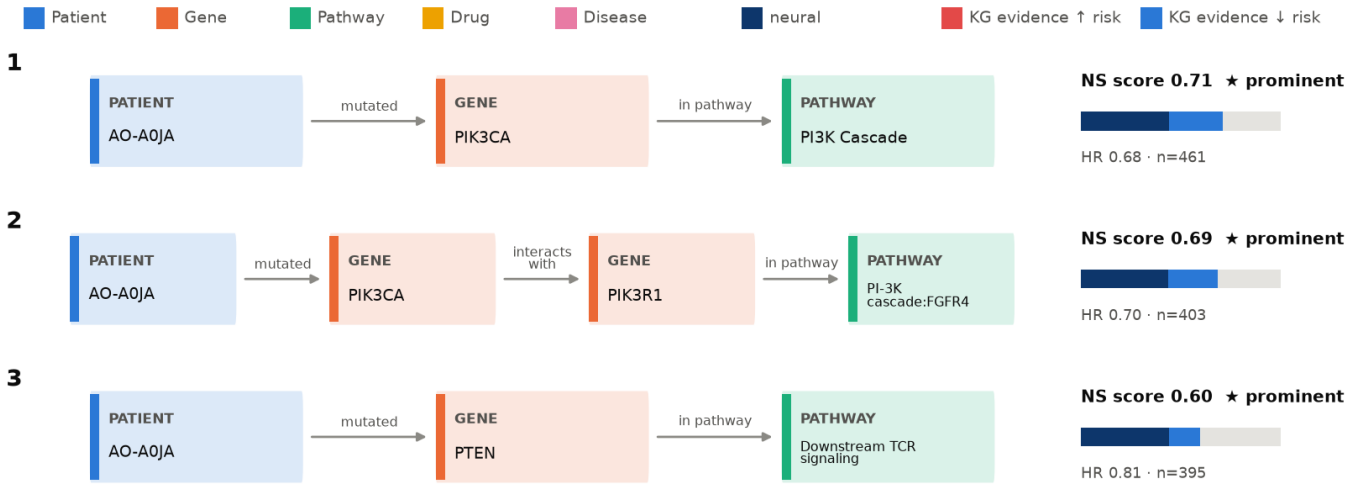
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 36-year-old patient has a LumA tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **70th risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI3K Cascade**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.68). Progression was recorded at 4.7 months; the model's estimate given progression was 32 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

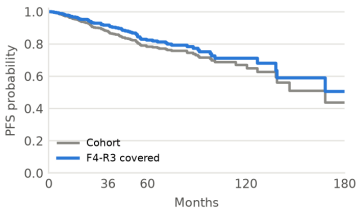


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AO-A0JA** mutated → **PIK3CA** in pathway → **PI3K Cascade**	73	↓ slower progression HR 0.68, z -2.0, n = 461 carriers of “PI3K Cascade”	**0.71**
2 ★	**TCGA-AO-A0JA** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR4**	73	↓ slower progression HR 0.70, z -1.8, n = 403 carriers of “PI-3K cascade:FGFR4”	**0.69**
3 ★	**TCGA-AO-A0JA** mutated → **PTEN** in pathway → **Downstream TCR signaling**	74	↓ slower progression HR 0.81, z -1.1, n = 395 carriers of “Downstream TCR signaling”	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

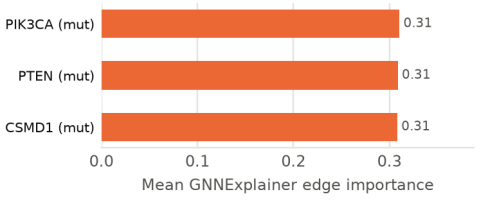
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.E542K	70	0.31
PTEN · driver	Mutation	p.Y88*	22	0.31
CSMD1	Mutation	p.D875H, p.S2505*	0	0.31



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1R3	70	III	LumA	0.79	no progression, followed 2.6 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A27L	49	III	LumA	0.73	no progression, followed 16.4 mo
TCGA-S3-A6ZH	29	III	LumA	0.70	no progression, followed 21.1 mo
TCGA-A7-A0D9	37	II	LumA	0.69	no progression, followed 37.4 mo
TCGA-BH-A0BC	60	III	LumA	0.69	no progression, followed 32.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A1LK · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**84**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**—**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-E2-A1LK-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **1.2** · MSIsensor score: **2.89** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

18% by 36 mo

12 mo 5% · 24 mo 11% · 60 mo 31%

Model-estimated progression time

≈ 8 months

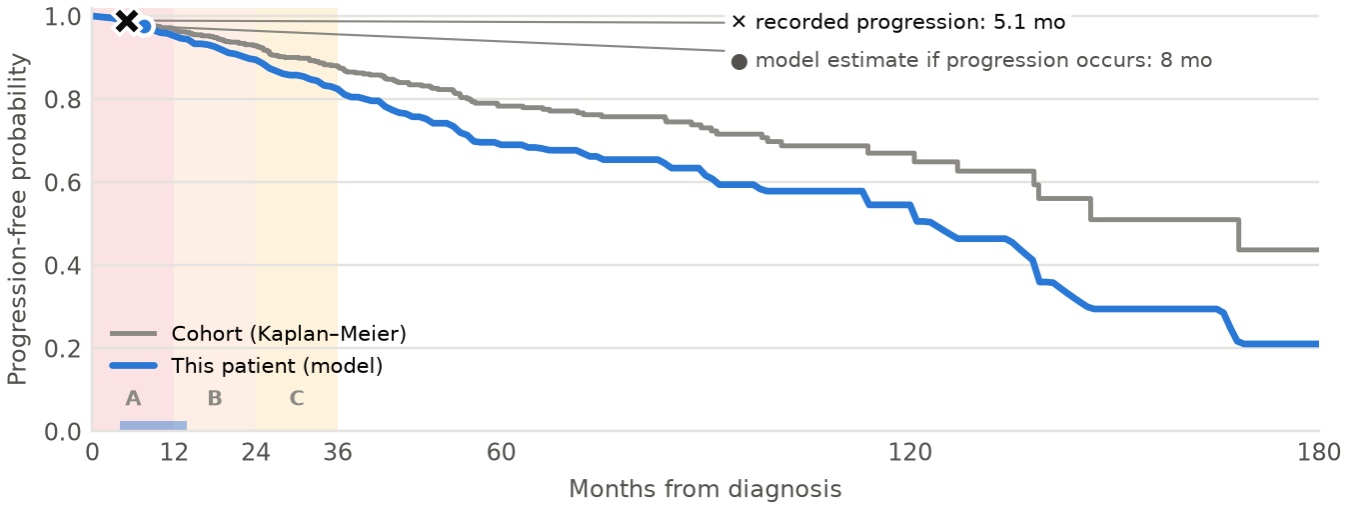
Window ≤ 12 months · IQR 4–14 mo if progression occurs

Recorded progression

5.1 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 2 mo



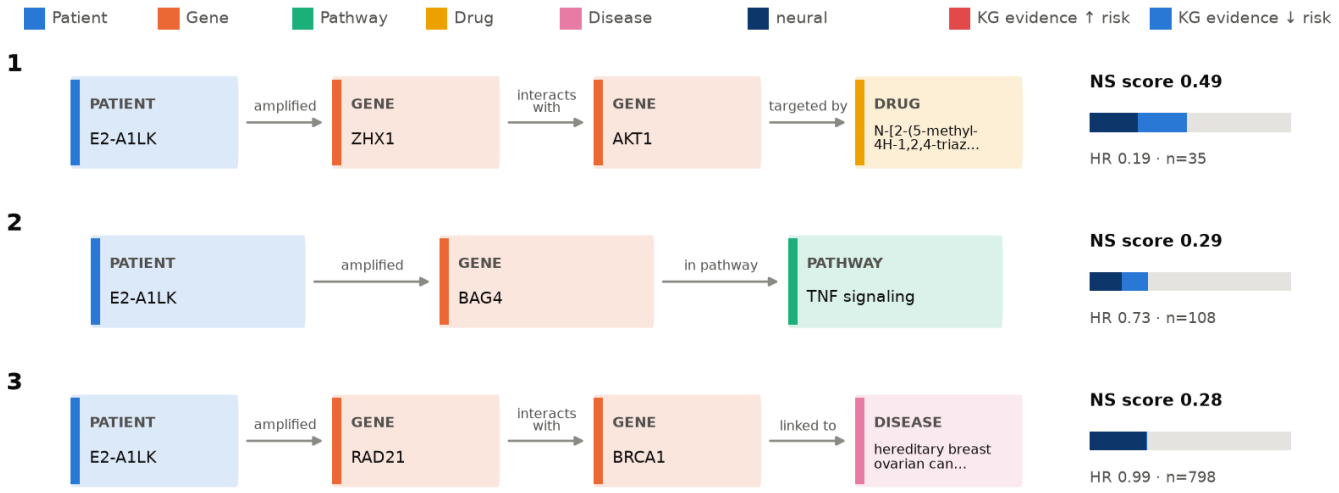
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 84-year-old patient has a Basal tumour, AJCC stage III (T4, N3, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5*) marks a group with a progression hazard 0.5× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ZHX1** to **N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.19). Progression was recorded at 5.1 months; the model's estimate given progression was 8 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

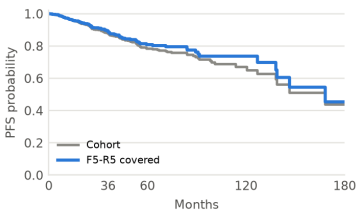


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-E2-A1LK** amplified → **ZHX1** interacts with → **AKT1** targeted by → **N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine**	40	↓ **slower progression** HR 0.19, z -1.9, n = 35 carriers of “N-[2-(5-methyl-4H-1,2,4-triazol-3-yl)phenyl]-7H-pyrrolo[2,3-d]pyrimidin-4-amine”	**0.49**
2	**TCGA-E2-A1LK** amplified → **BAG4** in pathway → **TNF signaling**	26	↓ **slower progression** HR 0.73, z -1.0, n = 108 carriers of “TNF signaling”	**0.29**
3	**TCGA-E2-A1LK** amplified → **RAD21** interacts with → **BRCA1** linked to → **hereditary breast ovarian cancer syndrome**	47	↓ **slower progression** HR 0.99, z -0.0, n = 798 carriers of “hereditary breast ovarian cancer syndrome”	**0.28**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

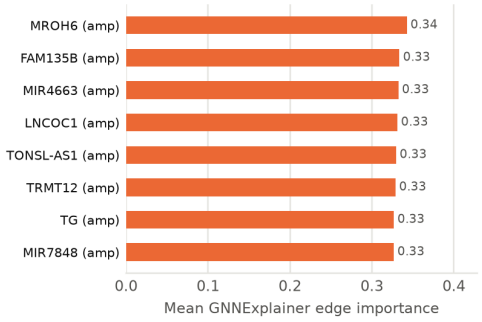
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
TP53 · driver	Mutation	p.A86Cfs*63	57	0.31
MROH6	High-level amplification	GISTIC ±2	0	0.34
FAM135B	High-level amplification	GISTIC ±2	0	0.33
MIR4663	High-level amplification	GISTIC ±2	0	0.33
LNCOC1	High-level amplification	GISTIC ±2	0	0.33
TONSL-AS1	High-level amplification	GISTIC ±2	0	0.33
TRMT12	High-level amplification	GISTIC ±2	1	0.33
TG	High-level amplification	GISTIC ±2	1	0.33
MIR7848	High-level amplification	GISTIC ±2	0	0.33



338 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840) … and 146 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YH	53	III	LumB	0.87	no progression, followed 21.7 mo
TCGA-C8-A131	82	III	Basal	0.85	no progression, followed 13.5 mo
TCGA-A8-A06P	63	III	LumA	0.85	no progression, followed 13.0 mo
TCGA-A2-A0CU	73	II	LumA	0.84	no progression, followed 5.2 mo
TCGA-A8-A09Q	83	III	LumB	0.84	no progression, followed 25.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**79**	Tumour type	**Breast Invasive Mixed Mucinous Carcinoma**
Sex	**Female**	OncoTree code	**IMMC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8480/3**

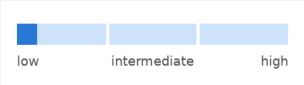
Sample: **TCGA-A7-A3RF-01** · Aneuploidy score: **0** · TMB (nonsyn., /Mb): **1.07** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

8th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 14%

Model-estimated progression time

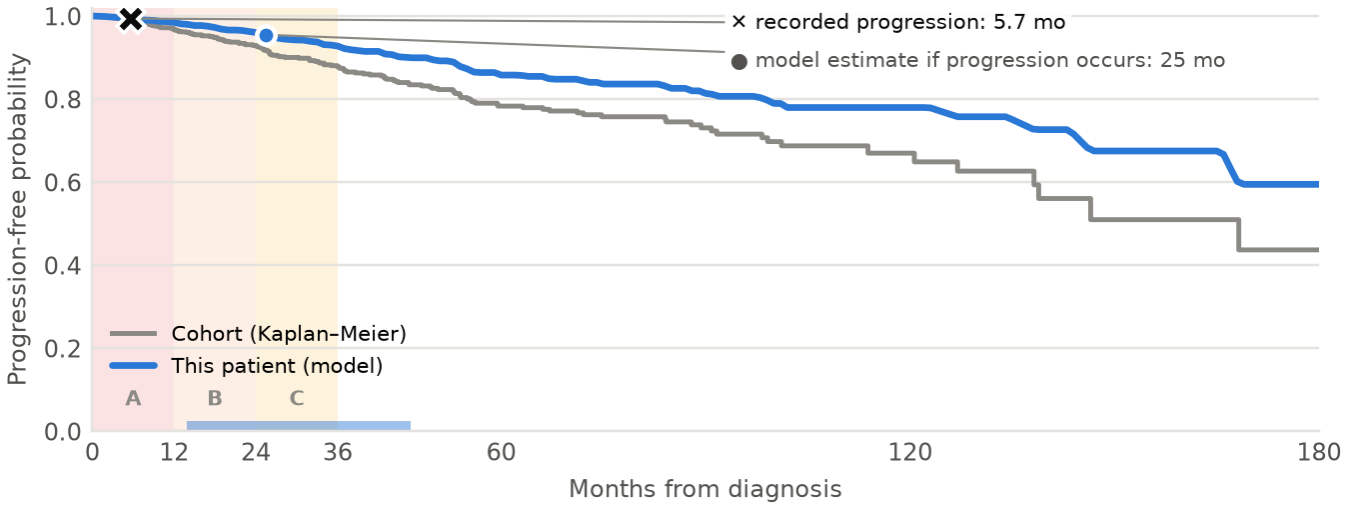
≈ 25 months

Window 24–36 months · IQR 14–47 mo
if progression occurs

Recorded progression

5.7 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 20 mo



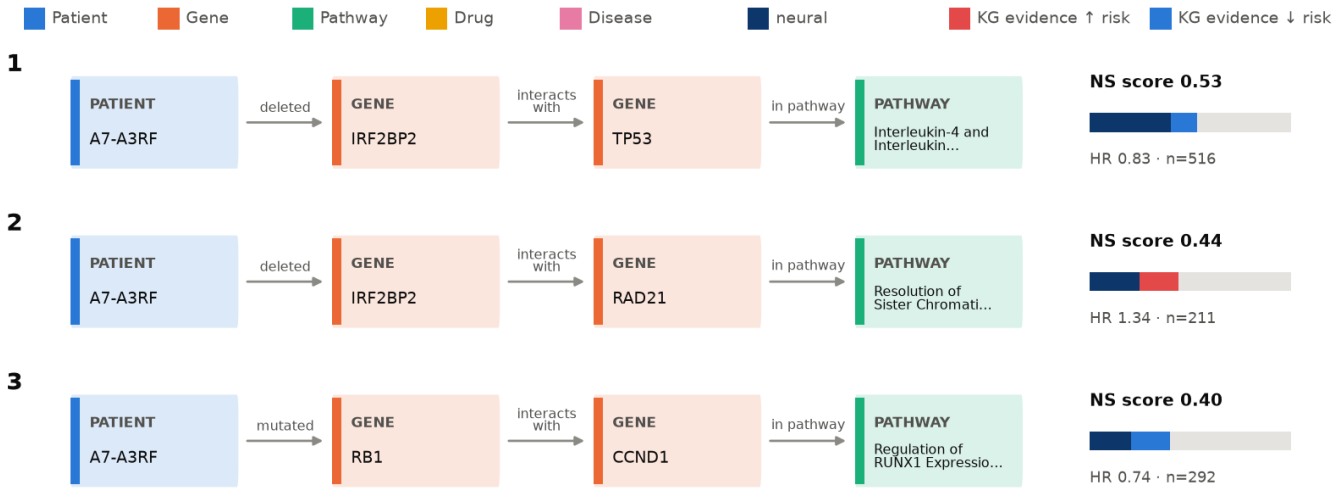
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 79-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **8th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **IRF2BP2** to **Interleukin-4 and Interleukin-13 signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.83). Progression was recorded at 5.7 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

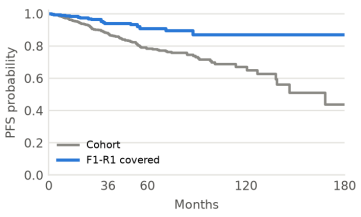


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A7-A3RF** deleted → **IRF2BP2** interacts with → **TP53** in pathway → **Interleukin-4 and Interleukin-13 signaling**	67	↓ **slower progression** HR 0.83, z -1.0, n = 516 carriers of “Interleukin-4 and Interleukin-13 signaling”	**0.53**
2	**TCGA-A7-A3RF** deleted → **IRF2BP2** interacts with → **RAD21** in pathway → **Resolution of Sister Chromatid Cohesion**	41	↑ **faster progression** HR 1.34, z +1.5, n = 211 carriers of “Resolution of Sister Chromatid Cohesion”	**0.44**
3	**TCGA-A7-A3RF** mutated → **RB1** interacts with → **CCND1** in pathway → **Regulation of RUNX1 Expression and Activity**	34	↓ **slower progression** HR 0.74, z -1.5, n = 292 carriers of “Regulation of RUNX1 Expression and Activity”	**0.40**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

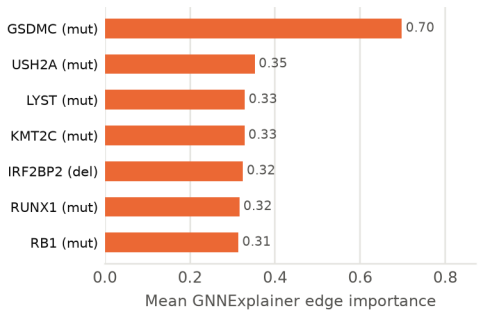
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Mutation	p.Q1937*	4	0.33
RUNX1 · driver	Mutation	3'UTR	18	0.32
RB1 · driver	Mutation	p.S866Afs*7	25	0.31
GSDMC	Mutation	p.R34C	0	0.70
USH2A	Mutation	p.S1051=	0	0.35
LYST	Mutation	p.P1758Q	0	0.33
IRF2BP2	Homozygous deletion	GISTIC ±2	0	0.32



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RB1 → Insulin human (DB00030); RB1 → Insulin pork (DB00071). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XV	84	II	LumA	0.79	no progression, followed 15.2 mo
TCGA-AO-A0J8	61	II	LumA	0.73	no progression, followed 22.4 mo
TCGA-AO-A125	72	II	LumA	0.73	no progression, followed 113.6 mo
TCGA-A2-A0EN	70	II	LumA	0.73	no progression, followed 134.4 mo
TCGA-A8-A0AD	83	I	LumA	0.71	no progression, followed 38.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-AC-A23H · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**90**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AC-A23H-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **142.67** · MSIsensor score: **0.54** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

80th

High risk tier · fused neuro-symbolic model



Probability of progression

14% by 36 mo

12 mo 3% · 24 mo 8% · 60 mo 27%

Model-estimated progression time

≈ 9 months

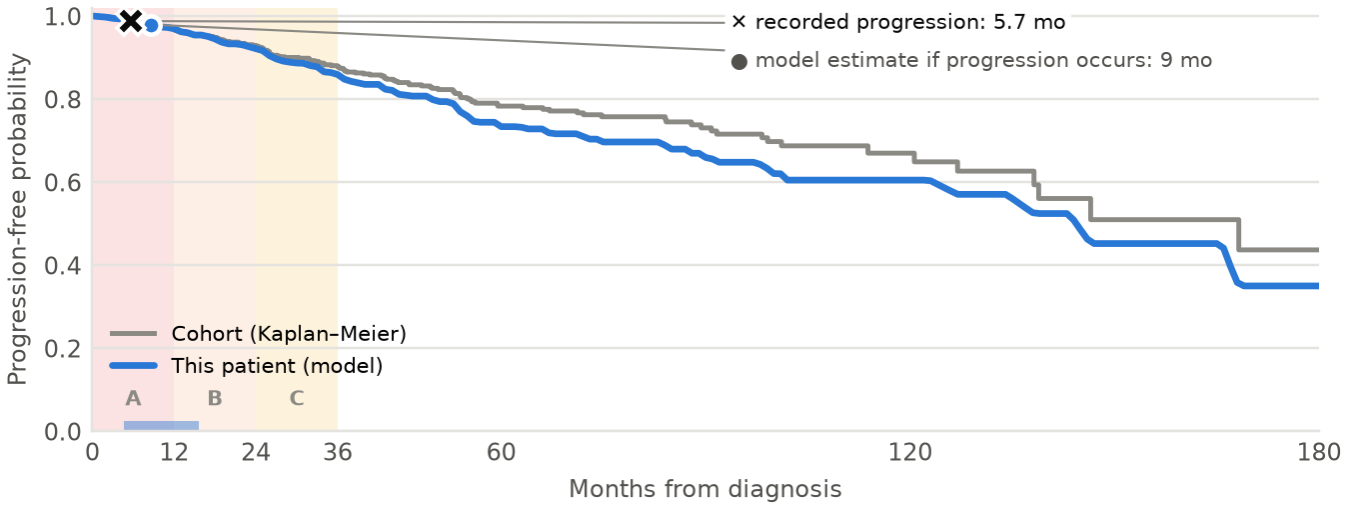
Window ≤ 12 months · IQR 5–16 mo if progression occurs

Recorded progression

5.7 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 3 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 90-year-old patient has a Her2 tumour, AJCC stage II (T2, missing, M0). The fused neuro-symbolic model places them in the **80th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PPFIA1** to **Acetylcholine Neurotransmitter Release Cycle**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.59). Progression was recorded at 5.7 months; the model's estimate given progression was 9 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

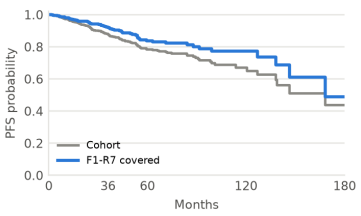


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AC-A23H** mutated → **PPFIA1** in pathway → **Acetylcholine Neurotransmitter Release Cycle**	100	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of “Acetylcholine Neurotransmitter Release Cycle”	**0.82**
2 ★	**TCGA-AC-A23H** mutated → **OR2G3** in pathway → **Olfactory Signaling Pathway**	94	↑ **faster progression** HR 1.30, z +1.2, n = 162 carriers of “Olfactory Signaling Pathway”	**0.72**
3 ★	**TCGA-AC-A23H** mutated → **NSMCE2** in pathway → **SUMOylation of DNA damage response and repair proteins**	58	↑ **faster progression** HR 1.70, z +2.8, n = 231 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.72**
4 ★	**TCGA-AC-A23H** mutated → **ENPP2** in pathway → **Vitamin B5 (pantothenate) metabolism**	96	↑ **faster progression** HR 1.26, z +0.9, n = 112 carriers of “Vitamin B5 (pantothenate) metabolism”	**0.70**
5 ★	**TCGA-AC-A23H** mutated → **TG** targeted by → **Iodine**	97	↑ **faster progression** HR 1.13, z +0.5, n = 125 carriers of “Iodine”	**0.65**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

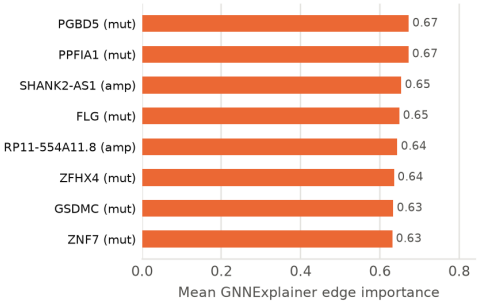
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R7 ↓ **lower risk**	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
FOXA1 · driver	Mutation	p.D226N	9	0.32
NF1 · driver	Mutation	p.S1896L	6	0.32
MAP3K1 · driver	Mutation	p.R283T	13	0.32
SF3B1 · driver	Mutation	p.L353=, p.L464F	4	0.32
ARID1A · driver	Mutation	p.M1273I	10	0.32
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.32
KMT2C · driver	Mutation	p.D3264N, p.D4344H, p.E3724K	4	0.32
BRCA2 · driver	Mutation	p.D1033H, p.D687H	14	0.31
ATM · driver	Mutation	p.I105=	29	0.31
PIK3CA · driver	Mutation	p.D603H, p.L456=, p.L989V	70	0.31
RB1 · driver	Mutation	p.G802=	25	0.31
ESR1 · driver	Mutation	p.L448=	141	0.31



177 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MAP3K1 → AZD-8330 (DB06061); MAP3K1 → Binimetinib (DB11967); MAP3K1 → Fostamatinib (DB12010); SF3B1 → H3B-8800 (DB14017); ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021) … and 279 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A12L	67	II	Her2	0.77	no progression, followed 11.9 mo
TCGA-AN-A046	68	II	LumA	0.75	no progression, followed 0.3 mo
TCGA-BH-A0BV	78	II	LumA	0.72	no progression, followed 49.9 mo
TCGA-A2-A0D1	76	II	Her2	0.71	no progression, followed 34.6 mo
TCGA-E2-A105	79	II	LumA	0.70	no progression, followed 43.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Lobular Carcinoma**
Sex	**Female**	OncoTree code	**ILC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**Yes**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8520/3**

Sample: **TCGA-Z7-A8R5-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

67th

High risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 23%

Model-estimated progression time

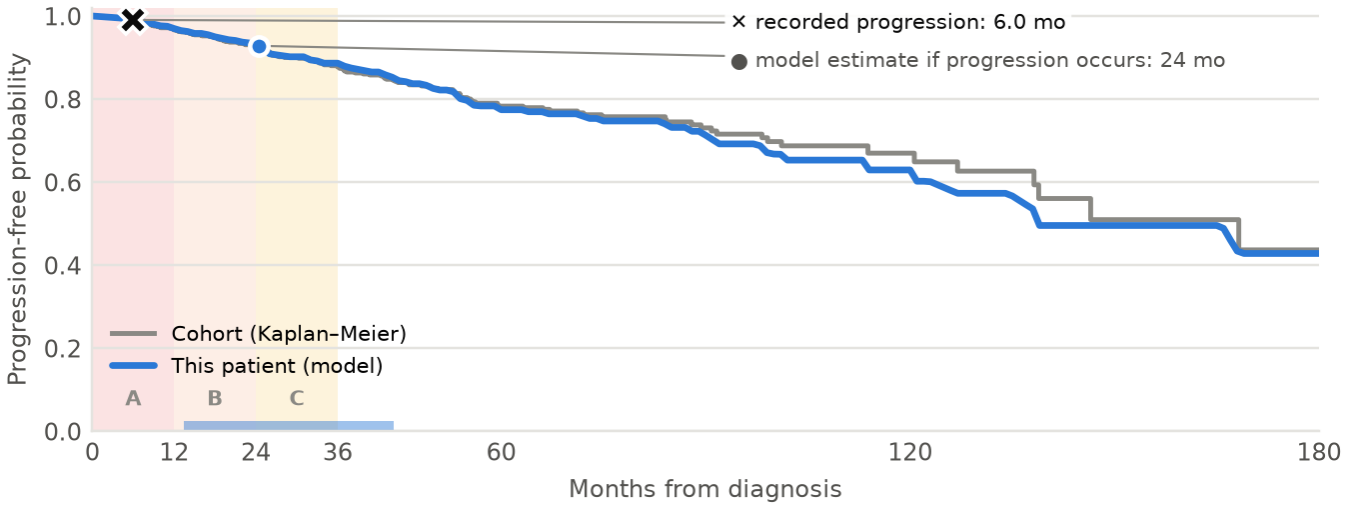
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

6.0 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 18 mo



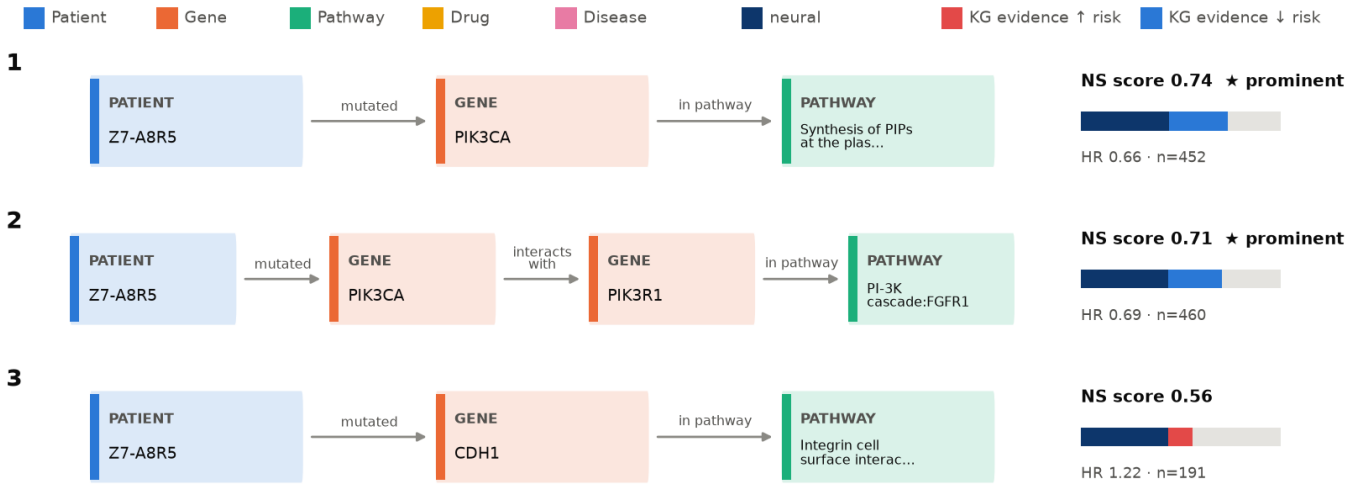
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **67th risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Synthesis of PIPs at the plasma membrane**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.66). Progression was recorded at 6.0 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

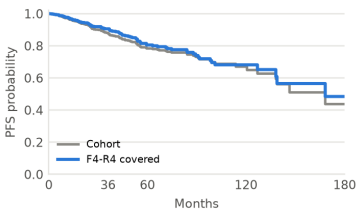


Only 2 of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-Z7-A8R5** mutated → **PIK3CA** in pathway → **Synthesis of PIPs at the plasma membrane**	74	↓ **slower progression** HR 0.66, z -2.2, n = 452 carriers of "Synthesis of PIPs at the plasma membrane"	**0.74**
2 ★	**TCGA-Z7-A8R5** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR1**	73	↓ **slower progression** HR 0.69, z -2.0, n = 460 carriers of "PI-3K cascade:FGFR1"	**0.71**
3	**TCGA-Z7-A8R5** mutated → **CDH1** in pathway → **Integrin cell surface interactions**	73	↑ **faster progression** HR 1.22, z +0.9, n = 191 carriers of "Integrin cell surface interactions"	**0.56**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

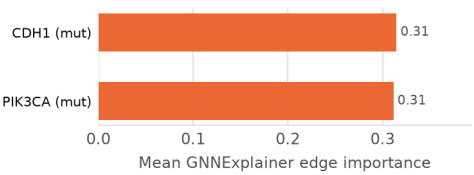
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.L71Sfs*12	17	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A1AL	60	III	LumA	0.89	no progression, followed 97.7 mo

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A425	70	III	LumA	0.79	progressed at 13.8 mo (group B)
TCGA-LD-A66U	44	II	LumA	0.78	no progression, followed 21.2 mo
TCGA-A2-A4S2	62	III	LumA	0.78	no progression, followed 21.1 mo
TCGA-A2-A0YL	48	III	LumA	0.77	no progression, followed 48.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A14Z · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**64**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE I**
Genetic ancestry	**EUR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

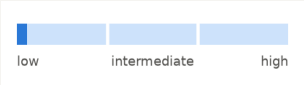
Sample: **TCGA-E2-A14Z-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.03** · MSIsensor score: **0.13** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

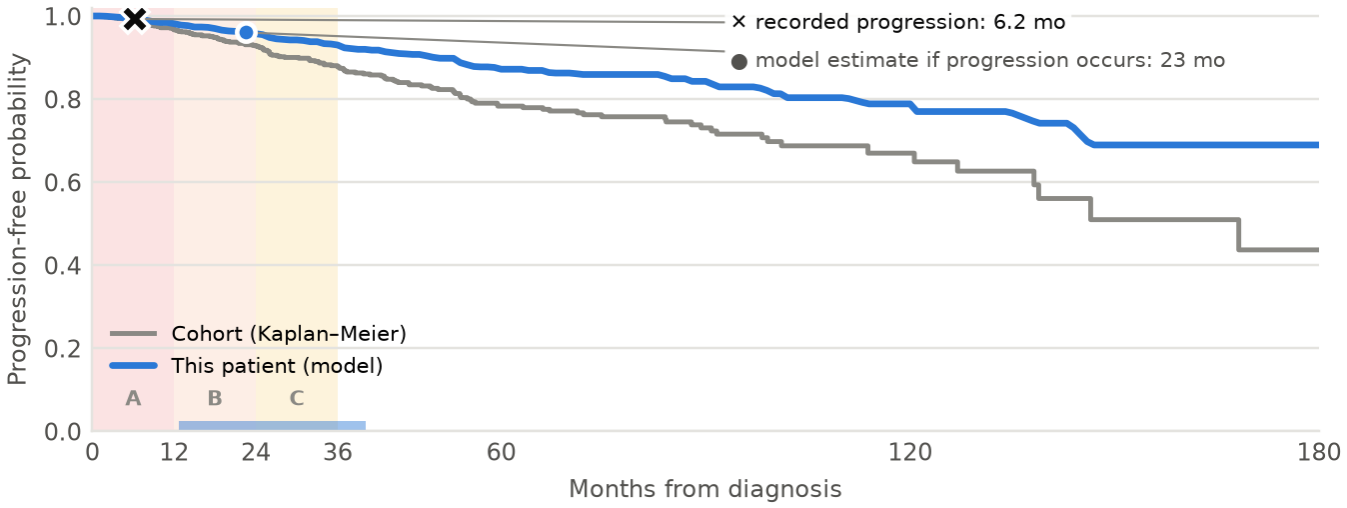
≈ 23 months

Window 12–24 months · IQR 13–40 mo
if progression occurs

Recorded progression

6.2 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 16 mo



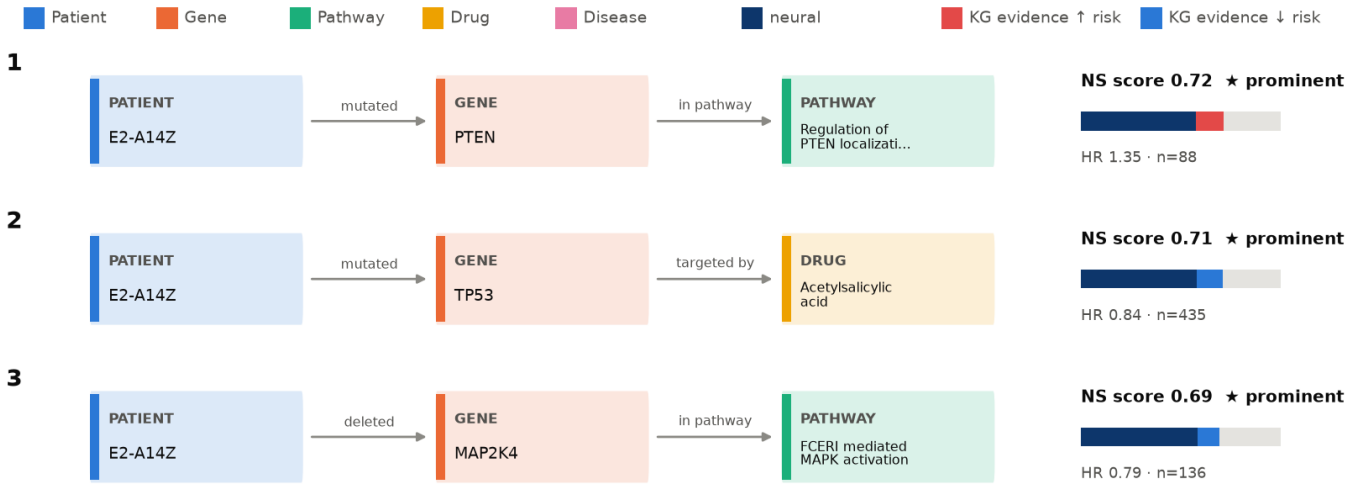
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 64-year-old patient has a LumA tumour, AJCC stage I (T1, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PTEN** to **Regulation of PTEN localization**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.35). Progression was recorded at 6.2 months; the model's estimate given progression was 23 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

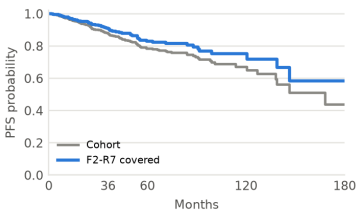


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A14Z** mutated → **PTEN** in pathway → **Regulation of PTEN localization**	96	↑ **faster progression** HR 1.35, z +1.1, n = 88 carriers of “Regulation of PTEN localization”	**0.72**
2 ★	**TCGA-E2-A14Z** mutated → **TP53** targeted by → **Acetylsalicylic acid**	97	↓ **slower progression** HR 0.84, z -1.0, n = 435 carriers of “Acetylsalicylic acid”	**0.71**
3 ★	**TCGA-E2-A14Z** deleted → **MAP2K4** in pathway → **FCERI mediated MAPK activation**	97	↓ **slower progression** HR 0.79, z -0.8, n = 136 carriers of “FCERI mediated MAPK activation”	**0.69**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

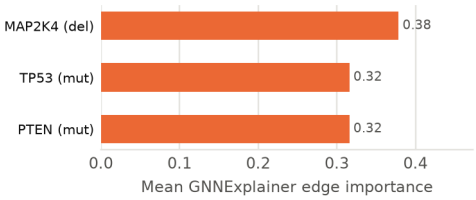
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R2 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	343 pts 24 events	**0.37** p 5.3e-06	not reached
F2-R3 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.78 AND Age < 83.5	317 pts 25 events	**0.41** p 3.7e-05	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MAP2K4 · driver	Homozygous deletion	GISTIC ±2	12	0.38
TP53 · driver	Mutation	p.R282W	57	0.32
PTEN · driver	Mutation	p.D252Rfs*42	22	0.32



3 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XA	64	I	LumA	0.94	no progression, followed 27.6 mo
TCGA-E2-A1IN	60	I	LumA	0.88	no progression, followed 22.2 mo
TCGA-AO-A03V	41	I	LumA	0.87	no progression, followed 44.4 mo
TCGA-A8-A06O	60	I	LumB	0.86	no progression, followed 13.0 mo
TCGA-GM-A2DK	58	I	LumA	0.84	no progression, followed 87.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-LL-A73Z · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**55**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR**	Pathologic T	**T1C**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

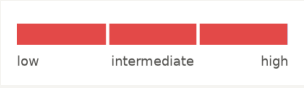
Sample: **TCGA-LL-A73Z-01** · Aneuploidy score: **11** · TMB (nonsyn., /Mb): **1.33** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

72% by 36 mo

12 mo 28% · 24 mo 49% · 60 mo 90%

Model-estimated progression time

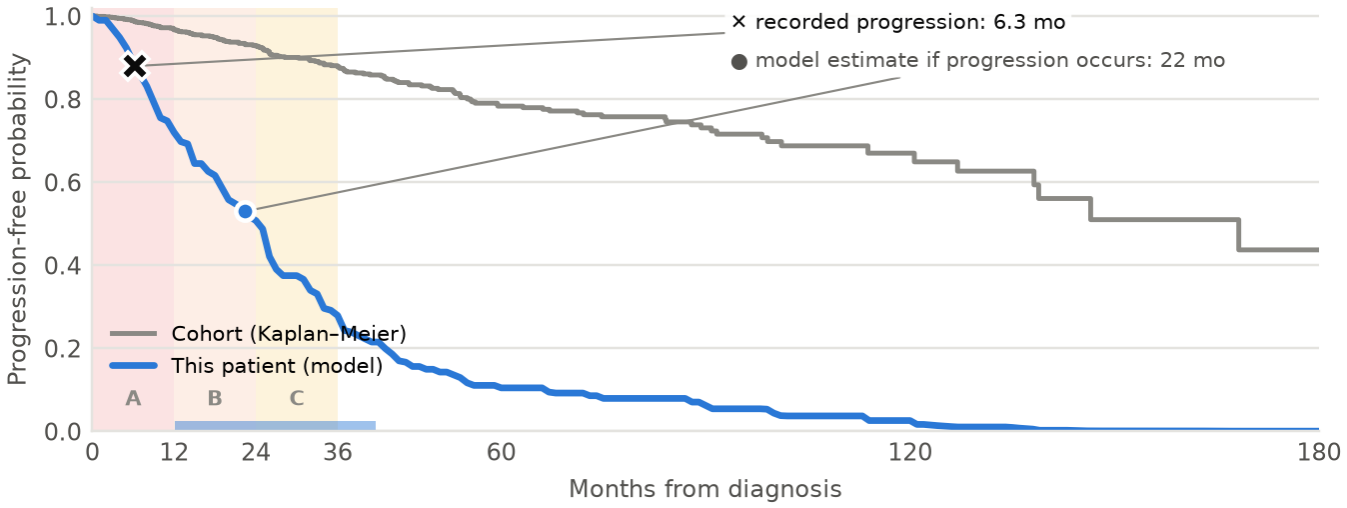
≈ 22 months

Window 12–24 months · IQR 12–42 mo
if progression occurs

Recorded progression

6.3 months

Group A · Early progression
✗ recorded window differs · |error| 16 mo



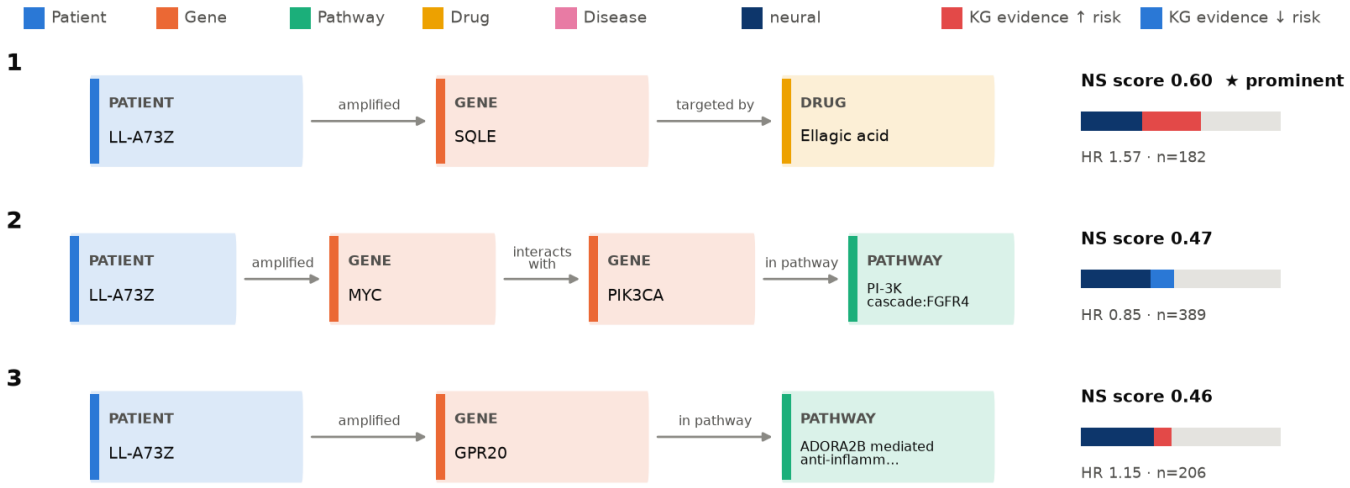
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 55-year-old patient has a Normal tumour, AJCC stage IV (T1, missing, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **SQLE** to **Ellagic acid**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.57). Progression was recorded at 6.3 months; the model's estimate given progression was 22 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

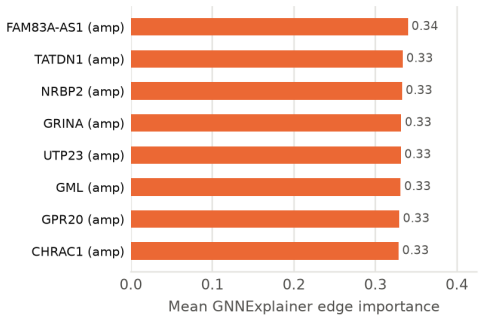
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-LL-A73Z** amplified → **SQLE** targeted by → **Ellagic acid**	51	↑ faster progression HR 1.57, z +2.2, n = 182 carriers of “Ellagic acid”	**0.60**
2	**TCGA-LL-A73Z** amplified → **MYC** interacts with → **PIK3CA** in pathway → **PI-3K cascade:FGFR4**	58	↓ slower progression HR 0.85, z -0.9, n = 389 carriers of “PI-3K cascade:FGFR4”	**0.47**
3	**TCGA-LL-A73Z** amplified → **GPR20** in pathway → **ADORA2B mediated anti-inflammatory cytokines production**	61	↑ faster progression HR 1.15, z +0.7, n = 206 carriers of “ADORA2B mediated anti-inflammatory cytokines production”	**0.46**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
CDH1 · driver	Mutation	p.D254Y	17	0.31
PTEN · driver	Mutation	p.V290Sfs*8	22	0.31
FAM83A-AS1	High-level amplification	GISTIC ±2	0	0.34
TATDN1	High-level amplification	GISTIC ±2	0	0.33
NRBP2	High-level amplification	GISTIC ±2	0	0.33
GRINA	High-level amplification	GISTIC ±2	0	0.33
UTP23	High-level amplification	GISTIC ±2	0	0.33
GML	High-level amplification	GISTIC ±2	0	0.33
GPR20	High-level amplification	GISTIC ±2	2	0.33
CHRAC1	High-level amplification	GISTIC ±2	0	0.33
MIR6893	High-level amplification	GISTIC ±2	0	0.33



235 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); PTEN → Phosphatidylethanolamine (DB04327); OPLAH → Glutamic acid (DB00142); GPT → Pyridoxal phosphate (DB00114); GPT → Glutamic acid (DB00142); GPT → Alanine (DB00160); GPT → Phenelzine (DB00780); TOP1MT → Irinotecan (DB00762); TOP1MT → Topotecan (DB01030) … and 73 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-EW-A1PF	50	I	missing	0.73	no progression, followed 14.4 mo
TCGA-B6-A0X1	48	II	Basal	0.72	progressed at 16.1 mo (group B)
TCGA-BH-A0C0	62	II	LumB	0.72	no progression, followed 41.8 mo
TCGA-EW-A1PD	61	II	missing	0.71	no progression, followed 13.9 mo
TCGA-EW-A1P1	68	III	Normal	0.70	progressed at 33.3 mo (group C)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E2-A10B · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

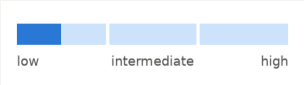
Sample: **TCGA-E2-A10B-01** · Aneuploidy score: **35** · TMB (nonsyn., /Mb): **1.53** · MSIsensor score: **0.98** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

16th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 14%

Model-estimated progression time

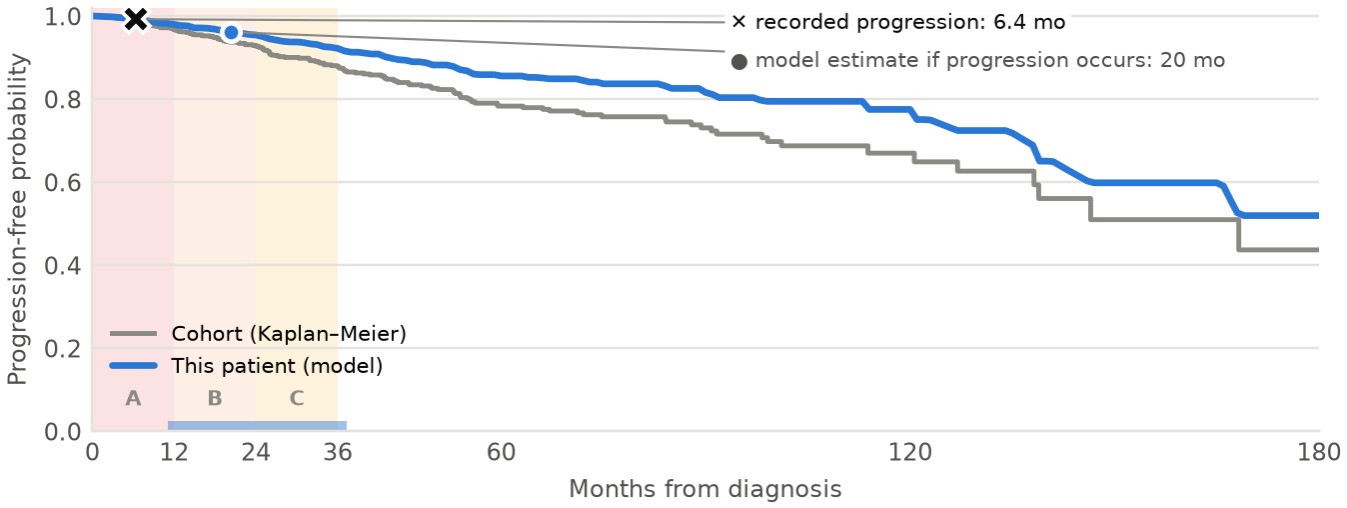
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

6.4 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 14 mo



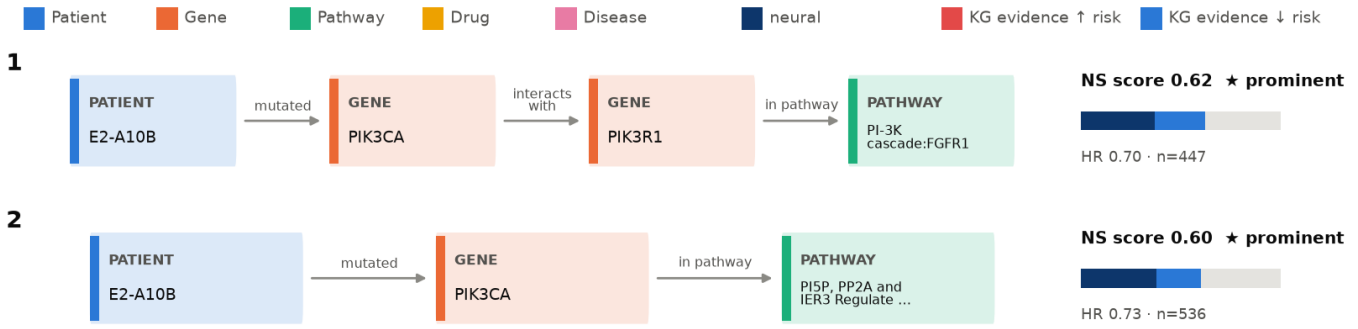
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a LumA tumour, AJCC stage II (T2, N1, M0). The fused neuro-symbolic model places them in the **16th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI-3K cascade:FGFR1**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 6.4 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

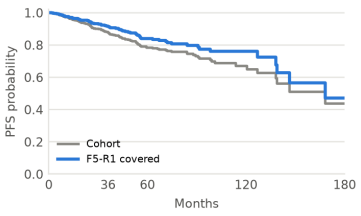


2 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A10B** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR1**	62	↓ **slower progression** HR 0.70, z -1.9, n = 447 carriers of "PI-3K cascade:FGFR1"	**0.62**
2 ★	**TCGA-E2-A10B** mutated → **PIK3CA** in pathway → **PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling**	63	↓ **slower progression** HR 0.73, z -1.7, n = 536 carriers of "PI5P, PP2A and IER3 Regulate PI3K/AKT Signaling"	**0.60**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

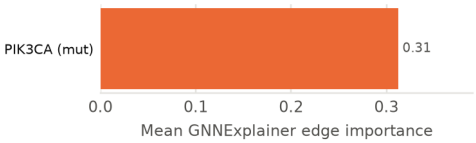
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PIK3CA · driver	Mutation	p.H1047R	70	0.31



1 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YC	59	II	LumA	0.99	progressed at 24.9 mo (group C)
TCGA-AR-A0TR	68	II	LumA	0.98	no progression, followed 5.3 mo
TCGA-EW-A1P5	77	II	LumA	0.96	no progression, followed 23.1 mo
TCGA-E9-A227	42	II	LumA	0.93	no progression, followed 32.1 mo
TCGA-BH-A0BJ	41	II	LumA	0.93	no progression, followed 21.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3B**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-EW-A1P8-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **1.23** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

87th

High risk tier · fused neuro-symbolic model



Probability of progression

15% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 29%

Model-estimated progression time

≈ 17 months

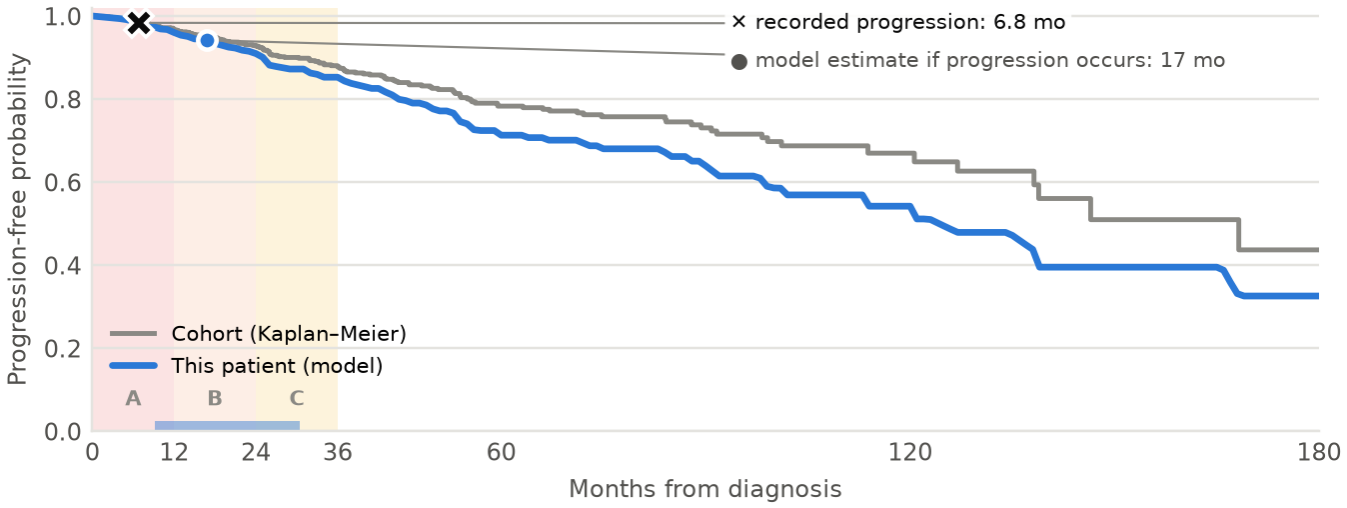
Window 12–24 months · IQR 9–30 mo if progression occurs

Recorded progression

6.8 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 10 mo



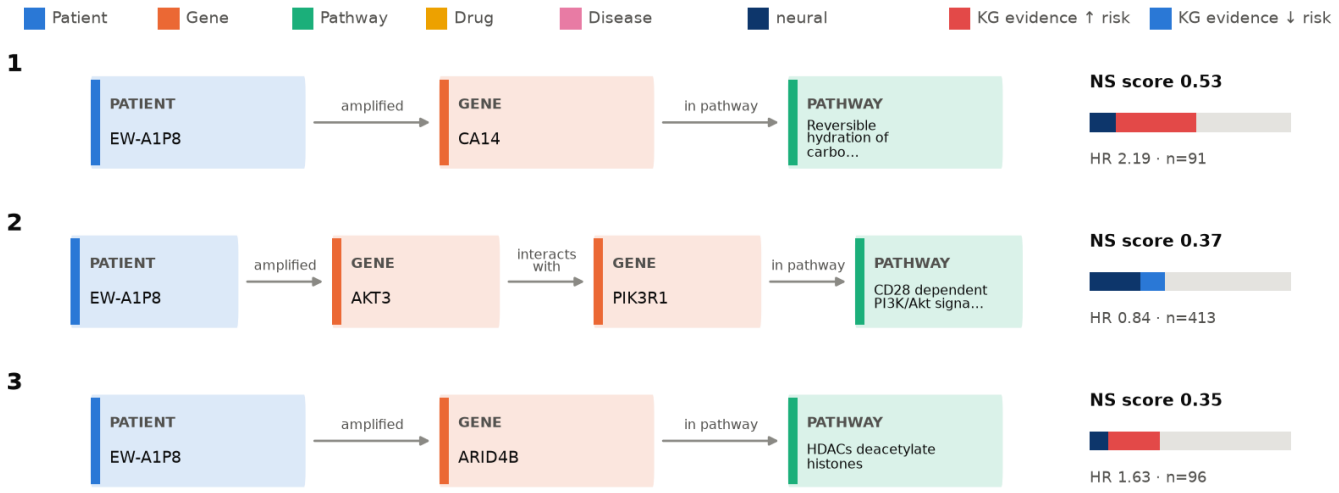
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a Basal tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **87th risk percentile** (high tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CA14** to **Reversible hydration of carbon dioxide**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.19). Progression was recorded at 6.8 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

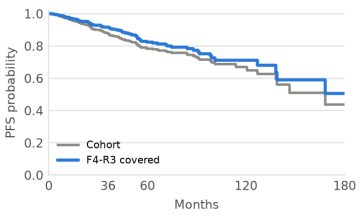


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-EW-A1P8** amplified → **CA14** in pathway → **Reversible hydration of carbon dioxide**	21	↑ **faster progression** HR 2.19, z +3.3, n = 91 carriers of “Reversible hydration of carbon dioxide”	**0.53**
2	**TCGA-EW-A1P8** amplified → **AKT3** interacts with → **PIK3R1** in pathway → **CD28 dependent PI3K/Akt signaling**	42	↓ **slower progression** HR 0.84, z -0.9, n = 413 carriers of “CD28 dependent PI3K/Akt signaling”	**0.37**
3	**TCGA-EW-A1P8** amplified → **ARID4B** in pathway → **HDACs deacetylate histones**	15	↑ **faster progression** HR 1.63, z +1.9, n = 96 carriers of “HDACs deacetylate histones”	**0.35**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

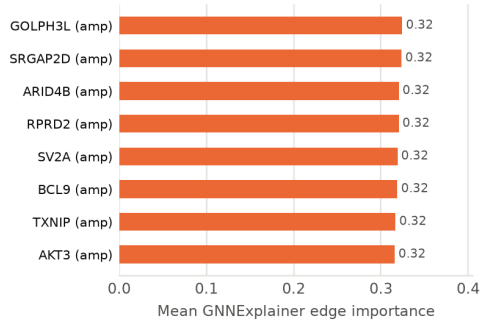
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ lower risk	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R4 ↓ lower risk	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ lower risk	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
TP53 · driver	Mutation	p.R282W	57	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
GOLPH3L	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
ARID4B	High-level amplification	GISTIC ±2	3	0.32
RPRD2	High-level amplification	GISTIC ±2	1	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32
BCL9	High-level amplification	GISTIC ±2	3	0.32
TXNIP	High-level amplification	GISTIC ±2	4	0.32
AKT3	High-level amplification	GISTIC ±2	24	0.32
PCNXL2	Mutation	p.R298P	0	0.32
ITGA10	High-level amplification	GISTIC ±2	3	0.31



27 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); SV2A → Levetiracetam (DB01202) … and 8 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-E9-A1ND	75	II	Basal	0.68	no progression, followed 41.6 mo
TCGA-BH-A0E0	38	III	Basal	0.68	no progression, followed 4.4 mo
TCGA-A8-A07U	66	III	Basal	0.67	no progression, followed 25.0 mo
TCGA-BH-A0B3	53	II	Basal	0.65	no progression, followed 39.6 mo
TCGA-A2-A0T2	66	IV	Basal	0.64	progressed at 8.4 mo (group A)

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A18T · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**70**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**—**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A18T-01** · Aneuploidy score: **22** · TMB (nonsyn., /Mb): **2.43** · MSIsensor score: **1.63** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

61st

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

10% by 36 mo

12 mo 3% · 24 mo 6% · 60 mo 20%

Model-estimated progression time

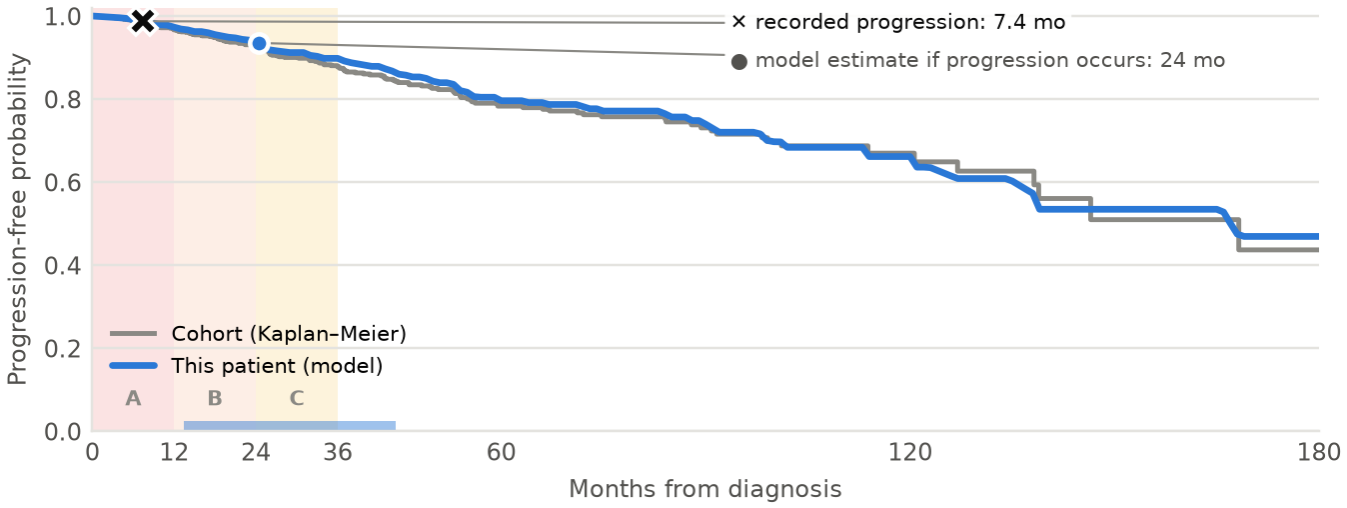
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

7.4 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 17 mo



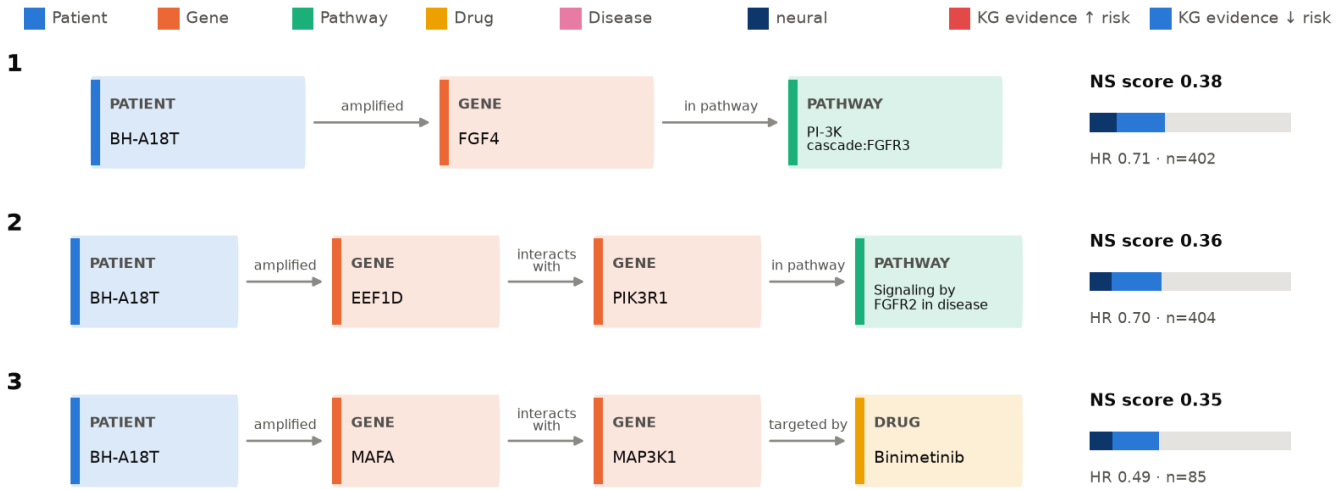
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 70-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **61st risk percentile** (intermediate tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FGF4** to **PI-3K cascade:FGFR3**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.71). Progression was recorded at 7.4 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

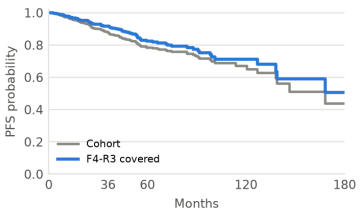


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-BH-A18T** amplified → **FGF4** in pathway → **PI-3K cascade:FGFR3**	23	↓ **slower progression** HR 0.71, z -1.8, n = 402 carriers of “PI-3K cascade:FGFR3”	**0.38**
2	**TCGA-BH-A18T** amplified → **EEF1D** interacts with → **PIK3R1** in pathway → **Signaling by FGFR2 in disease**	18	↓ **slower progression** HR 0.70, z -1.9, n = 404 carriers of “Signaling by FGFR2 in disease”	**0.36**
3	**TCGA-BH-A18T** amplified → **MAFA** interacts with → **MAP3K1** targeted by → **Binimetinib**	19	↓ **slower progression** HR 0.49, z -1.7, n = 85 carriers of “Binimetinib”	**0.35**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

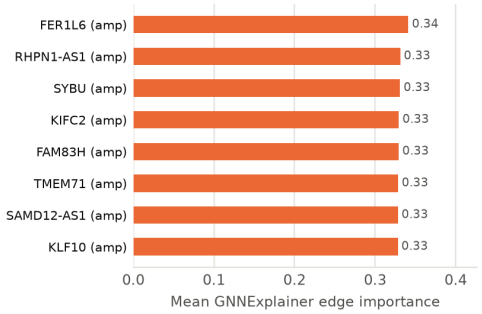
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ **lower risk**	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R1 ↓ **lower risk**	N stage = N0 AND TMB (nonsynonymous) ≥ 0.38 AND Age ≥ 40.5	355 pts 26 events	**0.39** p 7.7e-06	not reached
F4-R2 ↓ **lower risk**	N stage = N0 AND 34.5 ≤ Age < 83.5	386 pts 32 events	**0.43** p 2.5e-05	not reached
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ **lower risk**	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
TP53 · driver	Mutation	p.Y107D	57	0.32
MYC · driver	High-level amplification	GISTIC ±2	20	0.32
FOXA1 · driver	Mutation	p.A153V	9	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
FER1L6	High-level amplification	GISTIC ±2	0	0.34
RHPN1-AS1	High-level amplification	GISTIC ±2	0	0.33
SYBU	High-level amplification	GISTIC ±2	0	0.33
KIFC2	High-level amplification	GISTIC ±2	2	0.33
FAM83H	High-level amplification	GISTIC ±2	0	0.33
TMEM71	High-level amplification	GISTIC ±2	0	0.33
SAMD12-AS1	High-level amplification	GISTIC ±2	0	0.33



316 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 99 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AN-A0FX	52	II	Basal	0.89	no progression, followed 0.3 mo
TCGA-BH-A0DL	64	II	Basal	0.89	no progression, followed 78.3 mo
TCGA-E2-A150	48	II	Basal	0.89	no progression, followed 63.6 mo
TCGA-E2-A14R	62	II	Basal	0.87	progressed at 31.9 mo (group C)
TCGA-EW-A1P4	43	II	Basal	0.86	no progression, followed 29.8 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**48**	Tumour type	**Metaplastic Breast Cancer**
Sex	**Female**	OncoTree code	**MBC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**AMR**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8575/3**

Sample: **TCGA-AC-A2QJ-01** · Aneuploidy score: **23** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

91st

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 4% · 24 mo 10% · 60 mo 32%

Model-estimated progression time

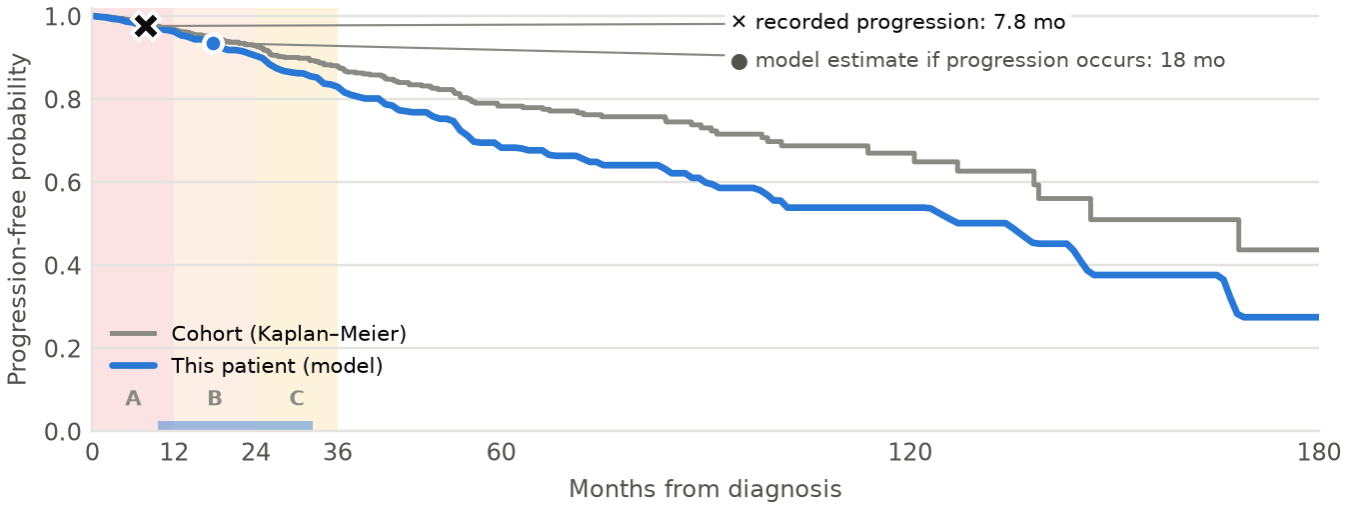
≈ 18 months

Window 12–24 months · IQR 10–32 mo
if progression occurs

Recorded progression

7.8 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 10 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 48-year-old patient has a subtype-unassigned tumour, AJCC stage III (T4, N0, M0). The fused neuro-symbolic model places them in the **91st risk percentile** (high tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. Progression was recorded at 7.8 months; the model's estimate given progression was 18 months.

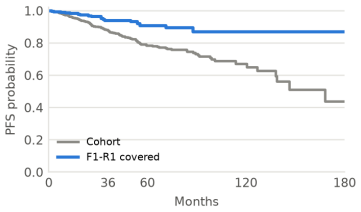
3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

No explanation path is available: none of this patient's altered panel genes has a stored PrimeKG pathway, drug, or breast-disease link (or the tumour carries no qualifying alteration). The risk estimate therefore rests on the clinical block and the symbolic rules below.

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

No mutation or high-level CNA on the 500-gene panel was recorded for this tumour.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A3J0	62	II	missing	0.65	no progression, followed 10.3 mo
TCGA-E9-A5UP	63	II	missing	0.65	no progression, followed 26.4 mo
TCGA-A7-A3IY	71	I	missing	0.62	no progression, followed 11.3 mo
TCGA-BH-A1FR	73	III	LumA	0.61	progressed at 54.0 mo (group D)
TCGA-A8-A07S	73	II	missing	0.61	no progression, followed 8.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A8-A091 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**61**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**—**	PAM50 subtype	**LumA**
Ethnicity	**—**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**Yes**	Pathologic N	**N0**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

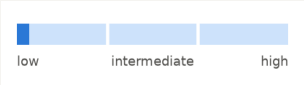
Sample: **TCGA-A8-A091-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.28** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

4th

Low risk tier · fused neuro-symbolic model



Probability of progression

7% by 36 mo

12 mo 1% · 24 mo 4% · 60 mo 13%

Model-estimated progression time

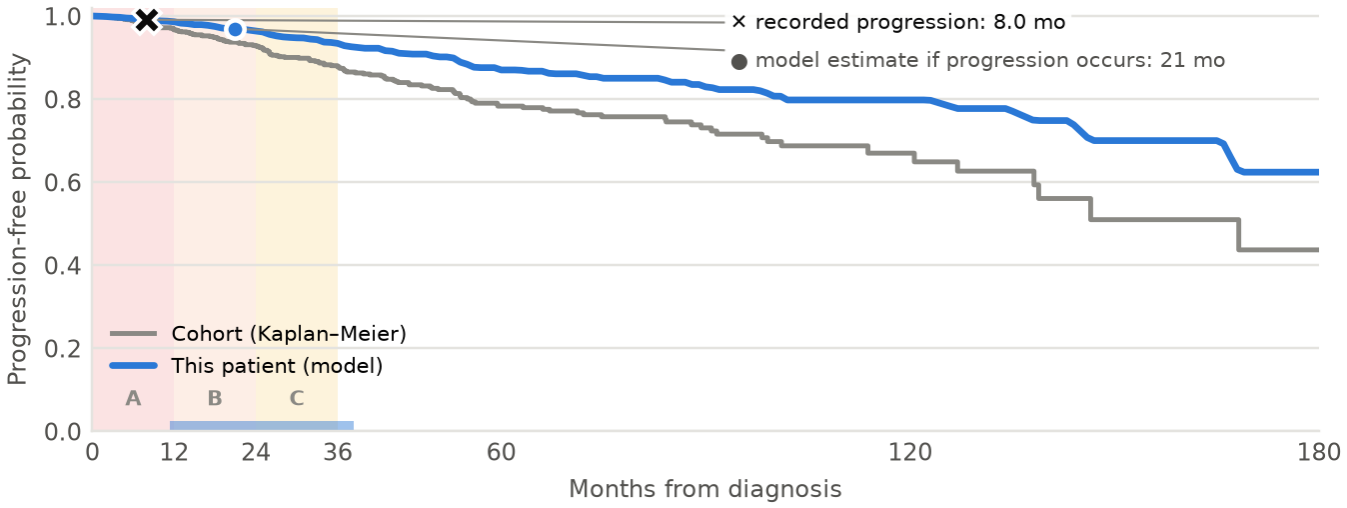
≈ 21 months

Window 12–24 months · IQR 11–38 mo
if progression occurs

Recorded progression

8.0 months

Group A · Early progression
✗ recorded window differs · |error| 13 mo



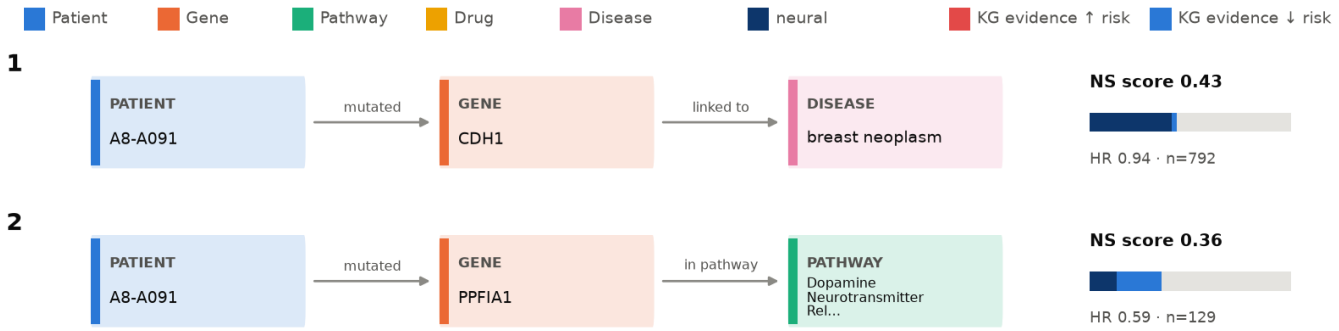
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 61-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **4th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **CDH1** to **breast neoplasm**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.94). Progression was recorded at 8.0 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

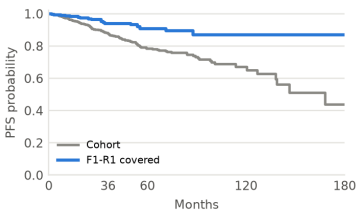


None of the 23 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-A8-A091** mutated → **CDH1** linked to → **breast neoplasm**	68	↓ **slower progression** HR 0.94, z -0.2, n = 792 carriers of “breast neoplasm”	**0.43**
2	**TCGA-A8-A091** mutated → **PPFIA1** in pathway → **Dopamine Neurotransmitter Release Cycle**	22	↓ **slower progression** HR 0.59, z -1.7, n = 129 carriers of “Dopamine Neurotransmitter Release Cycle”	**0.36**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

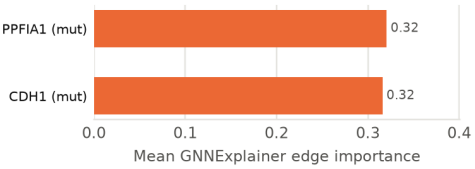
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.E243K	17	0.32
PPFIA1	Mutation	p.A638T	6	0.32



2 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AO-A12B	63	II	LumA	0.87	no progression, followed 98.3 mo
TCGA-B6-A2IU	62	II	LumA	0.86	progressed at 138.9 mo (group D)
TCGA-E9-A1R6	63	II	LumA	0.83	no progression, followed 11.1 mo
TCGA-GM-A2DM	57	II	LumA	0.82	no progression, followed 106.1 mo
TCGA-AR-A2LN	65	II	LumA	0.80	no progression, followed 38.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for

51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-A2-A0T2 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**66**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N3**
Radiation therapy	**No**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

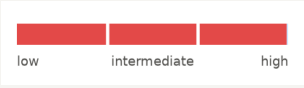
Sample: **TCGA-A2-A0T2-01** · Aneuploidy score: **34** · TMB (nonsyn., /Mb): **0.93** · MSIsensor score: **1.48** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

99th

High risk tier · fused neuro-symbolic model



Probability of progression

67% by 36 mo

12 mo 20% · 24 mo 45% · 60 mo 89%

Model-estimated progression time

≈ 8 months

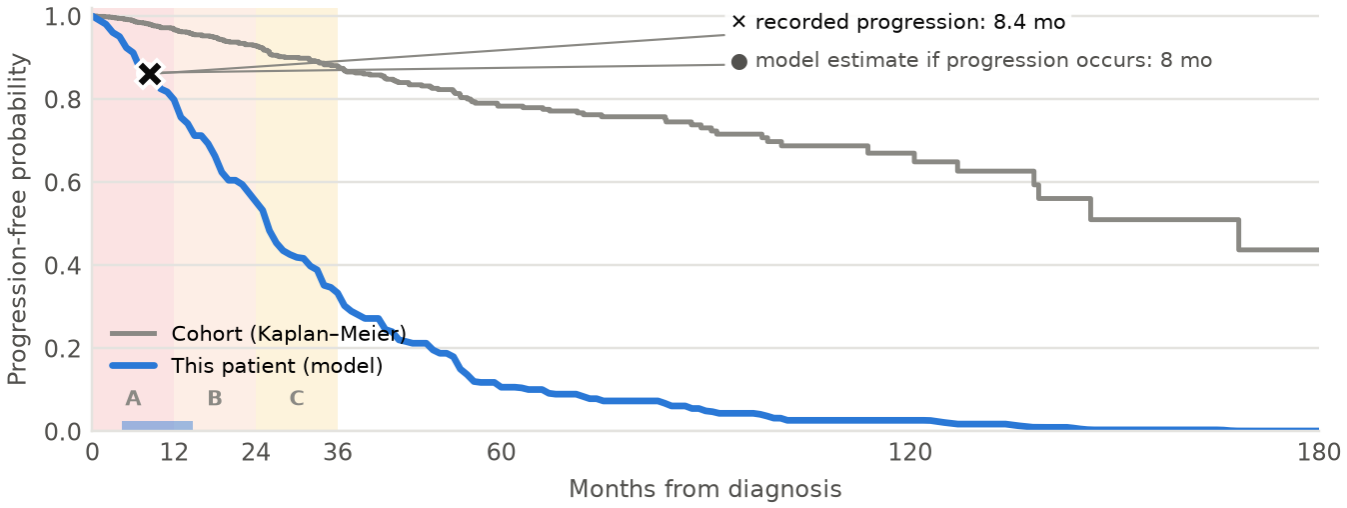
Window ≤ 12 months · IQR 4–15 mo if progression occurs

Recorded progression

8.4 months

Group A · Early progression

✓ **estimate in recorded window** · |error| 0 mo



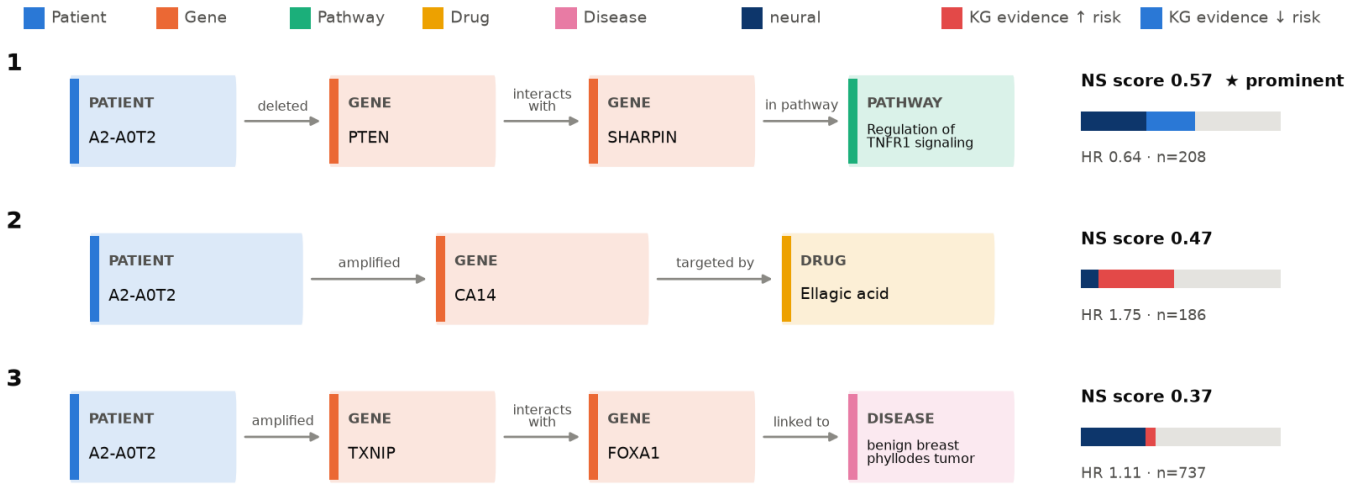
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 66-year-old patient has a Basal tumour, AJCC stage IV (T3, N3, M1). The fused neuro-symbolic model places them in the **99th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **PTEN** to **Regulation of TNFR1 signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.64). Progression was recorded at 8.4 months; the model's estimate given progression was 8 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

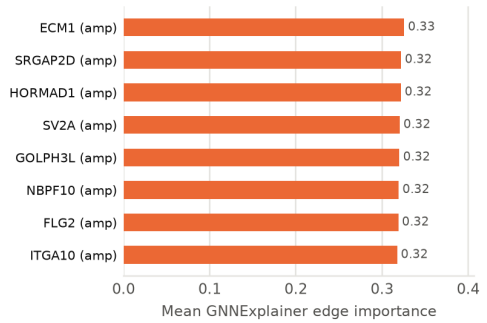
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A2-A0T2** deleted → **PTEN** interacts with → **SHARPIN** in pathway → **Regulation of TNFR1 signaling**	55	↓ **slower progression** HR 0.64, z -1.8, n = 208 carriers of “Regulation of TNFR1 signaling”	**0.57**
2	**TCGA-A2-A0T2** amplified → **CA14** targeted by → **Ellagic acid**	15	↑ **faster progression** HR 1.75, z +2.8, n = 186 carriers of “Ellagic acid”	**0.47**
3	**TCGA-A2-A0T2** amplified → **TXNIP** interacts with → **FOXA1** linked to → **benign breast phyllodes tumor**	54	↑ **faster progression** HR 1.11, z +0.4, n = 737 carriers of “benign breast phyllodes tumor”	**0.37**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	High-level amplification	GISTIC ±2	13	0.32
TP53 · driver	Mutation	p.Y220C	57	0.31
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
ECM1	High-level amplification	GISTIC ±2	1	0.33
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32
HORMAD1	High-level amplification	GISTIC ±2	0	0.32
SV2A	High-level amplification	GISTIC ±2	7	0.32
GOLPH3L	High-level amplification	GISTIC ±2	0	0.32
NBPF10	High-level amplification	GISTIC ±2	0	0.32
FLG2	High-level amplification	GISTIC ±2	1	0.32
ITGA10	High-level amplification	GISTIC ±2	3	0.32
ARNT	High-level amplification	GISTIC ±2	6	0.32



32 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PTEN → Phosphatidylethanolamine (DB04327); SV2A → Levetiracetam (DB01202) … and 20 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0E0	38	III	Basal	0.70	no progression, followed 4.4 mo
TCGA-EW-A1P8	58	III	Basal	0.64	progressed at 6.8 mo (group A)
TCGA-BH-A0B3	53	II	Basal	0.62	no progression, followed 39.6 mo
TCGA-BH-A0AV	52	I	Basal	0.62	no progression, followed 59.8 mo
TCGA-AC-A2BK	78	III	Basal	0.59	no progression, followed 73.1 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**88**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIB**
Genetic ancestry	**—**	Pathologic T	**T4B**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8541/3**

Sample: **TCGA-LL-A5YM-01** · Aneuploidy score: **16** · TMB (nonsyn., /Mb): **0.77** · MSIsensor score: **0.2** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

90th

High risk tier · fused neuro-symbolic model



Probability of progression

17% by 36 mo

12 mo 4% · 24 mo 9% · 60 mo 31%

Model-estimated progression time

≈ 12 months

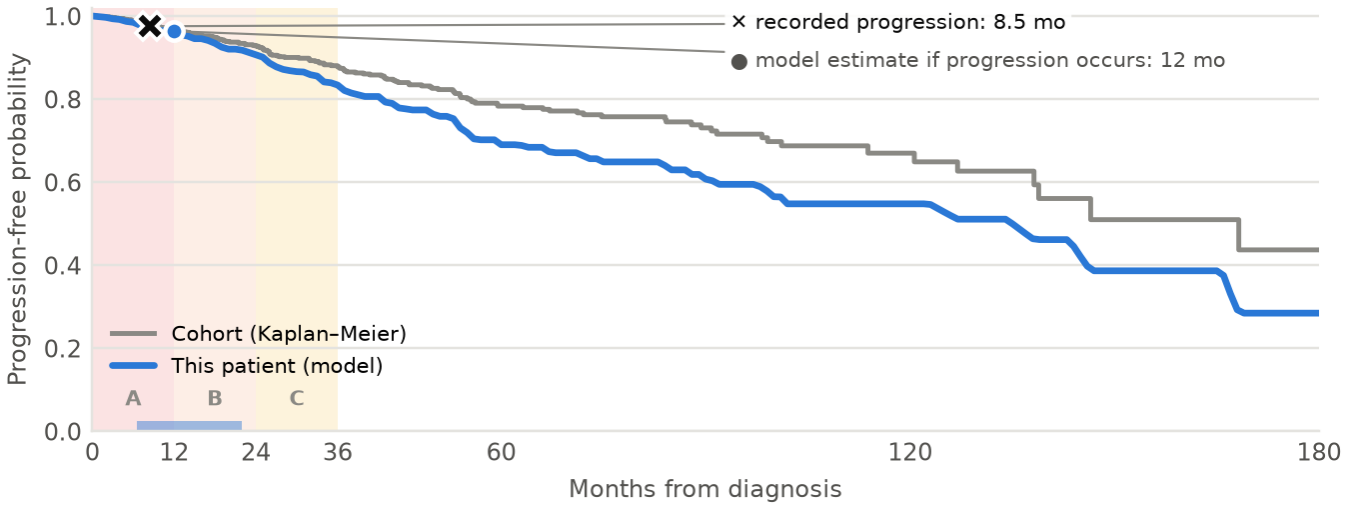
Window ≤ 12 months · IQR 7–22 mo if progression occurs

Recorded progression

8.5 months

■ **Group A** · Early progression

✓ **estimate in recorded window** · |error| 4 mo



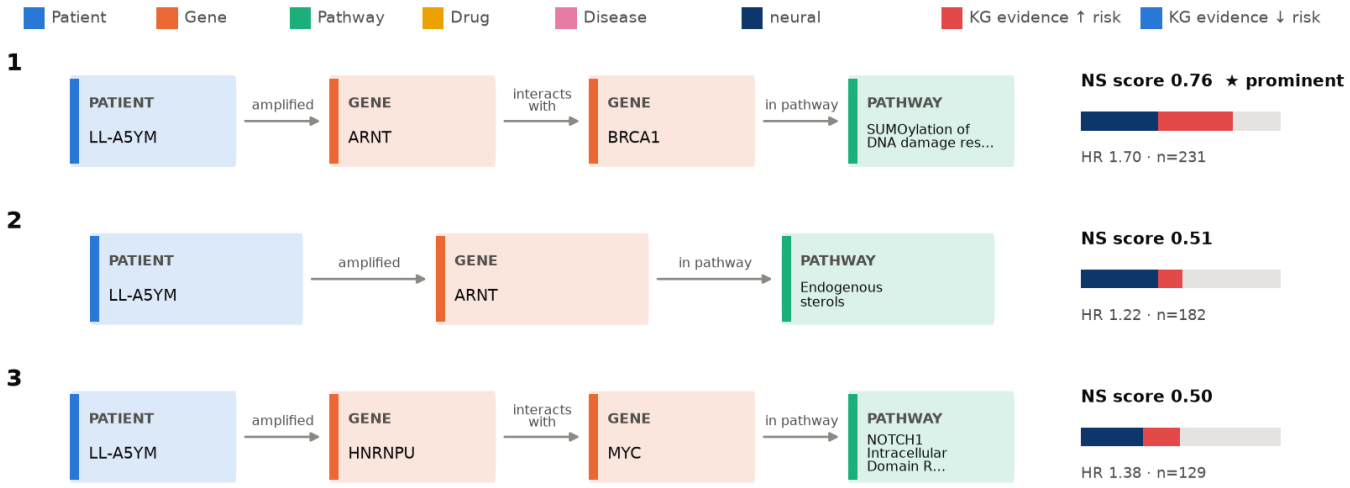
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 88-year-old patient has a LumB tumour, AJCC stage III (T4, missing, M0). The fused neuro-symbolic model places them in the **90th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ARNT** to **SUMOylation of DNA damage response and repair proteins**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.70). Progression was recorded at 8.5 months; the model's estimate given progression was 12 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

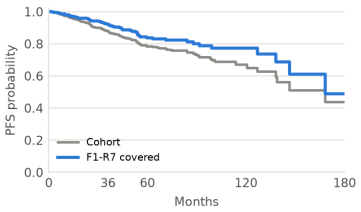


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-LL-A5YM** amplified → **ARNT** interacts with → **BRCA1** in pathway → **SUMOylation of DNA damage response and repair proteins**	65	↑ **faster progression** HR 1.70, z +2.8, n = 231 carriers of “SUMOylation of DNA damage response and repair proteins”	**0.76**
2	**TCGA-LL-A5YM** amplified → **ARNT** in pathway → **Endogenous sterols**	64	↑ **faster progression** HR 1.22, z +0.9, n = 182 carriers of “Endogenous sterols”	**0.51**
3	**TCGA-LL-A5YM** amplified → **HNRNPU** interacts with → **MYC** in pathway → **NOTCH1 Intracellular Domain Regulates Transcription**	52	↑ **faster progression** HR 1.38, z +1.4, n = 129 carriers of “NOTCH1 Intracellular Domain Regulates Transcription”	**0.50**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

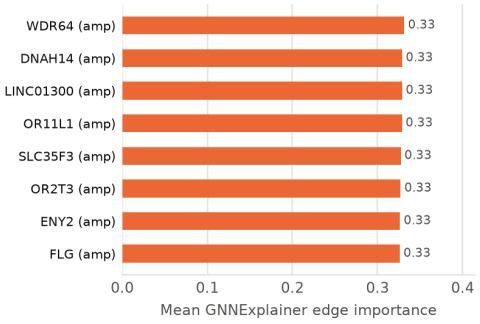
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R7 ↓ **lower risk**	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P432Tfs*41	13	0.31
MYC · driver	High-level amplification	GISTIC ±2	20	0.31
WDR64	High-level amplification	GISTIC ±2	0	0.33
DNAH14	High-level amplification	GISTIC ±2	0	0.33
LINC01300	High-level amplification	GISTIC ±2	0	0.33
OR11L1	High-level amplification	GISTIC ±2	1	0.33
SLC35F3	High-level amplification	GISTIC ±2	0	0.33
OR2T3	High-level amplification	GISTIC ±2	1	0.33
ENY2	High-level amplification	GISTIC ±2	1	0.33
FLG	High-level amplification	GISTIC ±2	1	0.33
FAM72C	High-level amplification	GISTIC ±2	0	0.33
PLD5	High-level amplification	GISTIC ±2	0	0.33



184 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

MYC → Acetylsalicylic acid (DB00945); MYC → Nadroparin (DB08813); NID1 → Urokinase (DB00013); KCNK9 → Doxapram (DB00561); KCNK9 → Halothane (DB01159); CA14 → Benzthiazide (DB00562); CA14 → Cyclothiazide (DB00606); CA14 → Acetazolamide (DB00819); CA14 → Zonisamide (DB00909); CA14 → Ellagic acid (DB08846) … and 60 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A1XG	86	III	LumA	0.81	no progression, followed 14.7 mo
TCGA-AN-A0FK	88	III	LumA	0.80	no progression, followed 7.0 mo
TCGA-A8-A09B	58	III	LumA	0.76	no progression, followed 12.0 mo
TCGA-A8-A06Y	66	II	LumA	0.74	progressed at 26.0 mo (group C)
TCGA-AN-A0XP	69	III	LumA	0.73	no progression, followed 0.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**39**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**Asian**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EAS**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

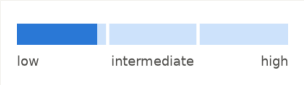
Sample: **TCGA-AO-A126-01** · Aneuploidy score: **4** · TMB (nonsyn., /Mb): **0.63** · MSIsensor score: **0.16** · AJCC edition: **—**

2 · Progression assessment

Model risk (cohort percentile)

30th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 16%

Model-estimated progression time

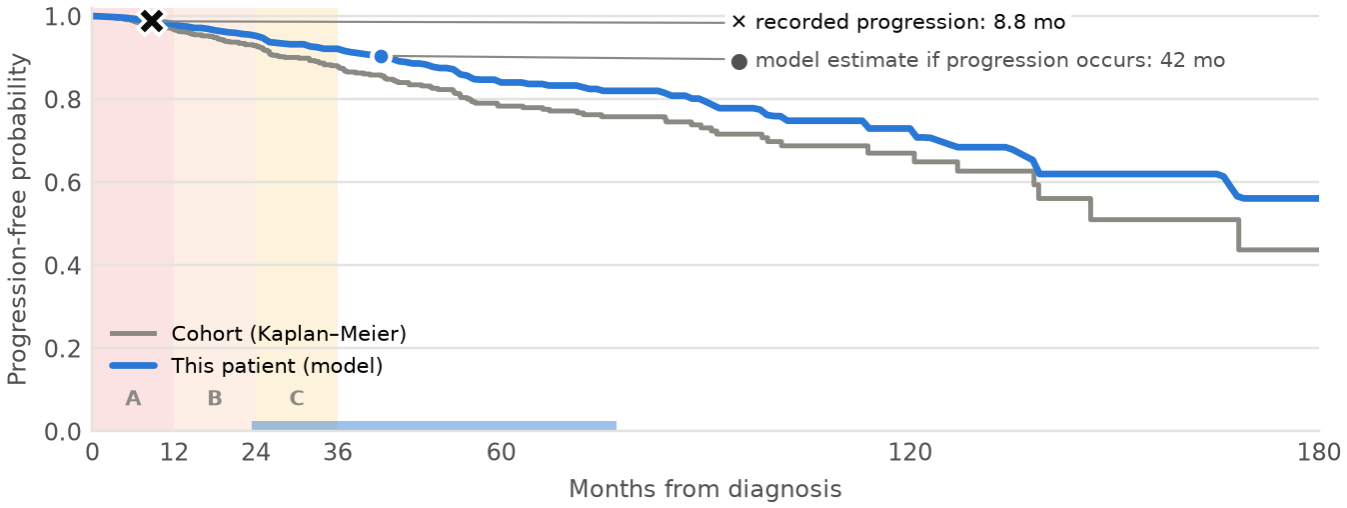
≈ 42 months

Window > 36 months · IQR 23–77 mo if progression occurs

Recorded progression

8.8 months

■ **Group A** · Early progression
✗ recorded window differs · |error| 34 mo



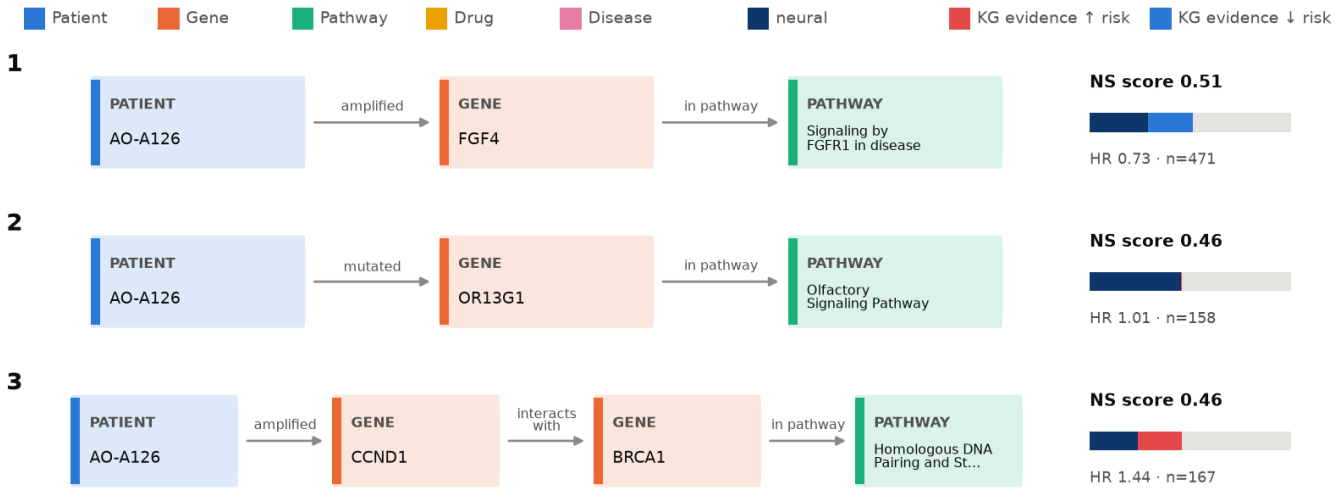
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 39-year-old patient has a LumA tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **30th risk percentile** (low tier). The most significant rule covering them (*Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **FGF4** to **Signaling by FGFR1 in disease**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.73). Progression was recorded at 8.8 months; the model's estimate given progression was 42 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

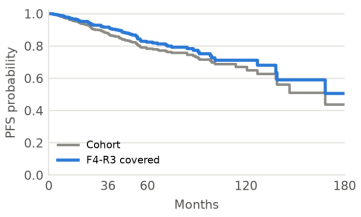


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AO-A126** amplified → **FGF4** in pathway → **Signaling by FGFR1 in disease**	48	↓ **slower progression** HR 0.73, z -1.7, n = 471 carriers of "Signaling by FGFR1 in disease"	**0.51**
2	**TCGA-AO-A126** mutated → **OR13G1** in pathway → **Olfactory Signaling Pathway**	75	↑ **faster progression** HR 1.01, z +0.1, n = 158 carriers of "Olfactory Signaling Pathway"	**0.46**
3	**TCGA-AO-A126** amplified → **CCND1** interacts with → **BRCA1** in pathway → **Homologous DNA Pairing and Strand Exchange**	40	↑ **faster progression** HR 1.44, z +1.6, n = 167 carriers of "Homologous DNA Pairing and Strand Exchange"	**0.46**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

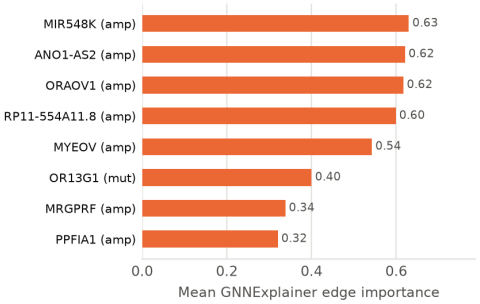
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R3 ↓ **lower risk**	Age < 83.5 AND M stage = M0 AND TMB (nonsynonymous) ≥ 0.20	662 pts 77 events	**0.36** p 2.3e-07	not reached
F4-R2 ↓ **lower risk**	N stage = N0 AND 34.5 ≤ Age < 83.5	386 pts 32 events	**0.43** p 2.5e-05	not reached
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo
F4-R5 ↓ **lower risk**	M stage = M0 AND Age ≥ 34.5	713 pts 89 events	**0.47** p 1.0e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
PIK3CA · driver	Mutation	p.E545K	70	0.32
MIR548K	High-level amplification	GISTIC ±2	0	0.63
ANO1-AS2	High-level amplification	GISTIC ±2	0	0.62
ORAOV1	High-level amplification	GISTIC ±2	0	0.62
RP11-554A11.8	High-level amplification	GISTIC ±2	0	0.60
MYEOV	High-level amplification	GISTIC ±2	0	0.54
OR13G1	Mutation	p.I42S	1	0.40
MRGPRF	High-level amplification	GISTIC ±2	0	0.34
PPFIA1	High-level amplification	GISTIC ±2	6	0.32
KCNK9	Mutation	p.H188=	4	0.32
FGF19	High-level amplification	GISTIC ±2	13	0.32



21 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015) … and 8 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A7-A26J	49	II	LumA	0.92	no progression, followed 20.6 mo
TCGA-E9-A1QZ	61	II	LumA	0.89	no progression, followed 24.8 mo
TCGA-C8-A12Y	44	II	LumA	0.88	no progression, followed 48.5 mo
TCGA-A8-A093	61	II	LumA	0.85	no progression, followed 18.0 mo
TCGA-BH-A18I	53	II	LumA	0.83	no progression, followed 35.9 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**50**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**Yes**	Pathologic N	**N2**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AR-A1AR-01** · Aneuploidy score: **17** · TMB (nonsyn., /Mb): **1.57** · MSIsensor score: **0.03** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

88th

High risk tier · fused neuro-symbolic model



Probability of progression

16% by 36 mo

12 mo 3% · 24 mo 9% · 60 mo 30%

Model-estimated progression time

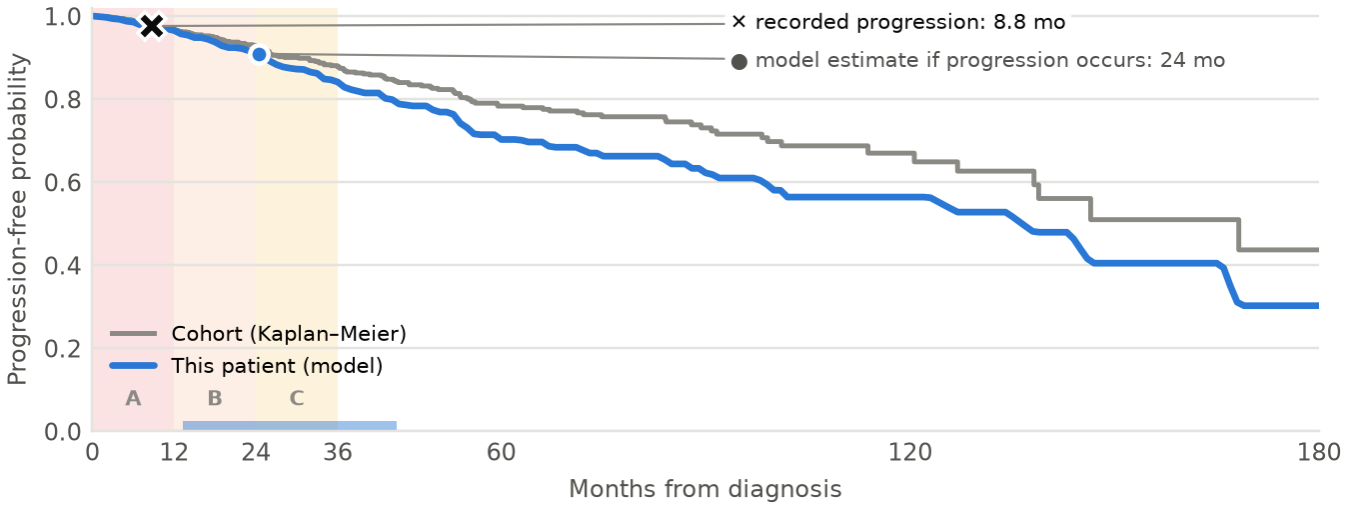
≈ 24 months

Window 24–36 months · IQR 13–45 mo
if progression occurs

Recorded progression

8.8 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 16 mo



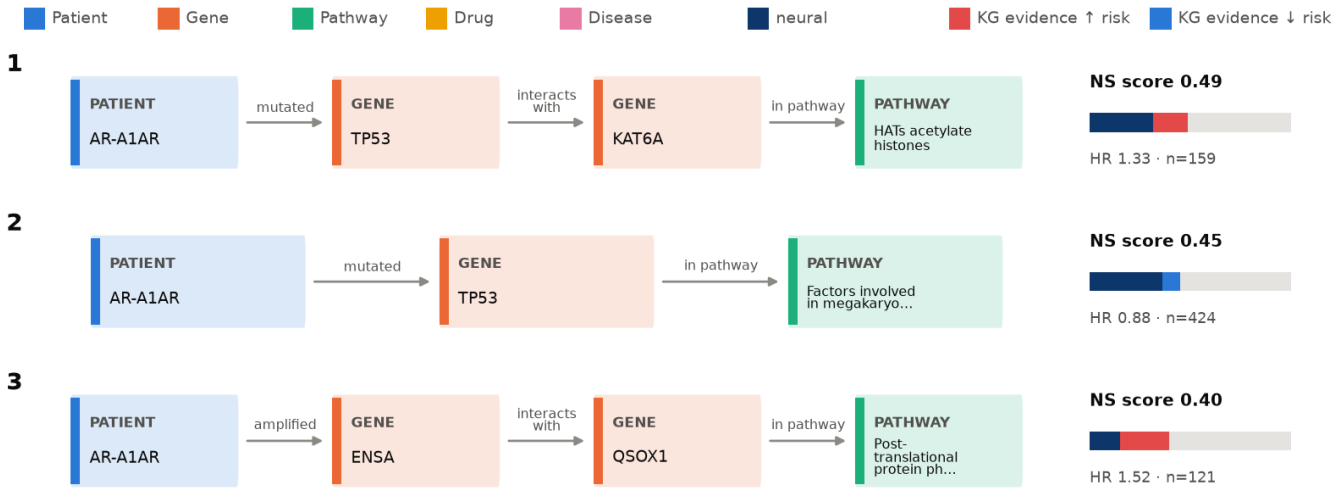
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 50-year-old patient has a Basal tumour, AJCC stage III (T1, N2, M0). The fused neuro-symbolic model places them in the **88th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **TP53** to **HATs acetylate histones**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.33). Progression was recorded at 8.8 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

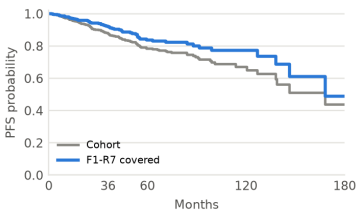


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A1AR** mutated → **TP53** interacts with → **KAT6A** in pathway → **HATs acetylate histones**	53	↑ **faster progression** HR 1.33, z +1.3, n = 159 carriers of “HATs acetylate histones”	**0.49**
2	**TCGA-AR-A1AR** mutated → **TP53** in pathway → **Factors involved in megakaryocyte development and platelet production**	60	↓ **slower progression** HR 0.88, z -0.7, n = 424 carriers of “Factors involved in megakaryocyte development and platelet production”	**0.45**
3	**TCGA-AR-A1AR** amplified → **ENSA** interacts with → **QSOX1** in pathway → **Post-translational protein phosphorylation**	25	↑ **faster progression** HR 1.52, z +1.8, n = 121 carriers of “Post-translational protein phosphorylation”	**0.40**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

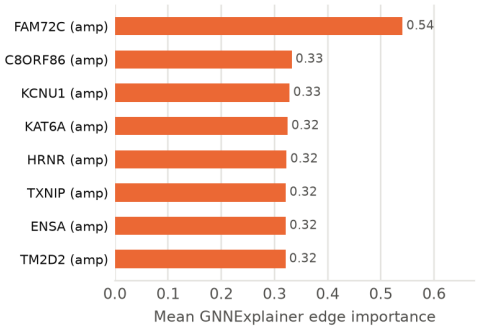
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R7 ↓ **lower risk**	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
KMT2C · driver	Homozygous deletion	GISTIC ±2	4	0.32
FGFR1 · driver	High-level amplification	GISTIC ±2	43	0.32
PTEN · driver	Homozygous deletion	GISTIC ±2	22	0.31
TP53 · driver	Mutation	p.S90Pfs*33	57	0.31
FAM72C	High-level amplification	GISTIC ±2	0	0.54
C8ORF86	High-level amplification	GISTIC ±2	0	0.33
KCNU1	High-level amplification	GISTIC ±2	1	0.33
KAT6A	High-level amplification	GISTIC ±2	2	0.32
HRNR	High-level amplification	GISTIC ±2	3	0.32
TXNIP	High-level amplification	GISTIC ±2	4	0.32
ENSA	High-level amplification	GISTIC ±2	1	0.32
TM2D2	High-level amplification	GISTIC ±2	0	0.32



62 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

FGFR1 → Palifermin (DB00039); FGFR1 → Sorafenib (DB00398); FGFR1 → Heparin (DB01109); FGFR1 → 3-[4-(1-formylpiperazin-4-yl)-benzylidenyl]-2-indolinone (DB02058); FGFR1 → XL999 (DB05014); FGFR1 → 3-(3-methoxybenzyl)-1H-pyrrolo[2,3-b]pyridine (DB07525); FGFR1 → (E)-[4-(3,5-Difluorophenyl)-3H-pyrrolo[2,3-b]pyridin-3-ylidene](3-methoxyphenyl)methanol (DB07840); FGFR1 → 3-[(3-(2-CARBOXYETHYL)-4-METHYLPYRROL-2-YL)METHYLENE]-2-INDOLINONE (DB08577); FGFR1 → Regorafenib (DB08896); FGFR1 → Ponatinib (DB08901) … and 66 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A0E6	69	I	Basal	0.79	no progression, followed 9.6 mo
TCGA-GM-A2DF	53	II	Basal	0.77	no progression, followed 70.8 mo
TCGA-AN-A0XN	68	III	LumA	0.76	no progression, followed 0.3 mo
TCGA-AR-A1AJ	83	I	Basal	0.75	no progression, followed 101.0 mo
TCGA-D8-A1XD	36	III	missing	0.69	no progression, followed 17.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-OL-A97C · Group A

Report date 14 Oct 2026

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**67**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**Black or African American**	PAM50 subtype	**Normal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIB**
Genetic ancestry	**AFR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N0**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**9020/3**

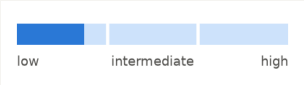
Sample: **TCGA-OL-A97C-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **2** · MSIsensor score: **0.05** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

25th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 4% · 60 mo 16%

Model-estimated progression time

≈ 17 months

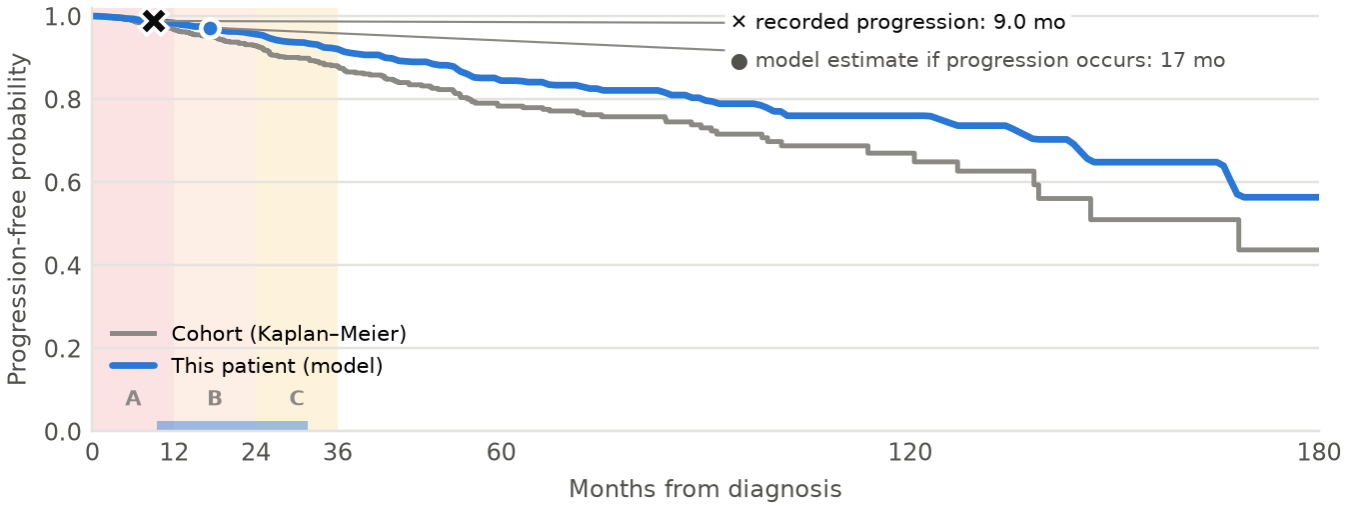
Window 12–24 months · IQR 9–32 mo if progression occurs

Recorded progression

9.0 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 8 mo



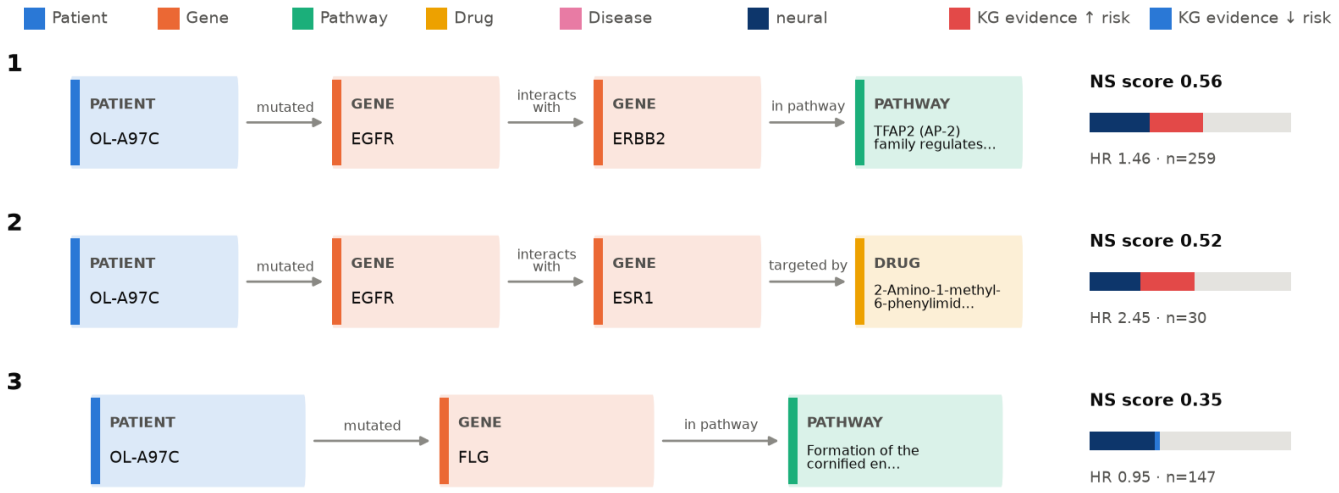
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 67-year-old patient has a Normal tumour, AJCC stage II (T3, N0, M0). The fused neuro-symbolic model places them in the **25th risk percentile** (low tier). The most significant rule covering them (*N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **EGFR** to **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.46). Progression was recorded at 9.0 months; the model's estimate given progression was 17 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

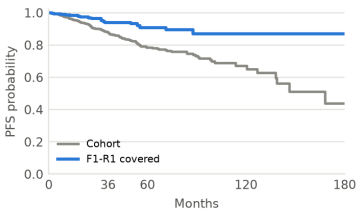


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-OL-A97C** mutated → **EGFR** interacts with → **ERBB2** in pathway → **TFAP2 (AP-2) family regulates transcription of growth factors and their receptors**	50	↑ **faster progression** HR 1.46, z +2.0, n = 259 carriers of "TFAP2 (AP-2) family regulates transcription of growth factors and their receptors"	**0.56**
2	**TCGA-OL-A97C** mutated → **EGFR** interacts with → **ESR1** targeted by → **2-Amino-1-methyl-6-phenylimidazo(4,5-b)pyridine**	42	↑ **faster progression** HR 2.45, z +2.0, n = 30 carriers of "2-Amino-1-methyl-6-phenylimidazo(4,5-b)pyridine"	**0.52**
3	**TCGA-OL-A97C** mutated → **FLG** in pathway → **Formation of the cornified envelope**	54	↓ **slower progression** HR 0.95, z -0.2, n = 147 carriers of "Formation of the cornified envelope"	**0.35**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

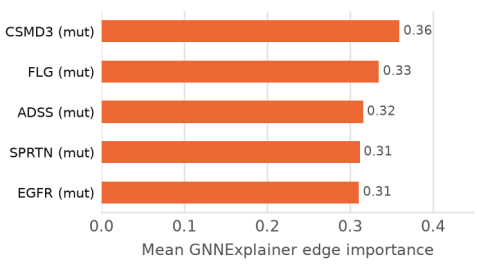
Rule	IF ...	Covered	HR vs rest	Median PFS
F1-R1 ↓ lower risk	N stage = N0 AND Age ≥ 41.5 AND TMB (nonsynonymous) < 8.45	365 pts 19 events	**0.26** p 4.0e-09	not reached
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo
F1-R2 ↓ lower risk	N stage = N0 AND Age ≥ 31.5	410 pts 29 events	**0.36** p 4.5e-07	not reached
F1-R8 ↓ lower risk	AJCC stage = II AND Age ≥ 39.5 AND TMB (nonsynonymous) < 141.95	452 pts 39 events	**0.47** p 7.9e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
EGFR · driver	Mutation	p.A289V	73	0.31
CSMD3	Mutation	p.L49V	0	0.36
FLG	Mutation	p.G2143E	1	0.33
ADSS	Mutation	3'UTR	0	0.32
SPRTN	Mutation	p.R140H	1	0.31



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

EGFR → Cetuximab (DB00002); EGFR → Lidocaine (DB00281); EGFR → Gefitinib (DB00317); EGFR → Erlotinib (DB00530); EGFR → Lapatinib (DB01259); EGFR → Panitumumab (DB01269); EGFR → Alvocidib (DB03496); EGFR → IGN311 (DB04988); EGFR → Matuzumab (DB05101); EGFR → Vandetanib (DB05294) … and 21 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-AR-A2LQ	59	II	Normal	0.73	no progression, followed 40.5 mo
TCGA-BH-A6R9	61	II	Normal	0.68	no progression, followed 5.3 mo
TCGA-AO-A1KO	46	II	Normal	0.67	no progression, followed 20.4 mo
TCGA-PL-A8LY	30	II	Normal	0.60	no progression, followed 0.3 mo
TCGA-AC-A7VC	56	II	Basal	0.58	no progression, followed 0.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**72**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumA**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T1**
Prior diagnosis	**Yes**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

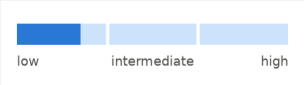
Sample: **TCGA-A7-A26H-01** · Aneuploidy score: **8** · TMB (nonsyn., /Mb): **3.67** · MSIsensor score: **0.18** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

24th

Low risk tier · fused neuro-symbolic model



Probability of progression

8% by 36 mo

12 mo 2% · 24 mo 5% · 60 mo 15%

Model-estimated progression time

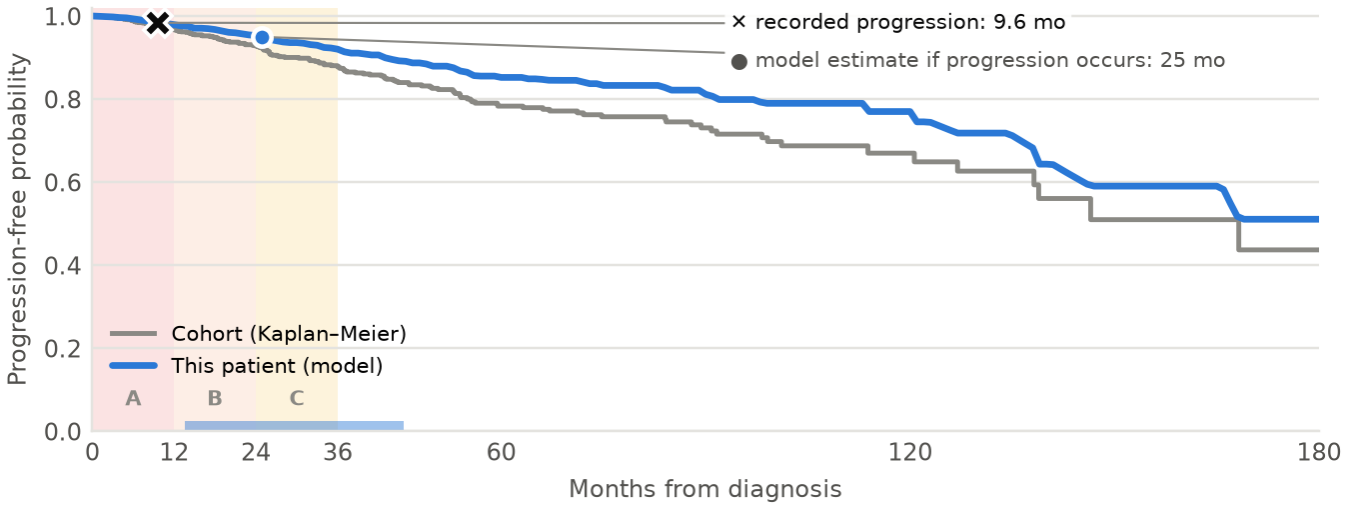
≈ 25 months

Window 24–36 months · IQR 14–46 mo
if progression occurs

Recorded progression

9.6 months

■ **Group A** · Early progression
✗ **recorded window differs** · |error| 15 mo



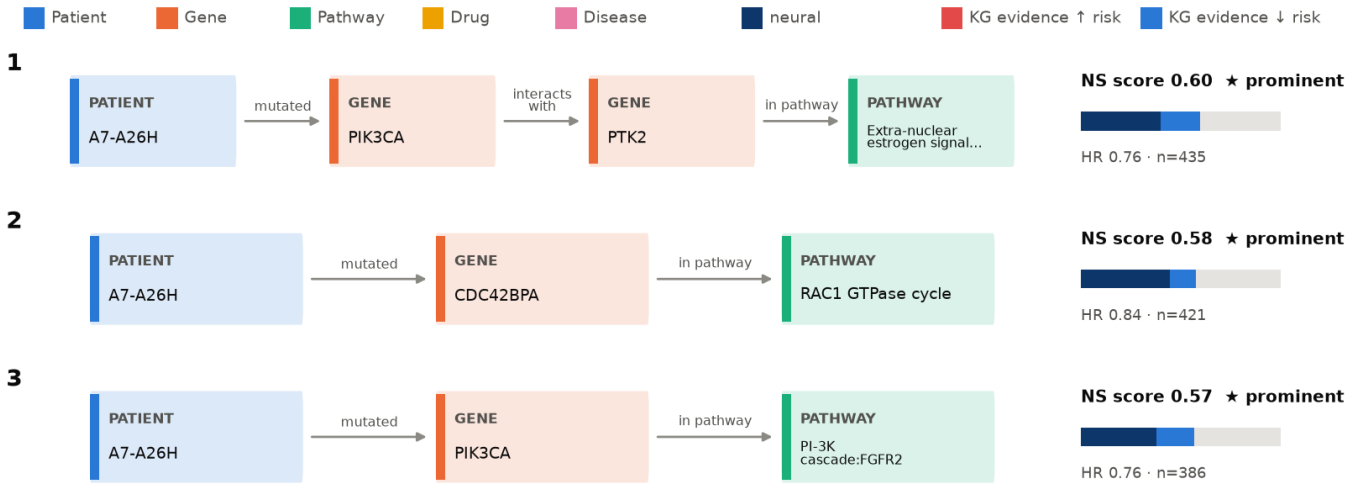
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 72-year-old patient has a LumA tumour, AJCC stage II (T1, N1, M0). The fused neuro-symbolic model places them in the **24th risk percentile** (low tier). The most significant rule covering them (*M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5*) marks a group with a progression hazard 0.3× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **Extra-nuclear estrogen signaling**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.76). Progression was recorded at 9.6 months; the model's estimate given progression was 25 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

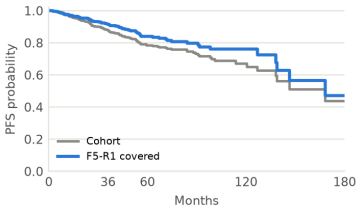


3 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-A7-A26H** mutated → **PIK3CA** interacts with → **PTK2** in pathway → **Extra-nuclear estrogen signaling**	67	↓ **slower progression** HR 0.76, z -1.5, n = 435 carriers of “Extra-nuclear estrogen signaling”	**0.60**
2 ★	**TCGA-A7-A26H** mutated → **CDC42BPA** in pathway → **RAC1 GTPase cycle**	74	↓ **slower progression** HR 0.84, z -1.0, n = 421 carriers of “RAC1 GTPase cycle”	**0.58**
3 ★	**TCGA-A7-A26H** mutated → **PIK3CA** in pathway → **PI-3K cascade:FGFR2**	63	↓ **slower progression** HR 0.76, z -1.4, n = 386 carriers of “PI-3K cascade:FGFR2”	**0.57**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

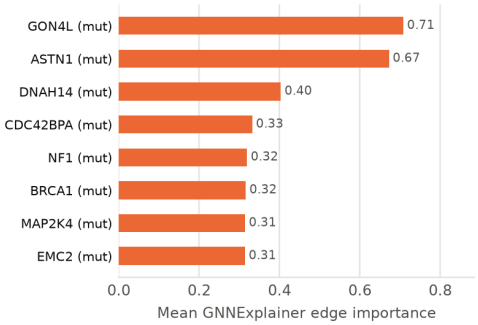
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R1 ↓ lower risk	M stage = M0 AND TMB (nonsynonymous) ≥ 0.22 AND Age < 83.5	647 pts 70 events	**0.32** p 2.6e-09	168 mo
F5-R2 ↓ lower risk	M stage = M0 AND 26.5 ≤ Age < 83.5	691 pts 79 events	**0.35** p 1.5e-07	168 mo
F5-R3 ↓ lower risk	AJCC stage = II AND 39.5 ≤ Age < 78.5	422 pts 31 events	**0.36** p 4.2e-07	168 mo
F5-R4 ↓ lower risk	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ lower risk	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
NF1 · driver	Mutation	p.L852=	6	0.32
BRCA1 · driver	Mutation	p.T688Vfs*12	29	0.32
MAP2K4 · driver	Mutation	p.L385P	12	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
GON4L	Mutation	p.E1520Q	0	0.71
ASTN1	Mutation	p.C699*	0	0.67
DNAH14	Mutation	p.*742*	0	0.40
CDC42BPA	Mutation	p.E889K	4	0.33
EMC2	Mutation	p.E57K	0	0.31



9 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-D8-A145	80	II	LumA	0.85	no progression, followed 13.5 mo
TCGA-BH-A0W5	77	II	LumA	0.84	no progression, followed 42.3 mo
TCGA-AO-A0JF	68	II	LumA	0.81	no progression, followed 65.1 mo
TCGA-E9-A1R0	58	II	LumA	0.81	no progression, followed 28.3 mo
TCGA-E2-A1L6	44	II	LumA	0.79	no progression, followed 54.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**68**	Tumour type	**Breast Invasive Carcinoma (NOS)**
Sex	**Female**	OncoTree code	**BRCNOS**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8575/3**

Sample: **TCGA-AR-A5QQ-01** · Aneuploidy score: **7** · TMB (nonsyn., /Mb): **2.23** · MSIsensor score: **0** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

92nd

High risk tier · fused neuro-symbolic model



Probability of progression

19% by 36 mo

12 mo 5% · 24 mo 12% · 60 mo 33%

Model-estimated progression time

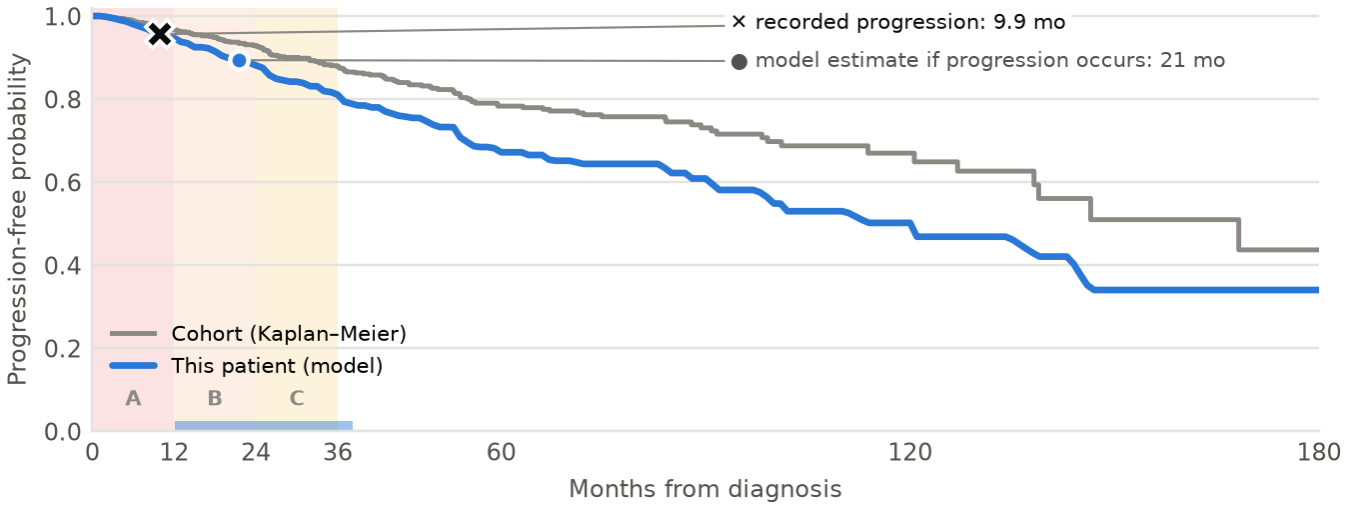
≈ 21 months

Window 12–24 months · IQR 12–38 mo
if progression occurs

Recorded progression

9.9 months

Group A · Early progression
✗ recorded window differs · |error| 12 mo



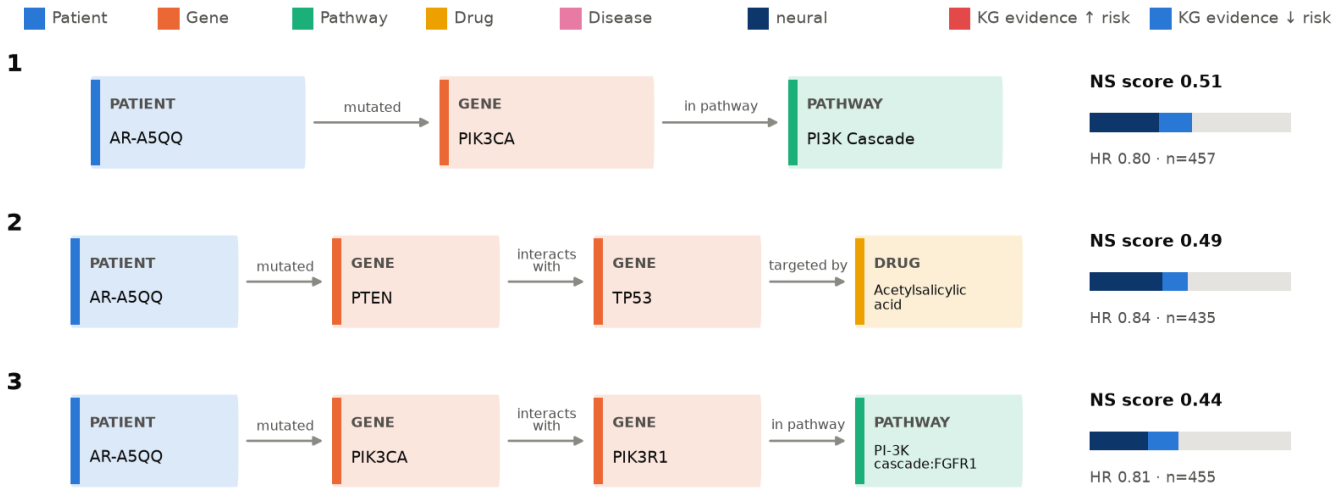
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 68-year-old patient has a Basal tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **92nd risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **PIK3CA** to **PI3K Cascade**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.80). Progression was recorded at 9.9 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

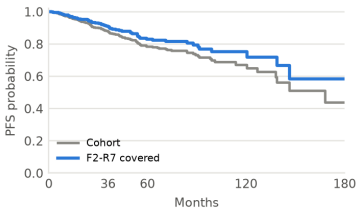


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-AR-A5QQ** mutated → **PIK3CA** in pathway → **PI3K Cascade**	57	↓ slower progression HR 0.80, z -1.2, n = 457 carriers of “PI3K Cascade”	**0.51**
2	**TCGA-AR-A5QQ** mutated → **PTEN** interacts with → **TP53** targeted by → **Acetylsalicylic acid**	60	↓ slower progression HR 0.84, z -1.0, n = 435 carriers of “Acetylsalicylic acid”	**0.49**
3	**TCGA-AR-A5QQ** mutated → **PIK3CA** interacts with → **PIK3R1** in pathway → **PI-3K cascade:FGFR1**	48	↓ slower progression HR 0.81, z -1.2, n = 455 carriers of “PI-3K cascade:FGFR1”	**0.44**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

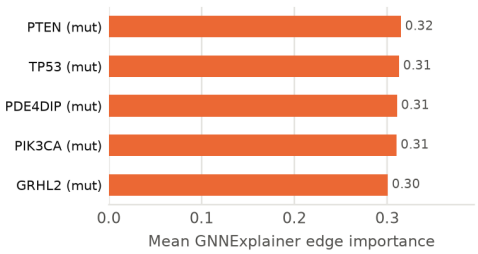
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
PTEN · driver	Mutation	p.R130Q	22	0.32
TP53 · driver	Mutation	p.R209Kfs*6	57	0.31
PIK3CA · driver	Mutation	p.H1047R	70	0.31
PDE4DIP	Mutation	p.H445R	0	0.31
GRHL2	Mutation	p.D336Y	0	0.30



5 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PTEN → Phosphatidylethanolamine (DB04327); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487); TP53 → Zinc chloride (DB14533); TP53 → Zinc sulfate, unspecified form (DB14548); PIK3CA → ATP (DB00171) … and 6 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-B6-A0IE	38	III	LumA	0.82	progressed at 27.4 mo (group C)
TCGA-C8-A133	65	III	LumA	0.68	no progression, followed 0.0 mo
TCGA-E9-A1NC	61	II	Basal	0.64	no progression, followed 39.6 mo
TCGA-A7-A26F	55	I	Basal	0.64	no progression, followed 24.3 mo
TCGA-BH-A0WA	82	I	Basal	0.62	no progression, followed 23.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-D8-A1Y1 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**80**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**LumB**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**No**	Pathologic M	**MX**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-D8-A1Y1-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **6.8** · MSIsensor score: **6.24** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

80th

High risk tier · fused neuro-symbolic model



Probability of progression

13% by 36 mo

12 mo 4% · 24 mo 8% · 60 mo 26%

Model-estimated progression time

≈ 13 months

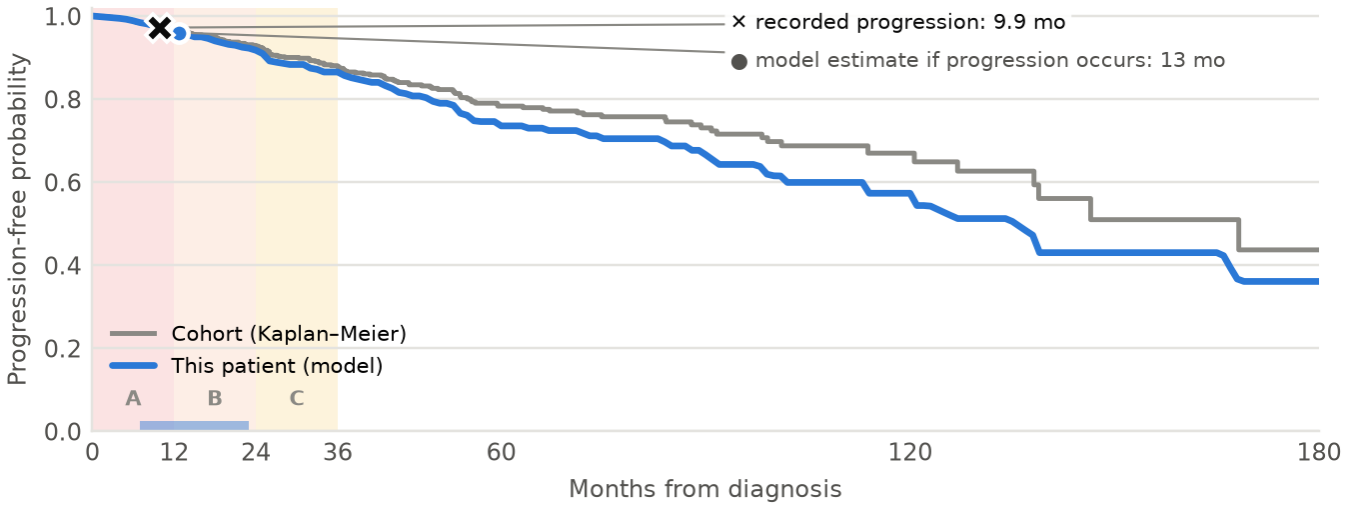
Window 12–24 months · IQR 7–23 mo if progression occurs

Recorded progression

9.9 months

Group A · Early progression

✗ recorded window differs · |error| 3 mo



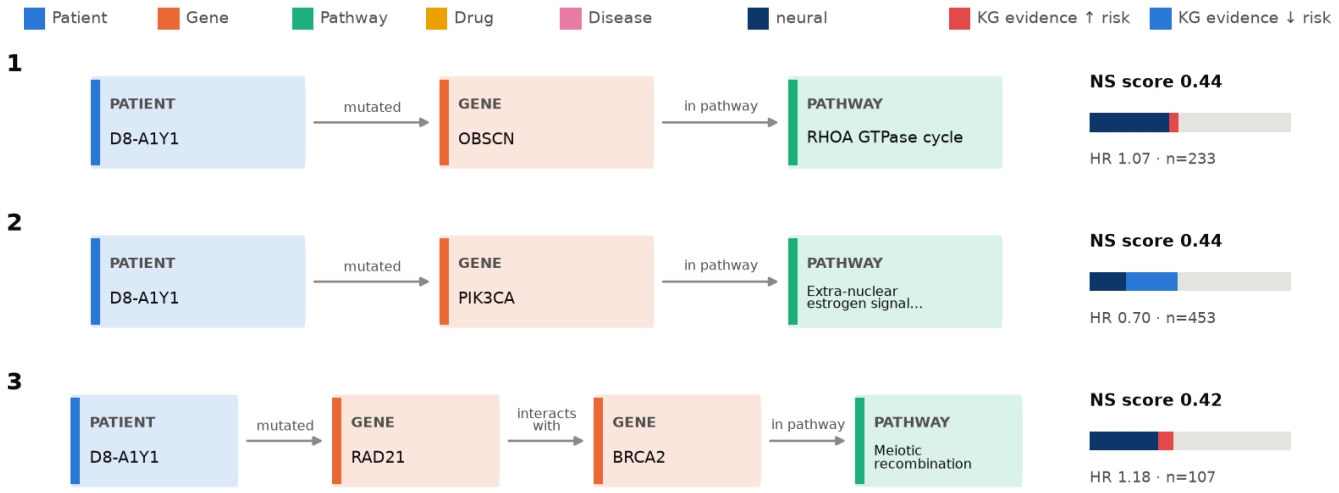
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 80-year-old patient has a LumB tumour, AJCC stage III (T3, N1, MX). The fused neuro-symbolic model places them in the **80th risk percentile** (high tier). The most significant rule covering them (*26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **OBSCN** to **RHOA GTPase cycle**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.07). Progression was recorded at 9.9 months; the model's estimate given progression was 13 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

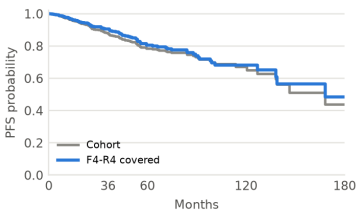


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-D8-A1Y1** mutated → **OBSCN** in pathway → **RHOA GTPase cycle**	66	↑ **faster progression** HR 1.07, z +0.3, n = 233 carriers of “RHOA GTPase cycle”	**0.44**
2	**TCGA-D8-A1Y1** mutated → **PIK3CA** in pathway → **Extra-nuclear estrogen signaling**	30	↓ **slower progression** HR 0.70, z -1.9, n = 453 carriers of “Extra-nuclear estrogen signaling”	**0.44**
3	**TCGA-D8-A1Y1** mutated → **RAD21** interacts with → **BRCA2** in pathway → **Meiotic recombination**	57	↑ **faster progression** HR 1.18, z +0.6, n = 107 carriers of “Meiotic recombination”	**0.42**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

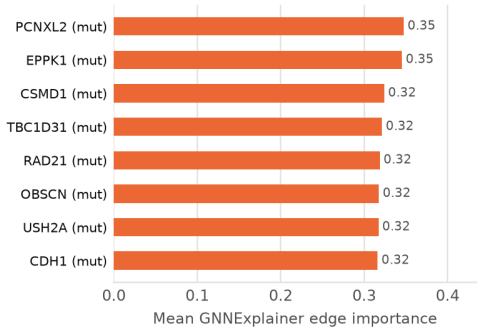
Rule	IF …	Covered	HR vs rest	Median PFS
F4-R4 ↓ **lower risk**	26.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	772 pts 95 events	**0.38** p 6.6e-05	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
CDH1 · driver	Mutation	p.X646_splice	17	0.32
MAP2K4 · driver	Mutation	p.D173Tfs*24	12	0.32
PIK3CA · driver	Mutation	p.E542K, p.F70=	70	0.31
PCNXL2	Mutation	p.T592M	0	0.35
EPPK1	Mutation	p.Q1841*	0	0.35
CSMD1	Mutation	p.G3126S	0	0.32
TBC1D31	Mutation	p.Q205R	0	0.32
RAD21	Mutation	p.A377G	7	0.32
OBSCN	Mutation	p.V4325M	8	0.32
USH2A	Mutation	p.D4968=	0	0.32
TAF2	Mutation	p.L828=	9	0.31
HMCN1	Mutation	p.P2507=	0	0.31



16 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

PIK3CA → ATP (DB00171); PIK3CA → Caffeine (DB00201); PIK3CA → XL765 (DB05241); PIK3CA → Wortmannin (DB08059); PIK3CA → Pilaralisib (DB11772); PIK3CA → Alpelisib (DB12015); PIK3CA → Copanlisib (DB12483); ATP1A4 → Rubidium Rb-82 (DB09479). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-Z7-A8R5	61	III	LumA	0.68	progressed at 6.0 mo (group A)
TCGA-A2-A1G4	71	III	LumB	0.63	no progression, followed 19.6 mo
TCGA-AR-A1AL	60	III	LumA	0.61	no progression, followed 97.7 mo
TCGA-BH-A1FJ	66	III	LumB	0.57	no progression, followed 63.4 mo
TCGA-EW-A1IZ	53	III	LumA	0.56	no progression, followed 18.2 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-E9-A243 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**52**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Basal**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIA**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N0 (I-)**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

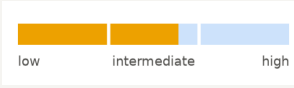
Sample: **TCGA-E9-A243-01** · Aneuploidy score: **21** · TMB (nonsyn., /Mb): **2.3** · MSIsensor score: **0.01** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

59th

Intermediate risk tier · fused neuro-symbolic model



Probability of progression

11% by 36 mo

12 mo 3% · 24 mo 7% · 60 mo 20%

Model-estimated progression time

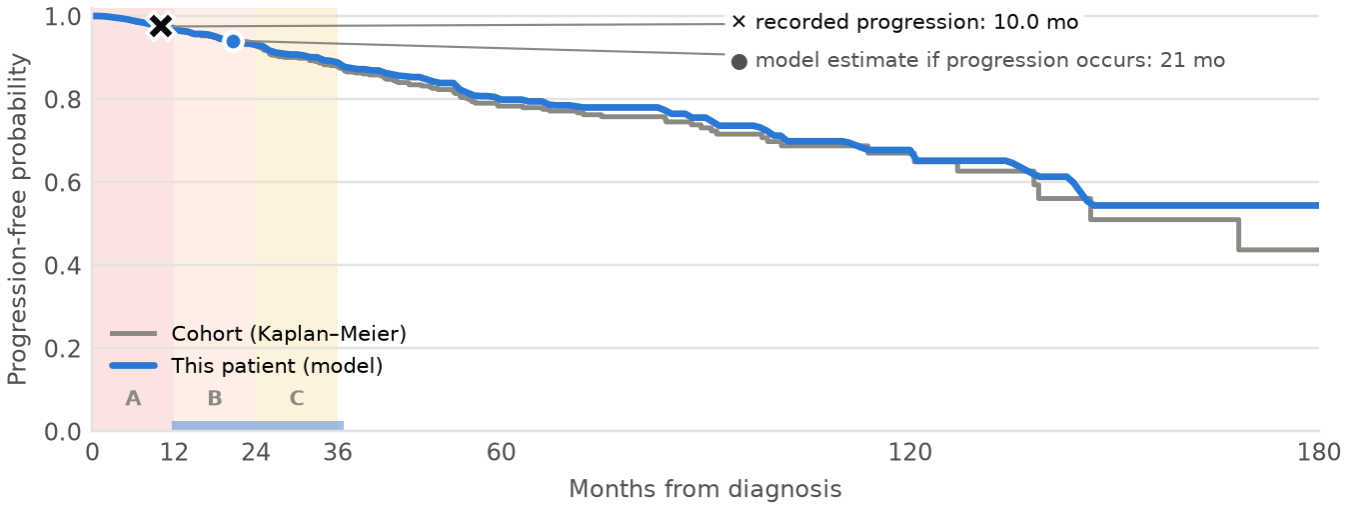
≈ 21 months

Window 12–24 months · IQR 12–37 mo
if progression occurs

Recorded progression

10.0 months

Group A · Early progression
✗ recorded window differs · |error| 11 mo



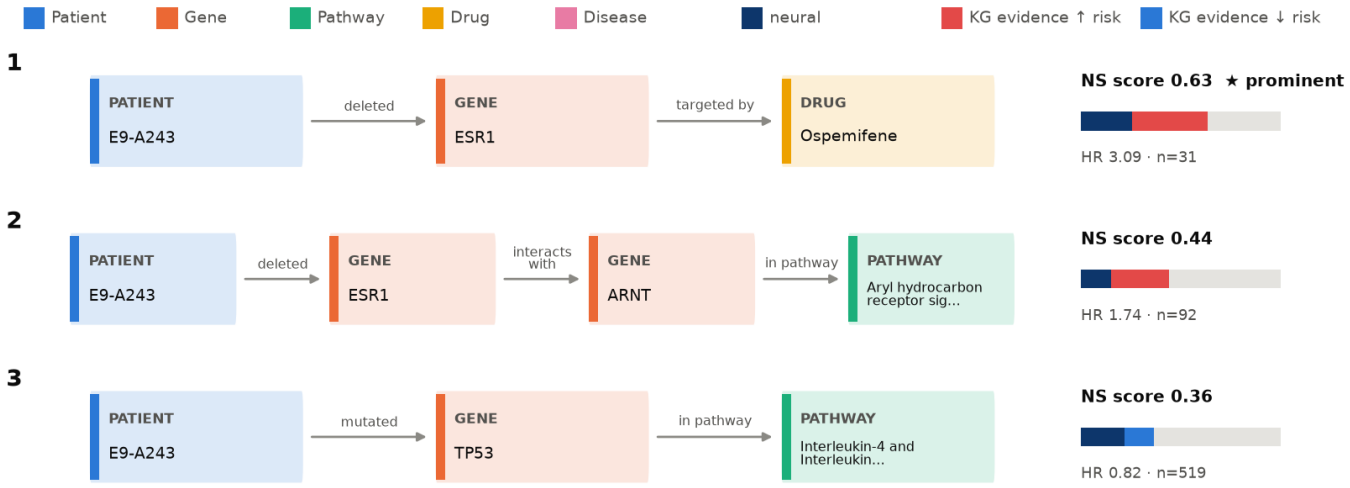
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 52-year-old patient has a Basal tumour, AJCC stage II (T2, N0, M0). The fused neuro-symbolic model places them in the **59th risk percentile** (intermediate tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 83.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ESR1** to **Ospemifene**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 3.09). Progression was recorded at 10.0 months; the model's estimate given progression was 21 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

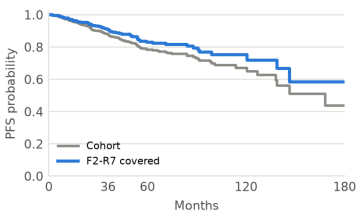


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E9-A243** deleted → **ESR1** targeted by → **Ospemifene**	42	↑ faster progression HR 3.09, z +2.8, n = 31 carriers of “Ospemifene”	**0.63**
2	**TCGA-E9-A243** deleted → **ESR1** interacts with → **ARNT** in pathway → **Aryl hydrocarbon receptor signalling**	25	↑ faster progression HR 1.74, z +2.2, n = 92 carriers of “Aryl hydrocarbon receptor signalling”	**0.44**
3	**TCGA-E9-A243** mutated → **TP53** in pathway → **Interleukin-4 and Interleukin-13 signaling**	37	↓ slower progression HR 0.82, z -1.1, n = 519 carriers of “Interleukin-4 and Interleukin-13 signaling”	**0.36**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

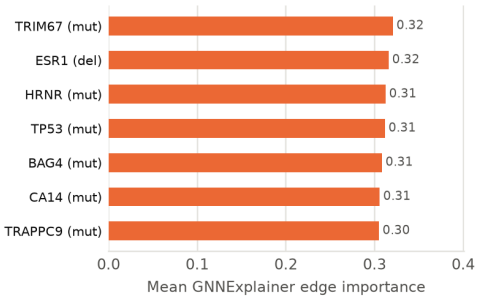
Rule	IF …	Covered	HR vs rest	Median PFS
F2-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 83.5	638 pts 66 events	**0.39** p 3.3e-07	not reached
F2-R2 ↓ lower risk	N stage = N0 AND 41.5 ≤ Age < 78.5	343 pts 24 events	**0.37** p 5.3e-06	not reached
F2-R3 ↓ lower risk	N stage = N0 AND TMB (nonsynonymous) ≥ 0.78 AND Age < 83.5	317 pts 25 events	**0.41** p 3.7e-05	not reached
F2-R8 ↓ lower risk	40.5 ≤ Age < 83.5 AND TMB (nonsynonymous) ≥ 0.20	705 pts 79 events	**0.47** p 1.3e-04	not reached



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ESR1 · driver	Homozygous deletion	GISTIC ±2	141	0.32
TP53 · driver	Mutation	p.W91*, p.Y220S	57	0.31
TRIM67	Mutation	p.D606N	0	0.32
HRNR	Mutation	p.Q396*	3	0.31
BAG4	Mutation	p.S303=	4	0.31
CA14	Mutation	p.V10E	6	0.31
TRAPPC9	Mutation	p.R553=	2	0.30



7 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ESR1 → Diethylstilbestrol (DB00255); ESR1 → Chlorotrianisene (DB00269); ESR1 → Conjugated estrogens (DB00286); ESR1 → Desogestrel (DB00304); ESR1 → Levonorgestrel (DB00367); ESR1 → Progesterone (DB00396); ESR1 → Lindane (DB00431); ESR1 → Raloxifene (DB00481); ESR1 → Toremifene (DB00539); ESR1 → Medroxyprogesterone acetate (DB00603) … and 123 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A3XX	49	II	Basal	0.87	no progression, followed 47.3 mo
TCGA-E2-A1B6	44	II	Basal	0.87	no progression, followed 28.5 mo
TCGA-AR-A1AI	47	II	Basal	0.86	progressed at 19.3 mo (group B)
TCGA-A7-A26I	65	II	Basal	0.85	no progression, followed 21.7 mo
TCGA-C8-A12V	55	II	Basal	0.85	no progression, followed 12.7 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**58**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Male**	OncoTree code	**IDC**
Race	**Black or African American**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**AFR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N1A**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-AC-A62V-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **2.27** · MSIsensor score: **0.07** · AJCC edition: **7TH**

2 · Progression assessment

Model risk (cohort percentile)

98th

High risk tier · fused neuro-symbolic model



Probability of progression

42% by 36 mo

12 mo 10% · 24 mo 25% · 60 mo 67%

Model-estimated progression time

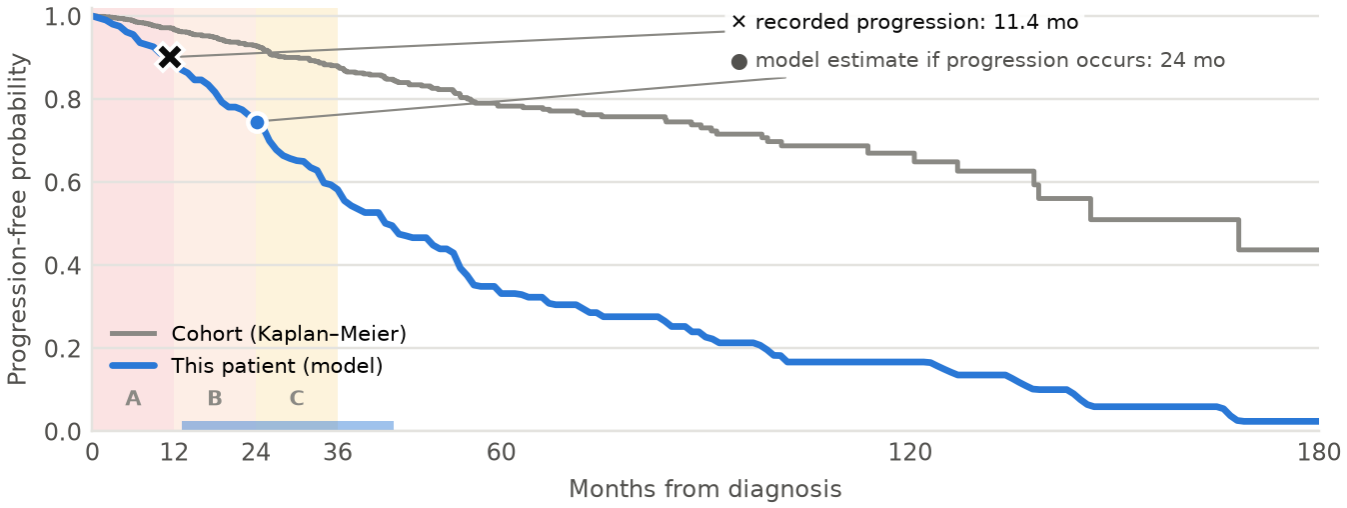
≈ 24 months

Window 24–36 months · IQR 13–44 mo
if progression occurs

Recorded progression

11.4 months

Group A · Early progression
✗ recorded window differs · |error| 13 mo



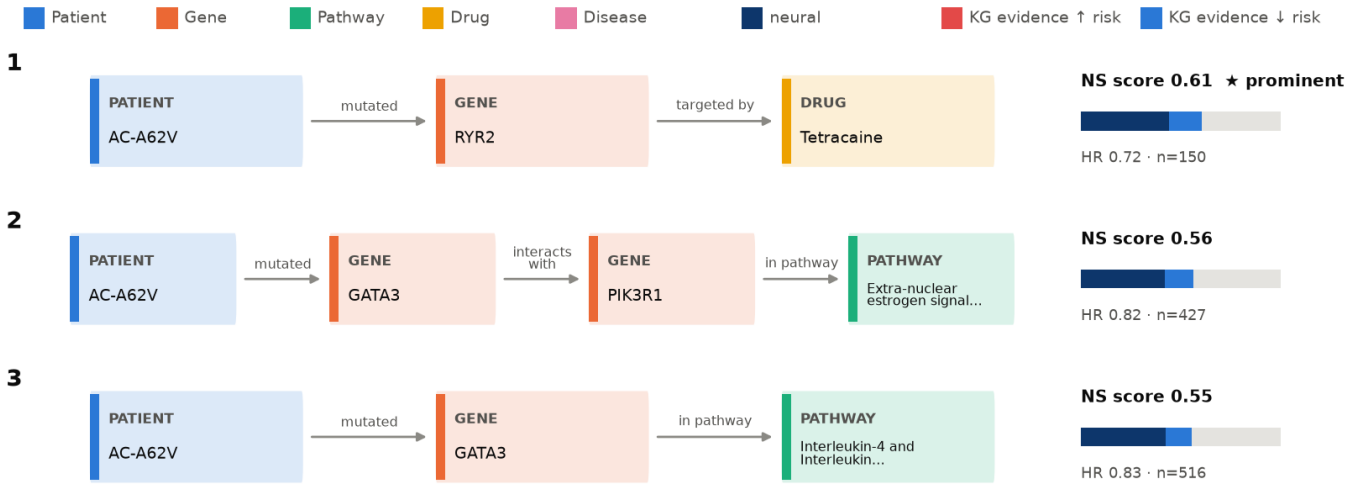
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 58-year-old patient has a subtype-unassigned tumour, AJCC stage IV (T2, N1, M1). The fused neuro-symbolic model places them in the **98th risk percentile** (high tier). The highest-scoring knowledge-graph path runs from **RYR2** to **Tetracaine**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.72). Progression was recorded at 11.4 months; the model's estimate given progression was 24 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)



Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

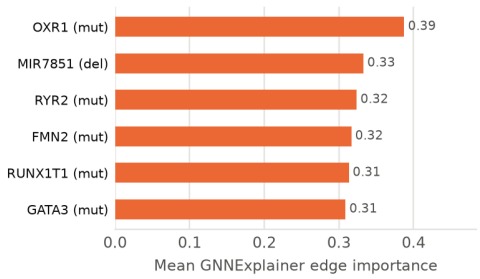
#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-AC-A62V** mutated → **RYR2** targeted by → **Tetracaine**	74	↓ slower progression HR 0.72, z -1.2, n = 150 carriers of “Tetracaine”	**0.61**
2	**TCGA-AC-A62V** mutated → **GATA3** interacts with → **PIK3R1** in pathway → **Extra-nuclear estrogen signaling**	70	↓ slower progression HR 0.82, z -1.1, n = 427 carriers of “Extra-nuclear estrogen signaling”	**0.56**
3	**TCGA-AC-A62V** mutated → **GATA3** in pathway → **Interleukin-4 and Interleukin-13 signaling**	71	↓ slower progression HR 0.83, z -1.0, n = 516 carriers of “Interleukin-4 and Interleukin-13 signaling”	**0.55**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

No informative rule (log-rank p < 0.05, covering ≤ 90% of the training fold) covers this patient; the rule model falls back to the training-fold Kaplan–Meier curve.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.R329Kfs*21	13	0.31
OXR1	Mutation	p.R662H	0	0.39
MIR7851	Homozygous deletion	GISTIC ±2	0	0.33
RYR2	Mutation	p.S4006=	7	0.32
FMN2	Mutation	p.R1581I	0	0.32
RUNX1T1	Mutation	3'UTR	0	0.31



6 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

RYR2 → Flecainide (DB01195); RYR2 → Tetracaine (DB09085); RYR2 → Calcium citrate (DB11093); RYR2 → Calcium Phosphate (DB11348); RYR2 → Calcium phosphate dihydrate (DB14481). These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-BH-A1FH	47	IV	LumA	0.82	progressed at 34.0 mo (group C)
TCGA-BH-A0HL	56	II	missing	0.69	no progression, followed 2.4 mo
TCGA-C8-A273	29	II	LumA	0.56	no progression, followed 16.9 mo
TCGA-AO-A03P	54	II	LumB	0.55	progressed at 63.2 mo (group D)
TCGA-AR-A1AV	68	II	missing	0.55	no progression, followed 61.3 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

■ **Group A** · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**71**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**—**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIC**
Genetic ancestry	**EUR**	Pathologic T	**T2**
Prior diagnosis	**No**	Pathologic N	**N3A**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

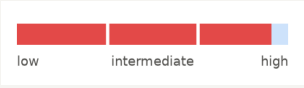
Sample: **TCGA-E2-A1LE-01** · Aneuploidy score: **13** · TMB (nonsyn., /Mb): **0** · MSIsensor score: **0.02** · AJCC edition: **6TH**

2 · Progression assessment

Model risk (cohort percentile)

94th

High risk tier · fused neuro-symbolic model



Probability of progression

21% by 36 mo

12 mo 6% · 24 mo 12% · 60 mo 35%

Model-estimated progression time

≈ 16 months

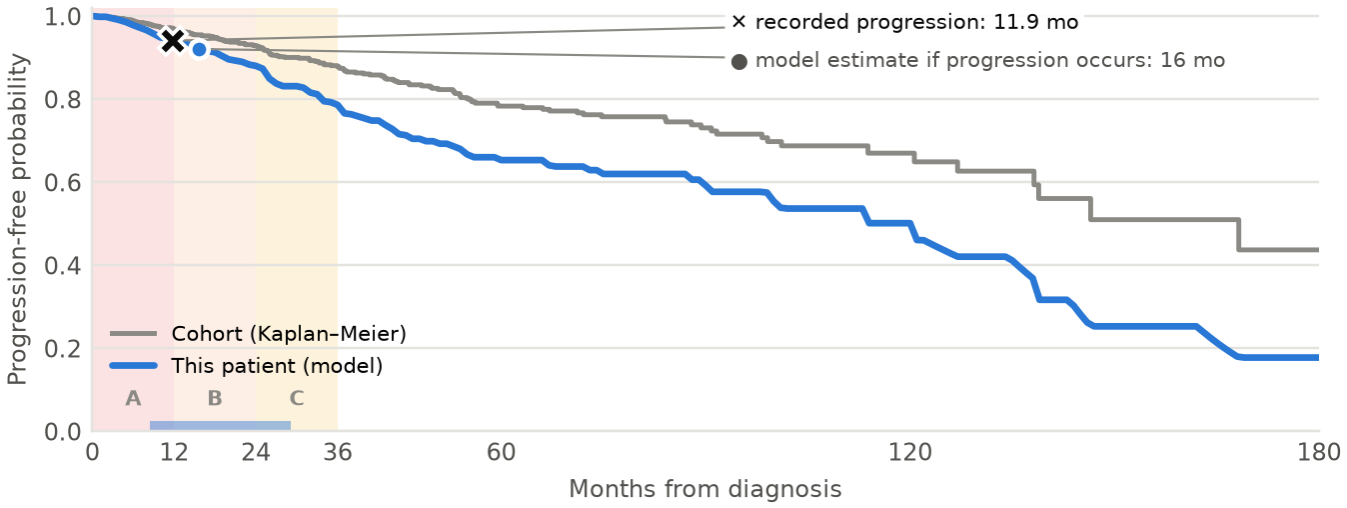
Window 12–24 months · IQR 8–29 mo if progression occurs

Recorded progression

11.9 months

■ **Group A** · Early progression

✗ **recorded window differs** · |error| 4 mo



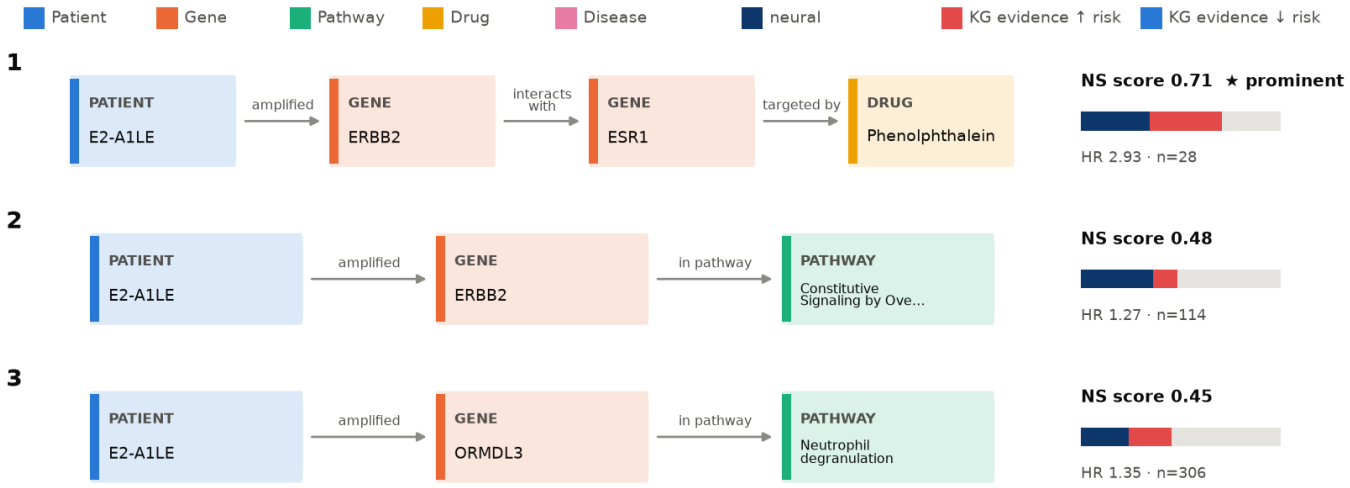
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 71-year-old patient has a subtype-unassigned tumour, AJCC stage III (T2, N3, M0). The fused neuro-symbolic model places them in the **94th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND 40.5 ≤ Age < 86.5*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **Phenolphthalein**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 2.93). Progression was recorded at 11.9 months; the model's estimate given progression was 16 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

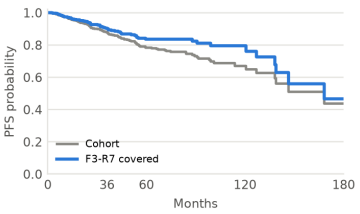


Only 1 of the 60 candidate paths clears the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 3 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-E2-A1LE** amplified → **ERBB2** interacts with → **ESR1** targeted by → **Phenolphthalein**	58	↑ **faster progression** HR 2.93, z +2.7, n = 28 carriers of “Phenolphthalein”	**0.71**
2	**TCGA-E2-A1LE** amplified → **ERBB2** in pathway → **Constitutive Signaling by Overexpressed ERBB2**	60	↑ **faster progression** HR 1.27, z +0.9, n = 114 carriers of “Constitutive Signaling by Overexpressed ERBB2”	**0.48**
3	**TCGA-E2-A1LE** amplified → **ORMDL3** in pathway → **Neutrophil degranulation**	40	↑ **faster progression** HR 1.35, z +1.6, n = 306 carriers of “Neutrophil degranulation”	**0.45**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

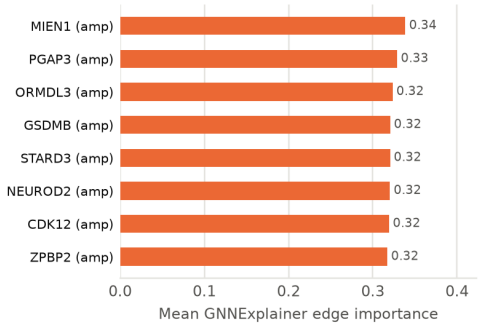
Rule	IF …	Covered	HR vs rest	Median PFS
F3-R7 ↓ lower risk	M stage = M0 AND 40.5 ≤ Age < 86.5	635 pts 66 events	**0.37** p 1.1e-07	168 mo
F3-R3 ↑ higher risk	N stage = N3 AND Aneuploidy score ≥ 2.5 AND TMB (nonsynonymous) < 5.58	47 pts 14 events	**4.25** p 1.1e-06	89 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
NF1 · driver	High-level amplification	GISTIC ±2	6	0.31
MIEN1	High-level amplification	GISTIC ±2	0	0.34
PGAP3	High-level amplification	GISTIC ±2	0	0.33
ORMDL3	High-level amplification	GISTIC ±2	2	0.32
GSDMB	High-level amplification	GISTIC ±2	4	0.32
STARD3	High-level amplification	GISTIC ±2	1	0.32
NEUROD2	High-level amplification	GISTIC ±2	0	0.32
CDK12	High-level amplification	GISTIC ±2	2	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32
PPP1R1B	High-level amplification	GISTIC ±2	1	0.32
CACNA1E	High-level amplification	GISTIC ±2	12	0.32



19 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 29 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-A2-A0YG	63	III	LumB	0.66	no progression, followed 21.9 mo
TCGA-A8-A09G	79	III	Her2	0.65	no progression, followed 0.0 mo
TCGA-D8-A3Z5	54	III	missing	0.63	no progression, followed 33.4 mo
TCGA-AO-A0JL	59	III	missing	0.63	no progression, followed 55.3 mo
TCGA-C8-A138	54	III	Her2	0.63	no progression, followed 12.5 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-B6-A0I9 · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**62**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IV**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**NX**
Radiation therapy	**Yes**	Pathologic M	**M1**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-B6-A0I9-01** · Aneuploidy score: **12** · TMB (nonsyn., /Mb): **2.4** · MSIsensor score: **0.58** · AJCC edition: **3RD**

2 · Progression assessment

Model risk (cohort percentile)

100th

High risk tier · fused neuro-symbolic model



Probability of progression

72% by 36 mo

12 mo 27% · 24 mo 52% · 60 mo 91%

Model-estimated progression time

≈ 14 months

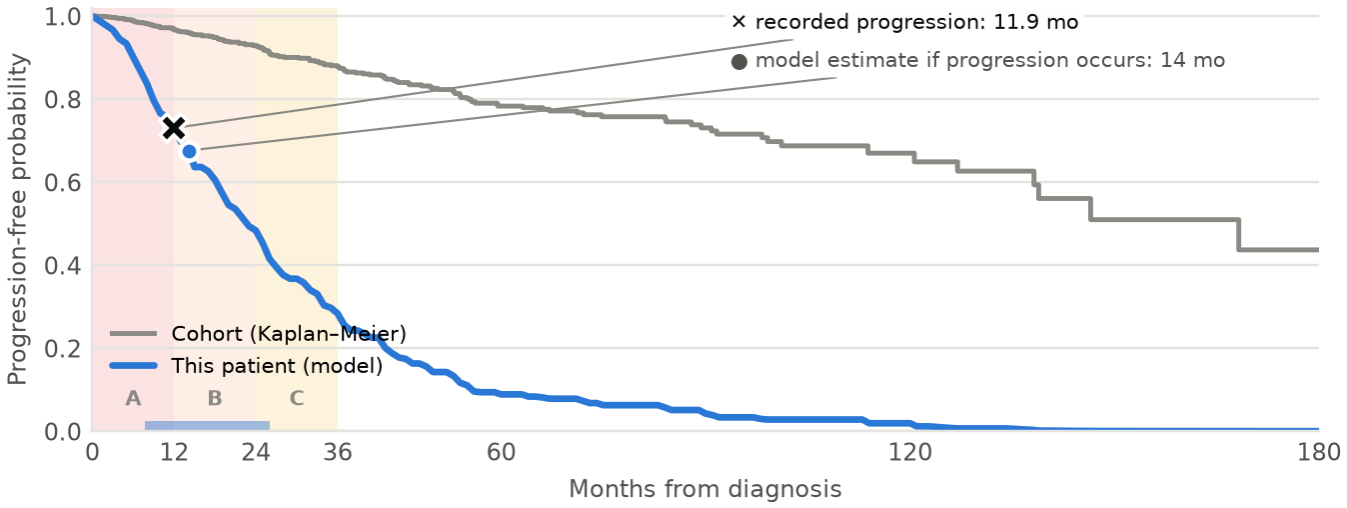
Window 12–24 months · IQR 8–26 mo if progression occurs

Recorded progression

11.9 months

Group A · Early progression

✗ recorded window differs · |error| 2 mo



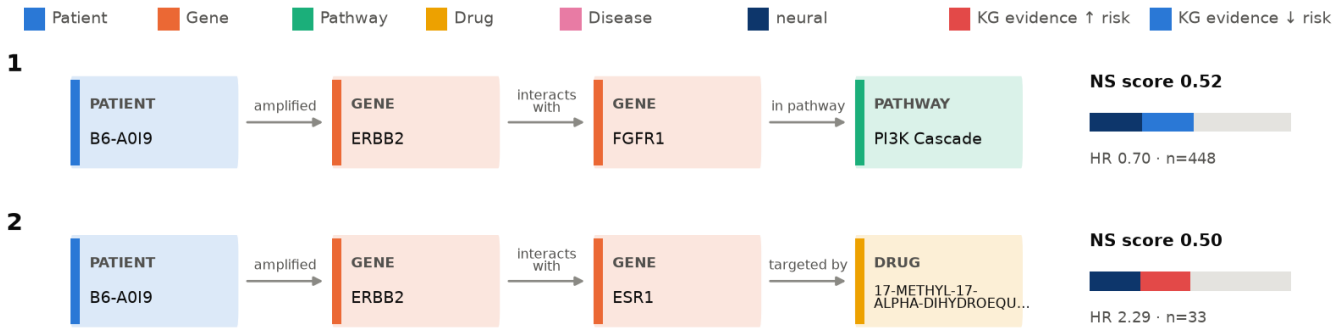
Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 62-year-old patient has a Her2 tumour, AJCC stage IV (T3, missing, M1). The fused neuro-symbolic model places them in the **100th risk percentile** (high tier). The most significant rule covering them (*TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5*) marks a group with a progression hazard 0.6× that of the remaining patients. The highest-scoring knowledge-graph path runs from **ERBB2** to **PI3K Cascade**; in the training folds, carriers of this PrimeKG fact showed slower progression (HR 0.70). Progression was recorded at 11.9 months; the model's estimate given progression was 14 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

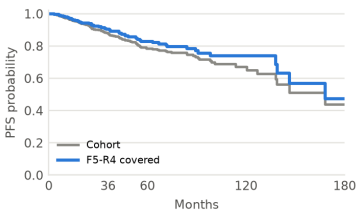


None of the 60 candidate paths clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile); the best remaining paths are shown so that 2 appear, and they are not marked ★. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1	**TCGA-B6-A0I9** amplified → **ERBB2** interacts with → **FGFR1** in pathway → **PI3K Cascade**	43	↓ **slower progression** HR 0.70, z -1.9, n = 448 carriers of "PI3K Cascade"	**0.52**
2	**TCGA-B6-A0I9** amplified → **ERBB2** interacts with → **ESR1** targeted by → **17-METHYL-17-ALPHA-DIHYDROEQUILENIN**	42	↑ **faster progression** HR 2.29, z +1.9, n = 33 carriers of "17-METHYL-17-ALPHA-DIHYDROEQUILENIN"	**0.50**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

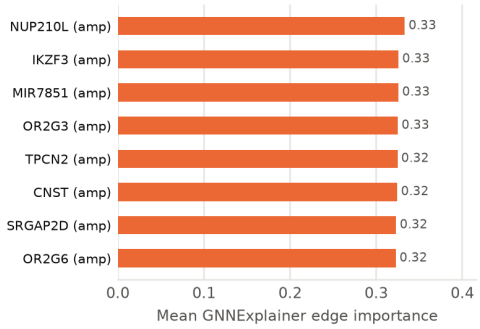
Rule	IF …	Covered	HR vs rest	Median PFS
F5-R4 ↓ **lower risk**	TMB (nonsynonymous) ≥ 0.82 AND Age < 78.5 AND Aneuploidy score < 29.5	567 pts 64 events	**0.56** p 2.1e-03	168 mo
F5-R5 ↓ **lower risk**	0.22 ≤ TMB (nonsynonymous) < 14.42 AND Age ≥ 39.5	719 pts 85 events	**0.54** p 4.6e-03	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
GATA3 · driver	Mutation	p.P408Afs*99	13	0.32
CCND1 · driver	High-level amplification	GISTIC ±2	26	0.32
TP53 · driver	Mutation	p.C135Y	57	0.31
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
EGFR · driver	Homozygous deletion	GISTIC ±2	73	0.31
NUP210L	High-level amplification	GISTIC ±2	0	0.33
IKZF3	High-level amplification	GISTIC ±2	0	0.33
MIR7851	High-level amplification	GISTIC ±2	0	0.33
OR2G3	High-level amplification	GISTIC ±2	1	0.33
TPCN2	High-level amplification	GISTIC ±2	1	0.32
CNST	High-level amplification	GISTIC ±2	0	0.32
SRGAP2D	High-level amplification	GISTIC ±2	0	0.32



148 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

CCND1 → Acetylsalicylic acid (DB00945); CCND1 → Arsenic trioxide (DB01169); CCND1 → Encorafenib (DB11718); CCND1 → Bryostatin 1 (DB11752); TP53 → Acetylsalicylic acid (DB00945); TP53 → Zinc (DB01593); TP53 → Triethyl phosphate (DB03347); TP53 → AZD 3355 (DB05404); TP53 → 1-(9-ethyl-9H-carbazol-3-yl)-N-methylmethanamine (DB08363); TP53 → Zinc acetate (DB14487) … and 130 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A12L	67	II	Her2	0.77	no progression, followed 11.9 mo
TCGA-A8-A09C	69	X	LumB	0.76	no progression, followed 1.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.74	no progression, followed 140.9 mo
TCGA-AN-A03X	74	II	LumA	0.74	no progression, followed 0.3 mo
TCGA-C8-A1HL	38	III	LumB	0.72	no progression, followed 10.4 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.

Neuro-Symbolic Progression Report

PrimeKG Patient Reasoning · Precision Oncology Research

Patient TCGA-BH-A1EV · Group A

Report date 14 Oct 2026

Group A · Early progression Breast invasive carcinoma · TCGA PanCancer Atlas 2018 · model: neurosymbolic cox

1 · Patient and tumour record

Age at diagnosis	**45**	Tumour type	**Breast Invasive Ductal Carcinoma**
Sex	**Female**	OncoTree code	**IDC**
Race	**White**	PAM50 subtype	**Her2**
Ethnicity	**Not Hispanic Or Latino**	AJCC pathologic stage	**STAGE IIIA**
Genetic ancestry	**EUR**	Pathologic T	**T3**
Prior diagnosis	**No**	Pathologic N	**N1B**
Radiation therapy	**Yes**	Pathologic M	**M0**
Neoadjuvant therapy	**No**	ICD-O-3 histology	**8500/3**

Sample: **TCGA-BH-A1EV-01** · Aneuploidy score: **18** · TMB (nonsyn., /Mb): **2.83** · MSIsensor score: **0.41** · AJCC edition: **5TH**

2 · Progression assessment

Model risk (cohort percentile)

96th

High risk tier · fused neuro-symbolic model



Probability of progression

24% by 36 mo

12 mo 5% · 24 mo 14% · 60 mo 42%

Model-estimated progression time

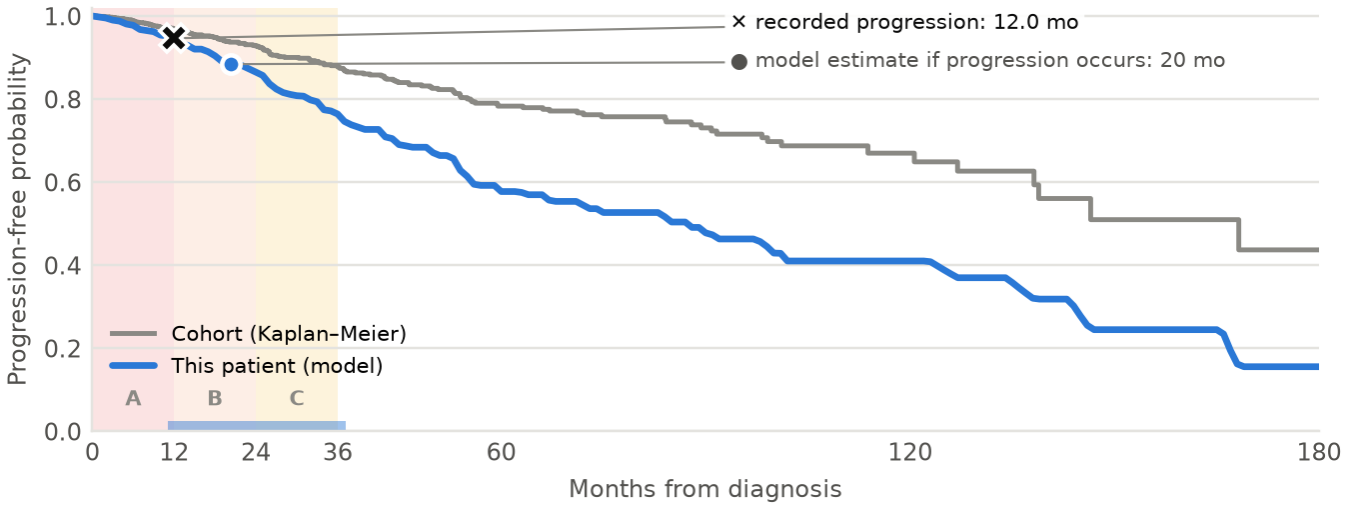
≈ 20 months

Window 12–24 months · IQR 11–37 mo
if progression occurs

Recorded progression

12.0 months

Group A · Early progression
✗ recorded window differs · |error| 8 mo



Blue: this patient's progression-free survival curve from the fused model, fitted on the four folds that exclude the patient. Grey: Kaplan–Meier curve of the whole cohort. Shaded bands mark the report groups A–C; the bar on the time axis is the model's interquartile range for the month of progression, given that progression occurs.

Summary

This 45-year-old patient has a Her2 tumour, AJCC stage III (T3, N1, M0). The fused neuro-symbolic model places them in the **96th risk percentile** (high tier). The most significant rule covering them (*M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02*) marks a group with a progression hazard 0.4× that of the remaining patients. The highest-scoring knowledge-graph path runs from **NFASC** to **Neurofascin interactions**; in the training folds, carriers of this PrimeKG fact showed faster progression (HR 1.51). Progression was recorded at 12.0 months; the model's estimate given progression was 20 months.

3 · Neuro-symbolic reasoning

3a · Top knowledge-graph paths (GNN-scored, PrimeKG-grounded)

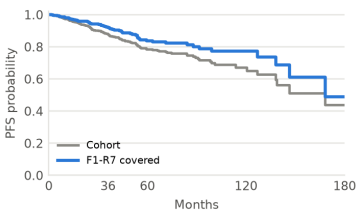


5 of 60 candidate paths shown; all clear the prominence cutoff (NS ≥ 0.57, the cohort's 80th percentile), capped at K = 5. NS score = 0.6 × neural percentile (GNNExplainer importance from the held-out fold model) + 0.4 × symbolic strength (|log-rank z| of the path's PrimeKG fact on the training folds, capped at 3).

#	Path	Neural pct	PrimeKG fact evidence (training folds)	NS
1 ★	**TCGA-BH-A1EV** amplified → **NFASC** in pathway → **Neurofascin interactions**	64	↑ **faster progression** HR 1.51, z +2.0, n = 173 carriers of “Neurofascin interactions”	**0.65**
2 ★	**TCGA-BH-A1EV** amplified → **CDK12** interacts with → **BRCA1** in pathway → **Meiotic synapsis**	66	↑ **faster progression** HR 1.44, z +1.6, n = 142 carriers of “Meiotic synapsis”	**0.61**
3 ★	**TCGA-BH-A1EV** amplified → **ATP1A4** in pathway → **Potential therapeutics for SARS**	60	↑ **faster progression** HR 1.52, z +1.8, n = 139 carriers of “Potential therapeutics for SARS”	**0.61**
4 ★	**TCGA-BH-A1EV** mutated → **SHANK2** in pathway → **Neurexins and neuroligins**	64	↓ **slower progression** HR 0.69, z -1.6, n = 210 carriers of “Neurexins and neuroligins”	**0.59**
5 ★	**TCGA-BH-A1EV** amplified → **CR1** in pathway → **Neutrophil degranulation**	61	↑ **faster progression** HR 1.36, z +1.7, n = 316 carriers of “Neutrophil degranulation”	**0.58**

3b · Symbolic survival rules that fire for this patient (RuleKit LR-Rules)

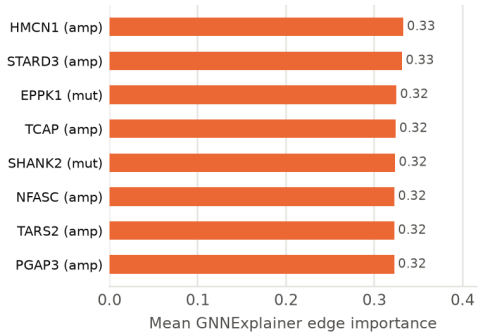
Rule	IF …	Covered	HR vs rest	Median PFS
F1-R7 ↓ lower risk	M stage = M0 AND Age ≥ 40.5 AND TMB (nonsynonymous) ≥ 0.02	609 pts 61 events	**0.36** p 1.8e-08	168 mo



Rules were learned by RuleKit's log-rank rule induction on the four folds that exclude this patient; counts, hazard ratios, and Kaplan–Meier medians are on that training data. [KG] marks a rule whose body contains a PrimeKG fact.

4 · Molecular profile and knowledge-graph context

Gene	Alteration	Detail	PrimeKG links	GNN imp.
ERBB2 · driver	High-level amplification	GISTIC ±2	55	0.31
BRCA2 · driver	High-level amplification	GISTIC ±2	14	0.31
HMCN1	High-level amplification	GISTIC ±2	0	0.33
STARD3	High-level amplification	GISTIC ±2	1	0.33
EPPK1	Mutation	p.L1474=	0	0.32
TCAP	High-level amplification	GISTIC ±2	1	0.32
SHANK2	Mutation	p.P1240T	1	0.32
NFASC	High-level amplification	GISTIC ±2	3	0.32
TARS2	High-level amplification	GISTIC ±2	2	0.32
PGAP3	High-level amplification	GISTIC ±2	0	0.32
MIR4728	High-level amplification	GISTIC ±2	0	0.32
ZPBP2	High-level amplification	GISTIC ±2	0	0.32



46 altered panel genes in total; known drivers first, then by GNN importance. CNA follows the graph policy: only homozygous deletion (–2) and high-level amplification (+2).

PrimeKG drug–protein links of the altered genes

ERBB2 → Trastuzumab (DB00072); ERBB2 → Lapatinib (DB01259); ERBB2 → IGN311 (DB04988); ERBB2 → Trastuzumab emtansine (DB05773); ERBB2 → Varlitinib (DB05944); ERBB2 → AV-412 (DB06021); ERBB2 → Pertuzumab (DB06366); ERBB2 → Afatinib (DB08916); ERBB2 → Tucatinib (DB11652); ERBB2 → Tesevatinib (DB11973) … and 36 more. These are knowledge-graph links (target, enzyme, carrier, or transporter), shown for research context only; they are not treatment recommendations.

Case-based context

Most similar patients	Age	Stage	Subtype	Similarity	Recorded outcome
TCGA-C8-A1HL	38	III	LumB	0.70	no progression, followed 10.4 mo
TCGA-A8-A07I	69	III	Her2	0.66	no progression, followed 14.0 mo
TCGA-B6-A0IQ	40	III	Basal	0.62	no progression, followed 140.9 mo
TCGA-B6-A0WX	40	III	Basal	0.62	progressed at 18.5 mo (group B)
TCGA-A8-A09B	58	III	LumA	0.62	no progression, followed 12.0 mo

Cosine similarity over stage, subtype, histology, age, driver alterations, and PrimeKG pathway facts. Outcomes are recorded follow-up, shown as context, not as a prediction.

How to read this report

The risk and the survival curve come from a Cox model on the clinical block (age, sex, PAM50 subtype, AJCC stage, M stage) plus the GNN risk and the RuleKit rule risk, all fitted on the four folds that exclude this patient. In cross-validation the GNN and rule terms carry close to zero weight once the clinical block is present, so the risk mostly reflects clinical stage, nodal status, age, and subtype. The knowledge-graph paths and rules explain which PrimeKG facts the graph model attended to and how patients sharing those facts fared; they are evidence, not causal claims. The progression-time estimate (loglogistic aft on progressors) is within 12 months of the recorded month for 51% of held-out progressors and in the recorded window for 34%; the recorded month is always shown beside it.